Derived manifolds

MIGOS seminar notes, University of Regensburg, 2026/27

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Most of this document is AI-generated

See *A note from the organizers* for more information.

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The program

Goals

One primary source of difficulties in the classical theory of manifolds are the singularities created by nontransverse intersections. The theory of derived manifolds provides a solution to this problem: it enlarges the class of smooth geometric objects to allow for well-behaved nontransverse intersections, preserving the transverse ones among manifolds.

In recent years, it is becoming clear that derived manifolds are a natural home for various moduli problems in geometry. One of the main goals of this seminar is to understand Steffens's theorem on derived representability for elliptic moduli problems, which says that the moduli space of solutions to an elliptic PDE is locally given by derived manifolds. Such moduli spaces are important in much of modern geometric topology, such as symplectic topology, Floer theory, complex geometry and the theory of 4-manifolds.

The seminar is intended to be accessible to first-year PhD students joining the CRC. In an attempt to keep the seminar as self-contained and self-motivated as possible, we devote a large part of the seminar to learning the background and tools from derived smooth geometry. Along the way, we will also encounter various higher categorical methods that make this theory work well, including the concepts of (homotopy) pullbacks in ∞-categories, derived ∞-categories, and (higher) stacks. Given the time limitations, these topics necessarily receive shorter treatment than they deserve, but we hope to illustrate why they are indispensible tools for the research of the CRC.

Each talk lasts 90 minutes, including questions from the audience.

Program

Talk 1, 13 October: Why derived manifolds?

Aim. Motivate derived manifolds through nontransverse intersections and elliptic moduli problems

Content. Prove that pullbacks of transverse intersection exist in the category of manifolds. Discuss the issues with non-transverse intersection. As a running example, compare a transverse intersection of two lines with a tangential intersection between a line and a parabola. Using more examples, explain how remembering the differential as a two-term complex solves some, but not all, problems. This shows that the correct home for such intersections is necessarily "derived". Give an informal discussion of the universal property of derived manifolds as a universal enlargement preserving transverse intersections.

End with giving a brief idea of moduli spaces of elliptic equations. Discuss the local finite-dimensional reduction of an elliptic equation, which allows one to present such moduli spaces locally as non-transverse intersections. Formulate the nonuniqueness and coherence problem for the resulting local presentations, again motivating derived manifolds.

Primary references. Spivak [Spi10, Section 1]; Steffens [Ste22, Chapter 1, especially Section 1.0.1]; Pardon [Par24, Sections 1–2].

Talk 2, 20 October: Introduction to C^∞-rings

Aim. Encode smooth manifolds through their rings of smooth functions.

Content. Define C^∞-rings and give their description as functors. Prove that $C^\infty(\mathbb{R}^n)$ is free on n generators. Prove Hadamard's lemma and discuss quotients of C^∞-rings. Define $C^\infty(X)$ for $X \subset \mathbb{R}^n$ an arbitrary closed subset. Prove that the resulting functor from closed subsets of euclidean spaces to C^∞-rings is fully faithful.

Primary references. Moerdijk and Reyes [MR91, Chapter I, Section 1].

Talk 3, 27 October: Geometric constructions in C^∞-rings

Aim. Understand the algebraic analogues of products and transverse pullbacks of manifolds.

Content. Introduce coproducts of C^∞-rings. Prove Lemma 2.1 in Moerdijk–Reyes, and deduce that $C^\infty(M)$ is finitely presented for every manifold M. Prove that $C^\infty(M \times N) \simeq C^\infty(M) \otimes C^\infty(N)$. Prove that transverse pullbacks in manifolds induce pushouts of C^∞-rings.

Primary references. Moerdijk and Reyes [MR91, Chapter I, Section 2].

Talk 4, 3 November: Derived manifolds

Aim. Understand the universal property of the ∞-category of derived manifolds.

Content. Formulate the 1-categorical universal property of finitely presented C^∞-rings. Use self-intersections to illustrate why ordinary pullbacks lose geometric information. Briefly introduce the idea behind ∞-categories and homotopy pullbacks, without going into a model. Define derived manifolds via the ∞-categorical analogue of the previous universal property. State the theorem of Carchedi–Steffens identifying them with finitely presented derived C^∞-rings, discussing as much of the proof as time permits.

Primary references. Carchedi and Steffens [CS19].

Talk 5, 10 November: Calculating derived intersections

Aim. Learn to calculate derived zero loci via explicit dg-algebra models

Content. Introduce C^∞-dg-algebras, state the C^∞-Dold–Kan correspondence, and explain how it identifies derived zero loci with homotopy pushouts of C^∞-dg-algebras. Construct the Koszul model for the derived zero locus of a smooth map $f\colon \mathbb{R}^n \to \mathbb{R}^r$ by resolving the evaluation map $C^\infty(\mathbb{R}^r) \to \mathbb{R}$. Explain the extension to families. Give examples, such as Spivak's Example 2.7 and the zero loci of the equations x, x^2, and 0, distinguishing the underlying zero set, classical truncation, and higher homotopy groups of the animated ring of functions.

Primary references. Steffens [Ste23, Definition 3.3.1, Theorem 3.3.12, and Examples 3.3.16–3.3.17]; Spivak [Spi10, Example 2.7].

Talk 6, 17 November: Tangent and cotangent complexes

Aim. Learn about (co)tangent complexes of derived manifolds.

Content. Define C^∞-derivations, the module of smooth Kähler differentials, and its relation to the cotangent bundle of a manifold. Explain the need for a cotangent *complex* in the derived setting. Introduce the derived ∞-category of real vector spaces, and discuss shifts and fiber/cofiber sequences, making the analogy to short exact sequences in abelian categories. Introduce derived square-zero extensions and C^∞-derivations, and define absolute and relative cotangent complexes. State base change and the transitivity cofiber sequence for cotangent complexes, and use that to calculate the tangent complex of the derived zero locus of a section $s\colon U \to E$ at a zero x:

$$[T_x U \xrightarrow{d_x s} E_x].$$

Interpret its homology in degrees 0 and -1 as deformation and obstruction spaces.

Primary references. Steffens [Ste22, Section 5]; Pascarella [Pas25]; Spivak [Spi10, Section 7].

Talk 7, 24 November: Quasi-smooth derived geometry and Kuranishi charts

Aim. Understand quasi-smooth derived C^∞-schemes and their local descriptions via Kuranishi charts.

Content. Introduce localizations of derived C^∞-rings, stating flatness as a black-box, and define the spectrum of a derived C^∞-ring. Define derived C^∞-schemes locally of finite presentation, and identify the affine objects with derived manifolds. Define quasi-smooth derived C^∞-schemes. Introduce Kuranishi charts (U, E, s) and their associated derived manifolds, and discuss the normal-form theorem: every quasi-smooth derived C^∞-scheme of finite presentation is locally presented by a Kuranishi chart. Illustrate the non-uniqueness of the presentation by discussing stabilization of charts.

Primary references. Steffens [Ste22, Chapter 1 and Sections 4.1.3 and 5.1.1].

Talk 8, 1 December: Derived bordism and fundamental classes

Aim. Understand the benefits of derived bordism over classical bordism.

Content. Following Spivak [Spi10], explain how quasi-smooth derived manifolds are good for doing intersection theory. Define derived cobordism over X. State proposition 3.3, recalling its proof for derived C^∞-rings. State proposition 3.4. Prove that the comparison map

$$\Omega(X) \xrightarrow{\cong} \Omega^{\mathrm{der}}(X).$$

is an isomorphism by transverse pertubation. Deduce the intersection formula

$$[A] \smile [B] = [A \times^{\mathrm{der}}_M B]$$

without transversality hypothesis. Discuss the derived self-intersection of the zero section of a vector bundle as the main example, recovering its Euler class.

Primary references. Spivak [Spi10, Sections 1–3].

Talk 9, 8 December: Stacks and local representability

Aim. Understand how stacks describe moduli problems.

Content. Define open coverings of derived manifolds and the resulting notion of (∞-)stacks on the category of derived manifolds via higher categorical descent. Briefly explain the comparison with smooth stacks. Define when such a derived C^∞-stack is representable by a derived C^∞-scheme, giving the derived zero locus of a section as a running example. Define effective epimorphisms and atlases. Prove the local criterion for representability, and explain how an open atlas produces a global representing derived C^∞-scheme. Introduce relative representability and its locality on the base.

Primary references. Steffens [Ste24, Section 2.1].

Talk 10, 15 December: Section stacks and derived solution stacks

Aim. Construct the moduli stack of solutions to a differential operator

Content. Motivate differential operators and their solutions (e.g. harmonic functions, critical points of nonlinear energies, pseudo-holomorphic curves), and explain how to see these as solutions for $P(u) = b$ for a map $P\colon \Gamma_N(X \to N) \to \Gamma_N(Y \to N)$ and a section b. Introduce jet stacks to define finite-order differential operators and their induced maps on sections. Explain the relative version for smooth families $M \to S$, and briefly discuss how descent allows S to be a stack. Define differential moduli problems and their solution stacks. Explain linearization and *elliptic* differential moduli problems. As a running example, use

$$P(u) = \Delta u + u^2$$

on a closed connected Riemannian manifold.

Primary references. Steffens [Ste24, Definition 3.2.1, Definition 3.3.7, Proposition 3.3.9, and Definitions 3.4.1–3.4.2]; Pardon [Par24, Section 4].

Talk 11, 12 January: Elliptic representability I: local reductions

Aim. Reduce the representability problem to section stacks of vector bundles

Content. State the seminar's main theorem: Steffens's relative elliptic representability theorem [Ste24, Theorem 3.4.3]. For most of the talk, fix $S = \mathrm{pt}$ and consider a moduli problem of the form $P(u) = b$ for a finite-order elliptic differential operator P and a section b. Use relative exponential maps and Lemma 3.4.4 to identify its solution stack locally around a solution u_0 with the solution stack of a moduli problem of the form $\widetilde{P}(u) = 0$ for $\widetilde{P}\colon Q \to \mathrm{Map}_N(N, E)$, where $E, F \to N$ are vector bundles and $Q \subseteq \mathrm{Map}_N(N, F)$ is an open substack. Explain how varying the chosen solution gives an effective open cover, so that local representability implies representability. Next, explain how the same coordinate construction reduces a moduli problem $P_1(u_1) = P_2(u_2)$ to this form. Finally, explain what changes when the base S is a general manifold or a smooth stack.

Primary references. Steffens [Ste24, Theorem 3.4.3, Proposition 3.2.19, Lemmas 3.2.20–3.2.21, Proposition 3.3.9, and Lemma 3.4.4]; Pardon [Par24, Introduction and the beginning of Section 5].

Talk 12, 19 January: Elliptic representability II: the local Kuranishi criterion

Aim. Explain the formal thickening argument that produces Kuranishi charts.

Content. Continue with $\widetilde{P}(u) = 0$ over $S = \mathrm{pt}$. For a finite-dimensional vector space V and a linear map $\iota\colon V \to \Gamma(N; E)$, introduce the augmented operator $\overline{P}(u, v) := \widetilde{P}(u) + \iota(v)$. Show that projection to V identifies the original solution stack with the derived pullback

$$\mathrm{Sol}(\widetilde{P}) \;\simeq\; \mathrm{Sol}(\overline{P}) \times^{\mathrm{der}}_{V} \{0\}.$$

Deduce that this gives a local Kuranishi presentation whenever the augmented solution stack is represented by a finite-dimensional manifold near the chosen solution.

Regard the Fréchet space of smooth sections as a smooth stack, and compare it to the derived section stack. Introduce submersive maps of stacks, and show that passing from smooth to derived stacks preserves pullbacks. Deduce that the augmented solution stack is a finite-dimensional manifold whenever the augmented operator is submersive on a suitable neighbourhood. Explain the relative version over S.

Primary references. Steffens [Ste24, Propositions 2.2.17–2.2.18 and Corollary 2.2.19, Propositions 3.1.7 and 3.1.9 and Lemma 3.2.11, the final pullback argument following the proof of claim $(*)$ in Theorem 3.4.3, Proposition 2.1.48 and Corollary 2.1.49].

Talk 13, 26 January: Elliptic representability III: the analytic input

Aim. Use nonlinear elliptic Fredholm analysis to make the augmented operator into a submersive map of smooth stacks

Content. Start with $S = \mathrm{pt}$. Introduce the Sobolev spaces of sections $H^{\ell}(E; N)$ and explain how a differential operator of order k extends to compatible maps

$$\widetilde{P}_{\ell}\colon Q_{k+\ell} \longrightarrow H^{\ell}(E; N)$$

for sufficiently large ℓ. Define Fredholm operators, and state as an analytic black-box that the Sobolev extensions of a linear elliptic operator on a compact manifold are Fredholm. Given a solution t of $\widetilde{P}$, use Fredholmness to choose a finite-dimensional augmentation whose derivative is surjective near $(t, 0)$.

For an ordinary manifold Z and a smooth family $h\colon Z \to \Gamma(N; E)$, apply the Banach implicit function theorem to the augmented equation $\widetilde{P}(u) + \iota(a) = h(z)$. Use linear and nonlinear elliptic regularity to show that the resulting solution manifold represents the pullback of smooth stacks and projects submersively to Z, and conclude the submersiveness of the augmented operator as required by the previous talk. Explain the relative version over S, and conclude the representability theorem. Illustrate the argument with $P(u) = \Delta u + u^2$ near $u = 0$.

Primary references. Steffens [Ste24, Lemmas 3.4.5–3.4.6, the analytic part of the proof of Theorem 3.4.3]

Talk 14, 2 February: Pseudo-holomorphic curves as a derived moduli problem

Aim. Apply the representability theorem to a key geometric application.

Content. Describe families of maps from smooth compact Riemann surfaces to an almost complex manifold and formulate the Cauchy–Riemann equation as a section of the bundle of

complex-antilinear maps. Explain the issue with the differential moduli problem in Steffens's Construction 4.0.1, and use local coordinates and a trivialization of the equation bundle to obtain an elliptic operator between sections of vector bundles. Identify its linearization and prove its ellipticity. Apply the representability theorem locally and use descent to obtain the derived moduli stack of pseudo-holomorphic curves. Write the relative deformation-obstruction complex and interpret its homology. Given time, briefly discuss compactification and applications to enumerative questions.

Primary references. Wendl [Wen14, Sections 2.1, 2.3–2.4, and 3.3–3.4]; Steffens [Ste24, Section 4].

A note from the organizers

These notes are AI-generated via Codex, model GPT-6 Astra. They were made in response to the program for the MIGOS seminar on derived manifolds held at the university of Regensburg in the winter semester 2026/27.

Each chapter is a draft of the corresponding seminar talk. We originally generated these notes to aid us in working out the details of the program and to ensure each talk was feasible. We have decided to publicly share them because they could be a valuable resource for participants for reviewing the content of the seminar, finding precise references and drafting their talks. Each chapter also has additional material which we expect will be beyond the scope of the talk, but which may be interesting to the participants.

That being said, it is important to keep in mind that most of this document was not created directly by us and its AI-generated nature should be taken into account at all times. While we have confirmed that there are no egregious errors and the structure of the chapters more or less follows the program, the text may still contain mathematical errors, misleading explanations, or mediocre writing. We recommend treating this document as one might treat the work of a smart but lazy fellow student: correct and insightful mathematics may be interlaced with sloppy or outright incorrect material. The primary source material listed in the program remains the ultimate authorative source, and this document merely aids in orientation.

We also ask the speakers in the seminar to think critically about the contents and organization of their talks, and avoid directly following the AI-generated write-up. One of the main purposes of a reading seminar like this is for the participants to learn various skills that are crucial for being a mathematician, like being able to write accurately yet concisely, to separate the crucial steps in a proof from routine verification, to condense a theory into its core concepts and ideas, and to effectively communicate the results to fellow mathematicians. While it might be tempting to outsource these things to AI, this would undermine the ultimate purpose of the seminar.

To repeat: we advise caution with using these notes. Everything beyond this point is AI-generated.

Conventions

Manifolds are finite-dimensional, Hausdorff, and second countable, and have no boundary unless stated otherwise. A closed manifold is compact and has no boundary. Throughout our derived ∞-categories we use chain complexes with homological grading: their differentials lower degree by one. The dg-algebra models for animated smooth rings are concentrated in nonnegative degrees, with $\pi_i = H_i$.

1 Why derived manifolds?

1.1 Transverse fiber products

Consider the following two intersections of curves in $\mathbb{R}^2$.

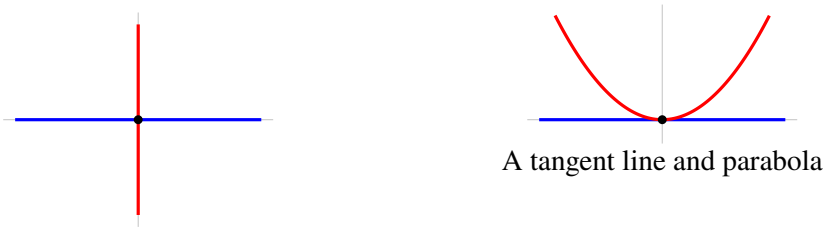


Figure 1.1: Two intersections with the same underlying set.

Set-theoretically, the answer is the same in both panels of Figure 1.1: the intersection consists only of the origin. Geometrically, the two answers behave very differently. The transverse intersection is stable locally under sufficiently small perturbations. For the tangent intersection, replace the parabola by

$$y = x^2 + \varepsilon.$$

Its intersection with the x-axis consists of two points when $\varepsilon < 0$, one point when $\varepsilon = 0$, and no points when $\varepsilon > 0$. For the fixed equation at $\varepsilon = 0$, we will find an infinitesimal solution direction which cannot extend to an actual curve of solutions. This is distinct from varying the equation: if ε is allowed to vary, then $x(\tau) = \tau$ and $\varepsilon(\tau) = -\tau^2$ do give a curve of solutions in the total family.

Question. What geometric object should replace a nontransverse intersection so that it remembers the defining equations, agrees with the ordinary intersection in the transverse case, and behaves well in families?

Derived manifolds provide such an object. Transversality gives the first test for the construction: a derived intersection must recover the ordinary manifold when the derivatives are independent. The nontransverse examples will show what additional information is needed. The same intersection-theoretic problem is the starting point of Spivak's construction of derived smooth manifolds [Spi10, Sections 1–3].

Let $f\colon X \to Z$ and $g\colon Y \to Z$ be smooth maps between smooth manifolds. Their fiber product as sets is

$$X \times_Z Y = \{(x, y) \in X \times Y \mid f(x) = g(y)\}.$$

Without an additional hypothesis, this subset of $X \times Y$ need not be a smooth manifold of the expected dimension. Transversality is the hypothesis which rules out this defect.

Definition 1.1 (Transverse maps)**.** Let $f\colon X \to Z$ and $g\colon Y \to Z$ be smooth maps. For $(x, y) \in X \times_Z Y$, with common image $z = f(x) = g(y)$, define the *linearized matching map*

$$\delta_{x,y} = D_x f - D_y g\colon T_x X \oplus T_y Y \longrightarrow T_z Z.$$

We say that f and g are *transverse* if $\delta_{x,y}$ is surjective for every $(x, y) \in X \times_Z Y$.

Equivalently, f and g are transverse if

$$\operatorname{im}(D_x f) + \operatorname{im}(D_y g) = T_z Z$$

whenever $f(x) = g(y) = z$. In particular, if $A, B \subseteq Z$ are embedded submanifolds and f and g are their inclusions, this condition becomes

$$T_z A + T_z B = T_z Z$$

at every $z \in A \cap B$.

We will use the following standard consequence of the regular value theorem.

Proposition 1.2 (Transverse preimages)**.** *Let $h\colon M \to N$ be a smooth map and let $S \subseteq N$ be an embedded submanifold. Suppose that*

$$\operatorname{im}(D_x h) + T_{h(x)} S = T_{h(x)} N$$

for every $x \in h^{-1}(S)$. Then $h^{-1}(S)$ is an embedded submanifold of M of codimension $\operatorname{codim}_N(S)$, and

$$T_x h^{-1}(S) = (D_x h)^{-1}(T_{h(x)} S)$$

for every $x \in h^{-1}(S)$.

This result is also called the transverse preimage theorem; see [Lee02, Chapter 8]. We take it as recalled differential topology and use it to prove the principal result of this section.

Theorem 1.3 (Transverse fiber product theorem)**.** *Let $f\colon X \to Z$ and $g\colon Y \to Z$ be transverse smooth maps. Then $X \times_Z Y$ is a smooth manifold. If it is nonempty, its dimension is*

$$\dim(X \times_Z Y) = \dim X + \dim Y - \dim Z.$$

For every $(x, y) \in X \times_Z Y$, there is a natural isomorphism

$$T_{(x,y)}(X \times_Z Y) \cong \ker\bigl(\delta_{x,y}\colon T_x X \oplus T_y Y \to T_z Z\bigr).$$

Proof. Consider the product map

$$h = f \times g\colon X \times Y \longrightarrow Z \times Z, \qquad h(x, y) = (f(x), g(y)).$$

If $\Delta_Z \subseteq Z \times Z$ denotes the diagonal, then

$$X \times_Z Y = h^{-1}(\Delta_Z).$$

At a point $(z, z) \in \Delta_Z$, its tangent space is

$$T_{(z,z)}\Delta_Z = \{(w, w) \mid w \in T_z Z\}.$$

The linear map

$$\pi_z \colon T_zZ \oplus T_zZ \longrightarrow T_zZ, \qquad \pi_z(a,b) = a - b,$$

is surjective and has kernel $T_{(z,z)}\Delta_Z$. Consequently, h is transverse to Δ_Z at (x,y) if and only if the composite

$$T_xX \oplus T_yY \xrightarrow{D_{(x,y)}h} T_zZ \oplus T_zZ \xrightarrow{\pi_z} T_zZ$$

is surjective. Here we use the elementary fact that a linear map to a vector space W is transverse to a subspace $U \subseteq W$ if and only if its composite with $W \to W/U$ is surjective. The displayed composite is precisely

$$(u,v) \longmapsto D_xf(u) - D_yg(v) = \delta_{x,y}(u,v).$$

Thus the transversality of f and g is equivalent to the transversality of h to Δ_Z.

By Proposition 1.2, the inverse image $h^{-1}(\Delta_Z)$ is a smooth submanifold of $X \times Y$. Since the diagonal has codimension $\dim Z$ in $Z \times Z$, this proves the dimension formula.

The tangent-space statement in Proposition 1.2 gives

$$\begin{aligned} T_{(x,y)}h^{-1}(\Delta_Z) &\cong \{(u,v) \in T_xX \oplus T_yY \mid (D_xf(u), D_yg(v)) \in T_{(z,z)}\Delta_Z\} \\ &= \{(u,v) \in T_xX \oplus T_yY \mid D_xf(u) = D_yg(v)\} \\ &= \ker(\delta_{x,y}), \end{aligned}$$

as asserted.

Finally, suppose a manifold W has smooth maps to X and Y whose composites to Z agree. Their product is a smooth map to $X \times Y$ with image in the embedded submanifold $h^{-1}(\Delta_Z)$, so its corestriction is smooth. It is the unique map compatible with the two projections. Thus the constructed manifold is the categorical pullback in manifolds. □

In particular, under the hypotheses of Theorem 1.3, there is a short exact sequence

$$0 \longrightarrow T_{(x,y)}(X \times_Z Y) \longrightarrow T_xX \oplus T_yY \xrightarrow{\delta_{x,y}} T_zZ \longrightarrow 0.$$

The question motivating the following examples is what remains meaningful when the last map is no longer surjective.

1.2 Nontransverse fiber products

For a transverse intersection, the kernel of the matching map is the tangent space of the intersection. Without transversality, this kernel still describes solutions of the linearized matching equation, but these need not extend to actual solutions. Attempting to extend a first-order solution to second order produces an error. In local coordinates, changing the second-order correction changes this error by an element of the image of the matching map. The remaining obstruction therefore takes values in its cokernel. We will see this explicitly in Example 1.6.

This gives a role to both the kernel, which describes first-order solutions, and the cokernel, in which obstruction classes take values. We package the matching map into a two-term complex.

Notation 1.4. For arbitrary smooth maps $f\colon X \to Z$ and $g\colon Y \to Z$, and a point $(x, y) \in X \times_Z Y$, write

$$\mathbb{T}_{x,y}(f, g) = \left[T_x X \oplus T_y Y \xrightarrow{\delta_{x,y}} T_z Z\right],$$

with the source in homological degree 0 and the target in degree -1. We call this the *linearized matching complex*. Thus

$$H_0\mathbb{T}_{x,y}(f, g) = \ker(\delta_{x,y}), \qquad H_{-1}\mathbb{T}_{x,y}(f, g) = \operatorname{coker}(\delta_{x,y}).$$

For now the complex is a way to keep the linear map together with its kernel and cokernel. Chapter 6 will identify it with the tangent complex of the derived intersection. The calculations here require only linear algebra.

We now return to the two intersections pictured in Figure 1.1. The following two examples calculate their linearized matching complexes. In both cases the set-theoretic intersection consists only of the origin, but the complexes are different.

Example 1.5 (Two transverse axes)**.** Parametrize the two coordinate axes by

$$i\colon \mathbb{R} \longrightarrow \mathbb{R}^2, \quad i(s) = (s, 0), \qquad k\colon \mathbb{R} \longrightarrow \mathbb{R}^2, \quad k(t) = (0, t).$$

Their intersection consists of the origin. The linearized matching map is

$$\delta_{0,0}\colon \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \qquad \delta_{0,0}(u, v) = (u, -v).$$

It is an isomorphism, so the maps are transverse and

$$H_0\mathbb{T}_{0,0}(i, k) = 0, \qquad H_{-1}\mathbb{T}_{0,0}(i, k) = 0.$$

Example 1.6 (A tangent line and parabola)**.** Parametrize the x-axis and the parabola $y = x^2$ by

$$i\colon \mathbb{R} \longrightarrow \mathbb{R}^2, \quad i(s) = (s, 0), \qquad j\colon \mathbb{R} \longrightarrow \mathbb{R}^2, \quad j(t) = (t, t^2).$$

A point (s, t) lies in the fiber product precisely when

$$s = t, \qquad t^2 = 0.$$

Thus the underlying fiber product is again a single point. At this point, however, the linearized matching map is

$$\delta_{0,0}\colon \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \qquad \delta_{0,0}(u, v) = (u - v, 0).$$

It follows that

$$H_0\mathbb{T}_{0,0}(i, j) = \{(u, u) \mid u \in \mathbb{R}\} \cong \mathbb{R}$$

and

$$H_{-1}\mathbb{T}_{0,0}(i, j) = \mathbb{R}^2/(\mathbb{R} \times \{0\}) \cong \mathbb{R}.$$

We refer provisionally to these as the infinitesimal deformation space and the obstruction space. Their alternating dimension is zero, which agrees with the expected dimension of the intersection.

The class represented by $(1,1)$ in degree 0 is a solution of the linearized matching equation. It is not a tangent vector to the ordinary intersection, which is a single point. To see why this first-order solution cannot extend to second order, write the matching equation as

$$F(s,t) = i(s) - j(t) = (s - t, -t^2) = 0.$$

An extension with initial velocity $(1,1)$ would have expansions

$$s(\tau) = \tau + a\tau^2 + O(\tau^3), \qquad t(\tau) = \tau + b\tau^2 + O(\tau^3).$$

Substitution gives

$$F(s(\tau), t(\tau)) = \tau^2(a - b, -1) + O(\tau^3).$$

The quadratic error is therefore

$$(a - b, -1) = (0, -1) + \delta_{0,0}(a, b).$$

Changing the second-order correction changes this error by an element of $\mathrm{im}(\delta_{0,0})$. Its class

$$[(0,-1)] \in \mathrm{coker}(\delta_{0,0}) = H_{-1}\mathbb{T}_{0,0}(i,j)$$

is nonzero, so no choice of a, b makes the equation hold through second order. In particular, $(1,1)$ cannot be the velocity of an actual curve of solutions.

The cokernel is the space in which this obstruction takes values; the obstruction class itself uses the quadratic term of F. It is not determined by the linearized complex alone. A nonzero cokernel does not by itself imply that a given first-order solution is obstructed.

The same example can be presented as a zero locus in one variable. Eliminating s gives the smooth function

$$q\colon \mathbb{R} \longrightarrow \mathbb{R}, \qquad q(t) = t^2.$$

Its linearized complex at the origin is

$$[\mathbb{R} \xrightarrow{D_0 q} \mathbb{R}] = [\mathbb{R} \xrightarrow{0} \mathbb{R}].$$

We can compare this directly with the matching complex in Example 1.6. Let

$$a = (1,1), \qquad b = (1,0)$$

be a basis of the source of $\delta_{0,0}$, and let e_1, e_2 be the standard basis of its target. Then

$$\delta_{0,0}(a) = 0, \qquad \delta_{0,0}(b) = e_1.$$

Consequently, the matching complex splits as

$$\mathbb{T}_{0,0}(i,j) \cong [\mathbb{R}b \xrightarrow{\sim} \mathbb{R}e_1] \oplus [\mathbb{R}a \xrightarrow{0} \mathbb{R}e_2].$$

The first summand is acyclic, while the second is isomorphic to the linearized complex of q.

This illustrates a change of presentation which should preserve the geometric object. Adding a variable s and an equation $s = 0$ to $q(t) = 0$ adds an acyclic summand $[\mathbb{R} \xrightarrow{1} \mathbb{R}]$ to its linearized complex: the new variable is uniquely eliminated by the new equation. In contrast, appending the equation $0 = 0$ without adding a variable contributes $[0 \to \mathbb{R}]$. Although the zero set is unchanged in both cases, only the first operation preserves the linearized deformation and obstruction spaces. Already the equations $t = 0$ and $t^2 = 0$ have the same zero set but different linearized complexes, respectively $[\mathbb{R} \xrightarrow{1} \mathbb{R}]$ and $[\mathbb{R} \xrightarrow{0} \mathbb{R}]$. Passing to the zero set therefore forgets information visible in the derivative. The next example shows that the derivative alone is still insufficient.

Example 1.7 (The same tangent complex from different equations)**.** The equations $x^2 = 0$ and $x^3 = 0$ both have zero set $\{0\}$ and linearized complex $[\mathbb{R} \xrightarrow{0} \mathbb{R}]$. Nevertheless their first-order solutions have different extension behavior. For an attempted extension

$$x(\tau) = \tau + a\tau^2 + O(\tau^3),$$

we obtain

$$x(\tau)^2 = \tau^2 + O(\tau^3), \qquad x(\tau)^3 = \tau^3 + O(\tau^4).$$

No choice of a makes the square vanish through second order, whereas the cubic does vanish through second order. Thus the first-order solution extends one order further for the cubic, although neither equation admits an actual curve of solutions with initial velocity 1. The nonlinear equation contains information beyond its tangent complex.

These examples show why the nonlinear equations matter, even when we know their zero set and their linearization. The next example raises a different issue: an equation can contribute to the linearized problem even when it imposes no condition at all on the set of solutions.

Example 1.8 (An excess self-intersection)**.** Compare an ordinary point with the self-intersection of the origin in $\mathbb{R}$, given by two maps $* \to \mathbb{R}$. The point has no variables and no equations. The self-intersection has no variables and one matching equation, which happens to vanish. Its ordinary pullback is again a point, but the two linearized complexes are respectively

$$[0 \to 0] \qquad \text{and} \qquad [0 \to \mathbb{R}].$$

The second has alternating dimension -1, which cannot be the dimension of an ordinary manifold. There is no nonzero first-order solution direction to obstruct. Rather, perturbing one of the two points away from the other produces a matching error which cannot be corrected, since there are no variables.

We therefore seek a geometric object which remembers this matching equation, even though it imposes no condition on the set of solutions. The derived self-intersection will be such an object. Its linearized complex will be $[0 \to \mathbb{R}]$, distinguishing it from an ordinary point. We will construct it in Chapter 5.

The preceding examples concern fixed equations. A one-parameter family shows why it is useful to work with the entire two-term complex rather than only its kernel or cokernel.

Example 1.9 (A family with jumping kernels)**.** For $t \in \mathbb{R}$, consider the linear map

$$L_t \colon \mathbb{R} \longrightarrow \mathbb{R}, \qquad L_t(u) = tu.$$

Its kernel and cokernel are

$$\ker L_t = \begin{cases} 0, & t \neq 0, \\ \mathbb{R}, & t = 0, \end{cases} \qquad \operatorname{coker} L_t = \begin{cases} 0, & t \neq 0, \\ \mathbb{R}, & t = 0. \end{cases}$$

Neither the kernels nor the cokernels form a vector bundle of constant rank over the parameter line. Let $E_0 = E_{-1} = \mathbb{R} \times \mathbb{R}$ be trivial line bundles over the first factor and define a bundle map

$$d \colon E_0 \longrightarrow E_{-1}, \qquad d(t, u) = (t, tu).$$

This gives a smooth two-term complex of vector bundles

$$[E_0 \xrightarrow{d} E_{-1}]$$

whose fiber over t is $[\mathbb{R} \xrightarrow{L_t} \mathbb{R}]$. The complex varies smoothly even though the dimensions of its homology groups jump. The alternating dimension

$$\dim \ker L_t - \dim \operatorname{coker} L_t = 0$$

is independent of t.

The total set of solutions is

$$\{(t, u) \in \mathbb{R}^2 \mid tu = 0\},$$

which is the union of the two coordinate axes. The additional branch over $t = 0$ records the jump in $\ker L_t$, and the two branches meet at the singular origin.

Here L_t linearizes the equation in u at $u = 0$, with t held fixed. Thus the displayed complex is relative to the parameter t. If we instead allow both t and u to vary, the derivative of $(t, u) \mapsto tu$ at the origin is zero, giving the absolute linearized complex $[\mathbb{R}^2 \xrightarrow{0} \mathbb{R}]$ of the total zero locus there.

The vector bundles and their differential give a single smooth object throughout the family. At each parameter, we take the corresponding linear map and calculate its kernel and cokernel. These can change dimension even though the vector bundles have fixed ranks.

We can now collect what the examples require from a replacement of singular fiber products.

(1) It should admit arbitrary pullbacks and recover the ordinary smooth fiber product when the maps are transverse.

(2) It should describe the nonlinear defining equations in a presentation-independent way, not merely their set of solutions or their first derivatives.

(3) It should organize deformation and obstruction data in families, even when the dimensions of the individual deformation and obstruction spaces jump.

The second requirement contains two lessons from the preceding calculations. The tangent line–parabola intersection and the zero locus of t^2 are two presentations of the same problem, so the desired object should not depend on the redundant variable. On the other hand, the zero set and the linearized complex are both too coarse: the first cannot distinguish t from t^2, and the second cannot distinguish t^2 from t^3.

Derived manifolds will provide a construction satisfying these requirements. It may be described via a universal property: they form a universal enlargement of manifolds in which finite homotopy limits exist and transverse pullbacks are preserved. We will explain these terms and state the precise universal property in Chapter 4. For now, the point is that there is a systematic construction of intersections, independent of a chosen system of coordinates or a redundant variable eliminated by an equation. The next chapters develop the algebraic tools needed to construct these objects and establish their properties.

1.3 Elliptic equations and finite-dimensional reduction

The final part of the seminar will prove *Steffens's relative elliptic representability theorem*. This is a principal application of the derived geometry we will develop: it gives a geometric structure to moduli problems defined by nonlinear elliptic partial differential equations. Such equations occur throughout differential geometry. For example, pseudo-holomorphic maps from a Riemann surface to an almost complex manifold satisfy an elliptic equation, and the last talk will apply the theorem to their moduli problem.

Informally, Steffens's theorem says that a family of elliptic equations on a proper family of manifolds has a solution stack represented, relative to its parameters, by a quasi-smooth derived C^∞-scheme [Ste24, Theorem 3.4.3]. Properness supplies the compactness needed for elliptic analysis. Quasi-smoothness means that, locally over the parameters, the solutions admit presentations as derived zero loci of sections of finite-rank vector bundles on finite-dimensional manifolds. Thus the finite-dimensional geometry of equations studied above also describes solutions of certain infinite-dimensional problems. The theorem asserts representability of the solution functor, so these local presentations describe an intrinsic moduli object, including its families and deformation theory.

Chapter 10 constructs the solution stacks to which the theorem applies. We state the precise theorem in Theorem 11.1 and prove it over Chapters 11 to 13. Here we explain the role of finite-dimensional reduction in that proof, starting with an equation on a fixed manifold.

Let M be a closed smooth manifold, let $\pi\colon V \to M$ be a smooth fiber bundle, and let $F \to M$ be a smooth vector bundle. Suppose that

$$P\colon \Gamma(V) \longrightarrow \Gamma(F)$$

is a nonlinear elliptic differential operator. We write

$$\mathrm{Sol}(P) = P^{-1}(0).$$

A point of $\mathrm{Sol}(P)$ is a section satisfying the geometric equation encoded by P. For the moment this notation denotes the ordinary set of solutions. The theorem concerns the derived solution stack, whose construction must also account for families of solutions and their defining equations.

For a concrete model, take $V = F = M \times \mathbb{R}$ over a closed connected Riemannian manifold. The semilinear equation

$$P(u) = \Delta u + u^2 = 0$$

uses the nonnegative Laplacian $\Delta = d^*d$ and defines a subset $\mathrm{Sol}(P) \subseteq C^\infty(M)$. It arises as the critical-point equation for the energy $\int_M (\frac{1}{2}|du|^2 + u^3/3)$. The zero function is a solution and its linearization is $D_0P = \Delta$. Since M is closed and connected, its kernel consists of the constant functions, and self-adjointness identifies its cokernel with the same one-dimensional space.

Nevertheless, the only real solution is $u = 0$: integrating the equation gives

$$0 = \int_M P(u) = \int_M u^2,$$

since the integral of Δu vanishes on a closed manifold. The constant function 1 is a first-order solution at zero. To extend it to second order, try

$$u(\tau) = \tau + \tau^2 v + O(\tau^3).$$

The coefficient of τ^2 in $P(u(\tau))$ is $\Delta v + 1$. Its integral is $\mathrm{vol}(M)$, so it cannot vanish for any v. Equivalently, the quadratic error represents the nonzero class of 1 in $\mathrm{coker}(\Delta)$. This is the same obstruction mechanism as for the tangent line and parabola. We will return to this example when we carry out a finite-dimensional reduction later in the seminar.

Why should such an equation admit the finite-dimensional local presentations promised by the theorem? Both $\Gamma(V)$ and $\Gamma(F)$ are infinite-dimensional. Ellipticity first gives a finiteness statement about the linearized equation; finite-dimensional reduction then extends this statement to the nonlinear equation near a solution.

Write

$$T^{\mathrm{vert}}V = \ker(D\pi \colon TV \longrightarrow TM)$$

for the vertical tangent bundle. At a solution u, the tangent space to the space of sections is

$$T_u\Gamma(V) \cong \Gamma(u^*T^{\mathrm{vert}}V).$$

If V is a vector bundle, then $T^{\mathrm{vert}}V \cong \pi^*V$, and this reduces to the familiar identification $T_u\Gamma(V) \cong \Gamma(V)$. The linearization of P at u is therefore the elliptic operator

$$D_uP \colon \Gamma(u^*T^{\mathrm{vert}}V) \longrightarrow \Gamma(F).$$

We therefore obtain the two-term complex

$$\mathbb{T}_uP = \left[\Gamma(u^*T^{\mathrm{vert}}V) \xrightarrow{D_uP} \Gamma(F)\right]$$

in homological degrees 0 and -1. Its homology is the infinitesimal deformation space and obstruction space of the equation at u:

$$H_0(\mathbb{T}_uP) = \ker(D_uP), \qquad H_{-1}(\mathbb{T}_uP) = \mathrm{coker}(D_uP).$$

Ellipticity and compactness imply that these spaces are finite-dimensional. As u varies, their dimensions can jump, so the complex $\mathbb{T}_uP$ is again more stable than its homology groups separately. This is the infinite-dimensional version of Example 1.9.[1]

The analytic step is to solve the nonlinear equation in the remaining directions, leaving only finitely many variables and equations. The following classical reduction previews that step. We quote it here under the usual analytic hypotheses: suitable Sobolev completions make P a smooth map of Banach manifolds with Fredholm linearization, and elliptic regularity identifies its nearby Sobolev solutions with smooth sections. These are the tools developed in Chapter 13.

Fact 1.10 (Local finite-dimensional reduction)**.** Let P satisfy the analytic hypotheses just stated, and let $u \in \mathrm{Sol}(P)$. There are finite-dimensional vector spaces K and C, an open neighborhood U of 0 in K, and a smooth map

$$\kappa \colon U \longrightarrow C, \qquad \kappa(0) = 0,$$

such that a neighborhood of u in $\mathrm{Sol}(P)$ is identified with $\kappa^{-1}(0)$. Moreover, at every $v \in \kappa^{-1}(0)$, the finite-dimensional complex

$$[K \xrightarrow{D_v\kappa} C]$$

is quasi-isomorphic to the complex $\mathbb{T}_{u_v}P$ at the corresponding solution u_v.

[1] This is the motivating linear picture in [Par24, Sections 1–2].

Here *quasi-isomorphic* means that the complexes can be connected by a zigzag of chain maps which induce isomorphisms on H_0 and H_{-1}. In particular, they have the same deformation and obstruction spaces.

The mechanism behind the fact is worth keeping in mind. After passing to Sobolev completions, $D_u P$ is a Fredholm operator. One chooses splittings which separate its kernel, image, and cokernel. The inverse function theorem solves the equation in the complementary directions, leaving a finite-dimensional obstruction map κ. Elliptic bootstrapping then shows that the resulting solutions are smooth. We quote this analytic argument; the precise conclusion above is [Ste22, Fact 1.0.0.1].

Thus the map κ gives a finite-dimensional presentation of the equation near the chosen solution. Its zero set describes nearby solutions, while its derivative recovers their deformation and obstruction spaces. The nonlinear terms of κ determine higher-order obstructions, as in the preceding examples $t \mapsto t^2$ and $t \mapsto t^3$. This explains why a derived zero locus is a plausible local model for the solution stack. To prove Steffens's theorem, we must establish this description for the solution functor itself, compatibly with parameters, and glue the resulting local models. The formal thickening argument in Chapter 12 and the analytic argument in Chapter 13 provide these local derived presentations.

1.4 From local zero loci to coherent moduli problems

The construction of κ depends on choices of Sobolev completions, splittings, finite-dimensional subspaces, and neighborhoods. Different choices can produce zero-locus presentations

$$\kappa \colon U \longrightarrow C \qquad \text{and} \qquad \kappa' \colon U' \longrightarrow C'$$

of different dimensions. An identification of their underlying zero sets says too little, just as equality of the zero sets of t and t^2 says too little.

There are therefore two global problems.

(1) One must specify when two local finite-dimensional reductions present the same geometric object.

(2) One must glue these local objects on overlaps, together with compatibilities between the comparison maps and all higher compatibilities.

Higher-categorical language records the comparison maps, homotopies between comparisons, and their iterated compatibilities. Later we will formulate the equation as a *derived solution functor*, which describes families of solutions together with additional information about the equations. Its definition belongs to Chapter 10; it is not simply the assignment of an ordinary solution set to each parameter. Identifying local finite-dimensional reductions with charts of this functor is an additional step beyond identifying their ordinary zero sets. If the functor is representable, its representing object is unique. This is the representability viewpoint discussed in [Par24, Section 1].

For elliptic equations, the derived comparison of the local reductions and descent will show that the solution stack is representable over its parameter stack by quasi-smooth derived C^∞-schemes. When the parameter object is a point, the solution stack itself is a quasi-smooth derived C^∞-scheme. Constructing virtual fundamental classes or enumerative invariants then requires further geometric input, such as orientations and compactness.

1.5 Further examples

An intersection can have an ordinary zero set of the wrong positive dimension. The zero map $\mathbb{R}^n \to \mathbb{R}^r$ has all of $\mathbb{R}^n$ as its zero set, while the expected dimension is $n - r$. Its linearized complex $[\mathbb{R}^n \xrightarrow{0} \mathbb{R}^r]$ displays both the variables and the redundant equations. The derived construction will distinguish this presentation from the manifold $\mathbb{R}^n$ with no equations imposed.

Exercise 1.11. Express the intersection of the graph of $f\colon U \to \mathbb{R}^r$ with the zero section of $U \times \mathbb{R}^r$ as a fibre product. Separate the acyclic summand of its linearized matching complex to obtain $[T_x U \to \mathbb{R}^r]$.

Exercise 1.12. For $m \geq 1$, find the first order at which the equations $x^m = 0$ and $x^{m+1} = 0$ can be distinguished by substituting $x(\tau) = \tau + O(\tau^2)$. As an algebraic reformulation, find the least N for which $x = \varepsilon$ satisfies one equation but not the other in the truncated polynomial ring $\mathbb{R}[\varepsilon]/(\varepsilon^N)$.

2 Introduction to C^∞-rings

A smooth equation contains information that its zero set alone does not record. For example, $t = 0$ and $t^2 = 0$ have the same solutions in $\mathbb{R}$, but their linearizations at the origin are different. Algebraic geometry suggests how to retain this distinction: associate to each equation the ring of functions modulo that equation. In these two cases, this gives

$$C^\infty(\mathbb{R})/(t) \qquad \text{and} \qquad C^\infty(\mathbb{R})/(t^2).$$

In the first quotient, the coordinate t vanishes; in the second, it survives as a nonzero element whose square is zero.

To develop this approach in smooth geometry, we first identify the algebraic structure carried by smooth functions. The ring $C^\infty(M)$ of a manifold M admits much more than addition and multiplication: every smooth map $f\colon \mathbb{R}^n \to \mathbb{R}$ defines an operation on n-tuples of smooth functions by pointwise evaluation,

$$(a_1, \dots, a_n) \longmapsto f(a_1, \dots, a_n), \qquad f(a_1, \dots, a_n)(x) = f(a_1(x), \dots, a_n(x)).$$

Addition and multiplication are the special cases $f(x, y) = x + y$ and $f(x, y) = xy$; smooth functional calculus also allows operations such as $a \mapsto \sin(a)$ and $a \mapsto \exp(a)$.

A C^∞-ring abstracts this structure: it is a set equipped with an operation for every smooth map $\mathbb{R}^n \to \mathbb{R}$, compatible with projections and composition. As we will see using Hadamard's lemma, these operations descend to quotients by arbitrary ideals. Thus the quotient rings above retain both the smooth functional calculus and the distinction between the two equations.

The operations formulation below follows Moerdijk and Reyes [MR91, Chapter I, Section 1]; a modern account, including the functor formulation, is given by Joyce [Joy12, Sections 2.1–2.2].

2.1 Rings with smooth operations

We give two formulations of the definition: one in terms of operations and their identities, and one in terms of functors on Euclidean spaces. The first makes the algebra explicit; the second will be useful when we pass to derived smooth rings.

Definition 2.1 (C^∞-ring)**.** A *C^∞-ring* is a set A equipped, for every $n \geq 0$ and every smooth map $f\colon \mathbb{R}^n \to \mathbb{R}$, with an operation

$$\Phi_f\colon A^n \longrightarrow A.$$

These operations satisfy the following two conditions.

(1) If $\pi_i\colon \mathbb{R}^n \to \mathbb{R}$ is the ith projection, then $\Phi_{\pi_i}(a_1, \ldots, a_n) = a_i$.

(2) If $f_1, \ldots, f_m\colon \mathbb{R}^n \to \mathbb{R}$ and $g\colon \mathbb{R}^m \to \mathbb{R}$ are smooth, then

$$\Phi_{g\circ(f_1,\ldots,f_m)} = \Phi_g \circ (\Phi_{f_1}, \ldots, \Phi_{f_m}).$$

The case $n = 0$ supplies an element of A for every constant in $\mathbb{R}$. The second axiom says that evaluating a composite smooth function gives the same answer as evaluating its pieces successively.

Example 2.2 (Smooth functions)**.** For every manifold M, the set $C^\infty(M)$ is a C^∞-ring under pointwise smooth functional calculus. More generally, the same construction works whenever a notion of smooth real-valued function has been specified and is closed under smooth substitution.

Definition 2.3 (C^∞-homomorphism)**.** A homomorphism of C^∞-rings is a map $\varphi\colon A \to B$ satisfying

$$\varphi\big(\Phi_f(a_1, \ldots, a_n)\big) = \Phi_f\big(\varphi(a_1), \ldots, \varphi(a_n)\big)$$

for every smooth $f\colon \mathbb{R}^n \to \mathbb{R}$ and every $(a_1, \ldots, a_n) \in A^n$.

If $u\colon M \to N$ is smooth, then pullback gives a homomorphism

$$u^*\colon C^\infty(N) \longrightarrow C^\infty(M), \qquad h \longmapsto h \circ u.$$

Thus smooth geometry enters the algebra contravariantly. This reversal of arrows will remain in force throughout the next several chapters.

Observation 2.4 (The underlying commutative algebra)**.** Every C^∞-ring has an underlying commutative $\mathbb{R}$-algebra. Addition, multiplication, scalar multiplication, zero, and one are obtained from the corresponding smooth maps between Euclidean spaces. Every C^∞-homomorphism is consequently a homomorphism of commutative $\mathbb{R}$-algebras.

Not every commutative $\mathbb{R}$-algebra admits such smooth operations. They impose algebraic conditions. For example, in every C^∞-ring and for every $a \in A$, the element $1 + a^2$ is invertible. Its inverse is

$$\Phi_g(a), \qquad g(x) = \frac{1}{1 + x^2},$$

because the identity $(1+x^2)g(x) = 1$ is an identity of smooth functions. Therefore the commutative $\mathbb{R}$-algebra $\mathbb{R}[t]$ cannot underlie a C^∞-ring: its element $1 + t^2$ has no inverse.

There is a compact categorical formulation of the same definition. Let Euc be the category whose objects are the Euclidean spaces $\mathbb{R}^n$, for $n \geq 0$, and whose morphisms are smooth maps. Its finite products are the usual products of Euclidean spaces.

Proposition 2.5 (Functor formulation)**.** *To give a C^∞-ring is equivalently to give a finite-product-preserving functor*

$$A\colon \mathrm{Euc} \longrightarrow \mathrm{Set}.$$

Under this correspondence, the underlying set of the ring is $A(\mathbb{R})$.

Proof. Suppose first that the operations of Definition 2.1 are given. Set $A(\mathbb{R}^n) = A^n$. If $f = (f_1, \dots, f_m) \colon \mathbb{R}^n \to \mathbb{R}^m$, define

$$A(f) = (\Phi_{f_1}, \dots, \Phi_{f_m}) \colon A^n \longrightarrow A^m.$$

The projection and composition axioms say precisely that this is a functor, and the construction visibly preserves finite products.

Conversely, let $A \colon \mathrm{Euc} \to \mathrm{Set}$ preserve finite products and put $A = A(\mathbb{R})$. The product comparison identifies $A(\mathbb{R}^n)$ with A^n. For a smooth map $f \colon \mathbb{R}^n \to \mathbb{R}$, the map $A(f)$ then defines $\Phi_f \colon A^n \to A$. Functoriality gives the two axioms in Definition 2.1. □

The category Euc, viewed as a category of operations, is an example of a Lawvere theory. The functor formulation will later allow us to replace sets by animae.

2.2 Generators and relations

To describe smooth equations algebraically, we need rings of functions in freely chosen variables and a way to impose relations among them. Smooth functions on $\mathbb{R}^n$ supply the free rings. Hadamard's lemma then shows that every ideal defines a quotient smooth ring. Together, these constructions give presentations and colimits in the category of C^∞-rings.

Let $x_1, \dots, x_n \in C^\infty(\mathbb{R}^n)$ be the coordinate functions. Every smooth function $f \in C^\infty(\mathbb{R}^n)$ is obtained by applying the operation Φ_f to these coordinates:

$$f = \Phi_f(x_1, \dots, x_n).$$

This tautological identity is the key to the universal property.

Theorem 2.6 (Free C^∞-rings)**.** *For every C^∞-ring A, evaluation on the coordinate functions induces a bijection*

$$\mathrm{Hom}_{C^\infty\text{-Ring}}\big(C^\infty(\mathbb{R}^n), A\big) \longrightarrow A^n, \qquad \varphi \longmapsto \big(\varphi(x_1), \dots, \varphi(x_n)\big).$$

Thus $C^\infty(\mathbb{R}^n)$ is the free C^∞-ring on $x_1, \dots, x_n$.

This is Moerdijk and Reyes [MR91, Chapter I, Proposition 1.1].

Proof. Given $(a_1, \dots, a_n) \in A^n$, define

$$\mathrm{ev}_{a_1,\dots,a_n} \colon C^\infty(\mathbb{R}^n) \longrightarrow A, \qquad f \longmapsto \Phi_f(a_1, \dots, a_n).$$

The composition axiom for smooth functional calculus says exactly that this map preserves every smooth operation, so it is a C^∞-homomorphism. It sends x_i to a_i.

Conversely, if $\varphi \colon C^\infty(\mathbb{R}^n) \to A$ is a C^∞-homomorphism, then

$$\varphi(f) = \varphi\big(\Phi_f(x_1, \dots, x_n)\big) = \Phi_f\big(\varphi(x_1), \dots, \varphi(x_n)\big).$$

Hence φ is uniquely determined by the images of the coordinate functions, and the two constructions are inverse. □

Smooth functions on $\mathbb{R}^n$ therefore play the role that polynomials in n variables play in ordinary commutative algebra. Here the allowed operations include every smooth map $\mathbb{R}^n \to \mathbb{R}$, so the coordinate generators produce all smooth functions through smooth substitution.

To use quotient rings, we must show that an ordinary ideal is respected by all smooth operations. The required fact is a multivariable form of the fundamental theorem of calculus.

Lemma 2.7 (Hadamard's lemma)**.** *For every smooth map $f\colon \mathbb{R}^n \to \mathbb{R}$, there are smooth maps $g_1, \dots, g_n\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ such that*

$$f(y) - f(x) = \sum_{i=1}^{n} (y_i - x_i) g_i(x, y).$$

Proof. Put

$$g_i(x, y) = \int_0^1 \frac{\partial f}{\partial x_i}\big(x + s(y - x)\big)\, ds.$$

Differentiation under the integral sign shows that g_i is smooth. The fundamental theorem of calculus applied to $s \mapsto f(x + s(y - x))$ gives

$$f(y) - f(x) = \int_0^1 \frac{d}{ds} f(x + s(y - x))\, ds = \sum_{i=1}^{n} (y_i - x_i) g_i(x, y).$$

□

Proposition 2.8 (Quotient C^∞-rings)**.** *Let A be a C^∞-ring, and let $I \subseteq A$ be an ideal in its underlying commutative algebra. There is a unique C^∞-ring structure on A/I for which the quotient map $A \to A/I$ is a C^∞-homomorphism.*

This is Moerdijk and Reyes [MR91, Chapter I, Proposition 1.2].

Proof. The only possible definition is

$$\Phi_f(\overline{a}_1, \dots, \overline{a}_n) = \overline{\Phi_f(a_1, \dots, a_n)}.$$

We check that it is independent of the representatives. Suppose that $b_i - a_i \in I$ for every i. Apply Lemma 2.7 inside A to obtain

$$\Phi_f(b_1, \dots, b_n) - \Phi_f(a_1, \dots, a_n) = \sum_{i=1}^{n} (b_i - a_i) \Phi_{g_i}(a_1, \dots, a_n, b_1, \dots, b_n).$$

The right-hand side lies in I. Thus the operations are well defined. Their projection and composition axioms descend from those in A. Uniqueness is forced by the requirement that the quotient map preserve every operation. □

This result is special to smooth functional calculus. It says that every ordinary ideal in the underlying algebra is already compatible with all the smooth operations.

We can now express finiteness in terms of generators and relations. The generators are allowed to enter arbitrary smooth operations, rather than just polynomial expressions.

Definition 2.9 (Finite generation and finite presentation)**.** A C^∞-ring A is *finitely generated* if there are $a_1, \ldots, a_n \in A$ such that no proper C^∞-subring contains all the a_i. It is *finitely presented* if it is isomorphic to

$$C^\infty(\mathbb{R}^n)/(f_1, \ldots, f_k)$$

for some finite collections of generators and relations.

Proposition 2.10. *A C^∞-ring is finitely generated if and only if it is isomorphic to $C^\infty(\mathbb{R}^n)/I$ for some n and some ideal $I \subseteq C^\infty(\mathbb{R}^n)$.*

Proof. If $a_1, \ldots, a_n$ generate A, the universal property in Theorem 2.6 gives a homomorphism $C^\infty(\mathbb{R}^n) \to A$ sending x_i to a_i. Its image is a C^∞-subring containing the generators, so it is surjective. By Proposition 2.8, the induced isomorphism of commutative algebras $C^\infty(\mathbb{R}^n)/I \to A$ is an isomorphism of C^∞-rings. The converse follows because the images of the coordinate functions generate every such quotient. □

Finite generation here means generation under every smooth operation. It does not imply that the underlying commutative $\mathbb{R}$-algebra is finitely generated. Likewise, finite presentation is a genuine additional condition: smooth function rings are not Noetherian. The next chapter will return to finite presentation for manifolds.

Limits of C^∞-rings are computed on underlying sets, with each smooth operation acting componentwise. For example, the product ring $A \times B$ has $\Phi_f((a_1, b_1), \ldots, (a_n, b_n)) = (\Phi_f(a_1, \ldots, a_n), \Phi_f(b_1, \ldots, b_n))$. The one-element ring is terminal, and $\mathbb{R} = C^\infty(\mathbb{R}^0)$ is initial.

There is a free smooth ring on any set I of generators. Its elements are smooth expressions in finitely many of those generators, with expressions identified when they define the same smooth function after adding unused variables. Equivalently, it is the filtered colimit of $C^\infty(\mathbb{R}^J)$ over finite subsets $J \subseteq I$. The finite-generator case is Theorem 2.6.

By Proposition 2.8, quotients by arbitrary ideals carry smooth operations. Combining free objects and quotients constructs all colimits: introduce generators for the elements of the rings in a diagram, and impose their operation relations and the identifications specified by the diagram. Thus the category of smooth rings is complete and cocomplete. Chapter 3 calculates its coproducts more explicitly.

2.3 Taylor expansions and infinitesimal information

We return to the quotient rings from the chapter opening. Taylor's formula will identify the information retained by $t^2 = 0$ and show why real points do not determine a smooth ring.

Example 2.11 (A point and a double point)**.** We can now compute the two quotients from the opening. In one variable, Lemma 2.7 gives

$$f(t) = f(0) + tg(t)$$

for some smooth g. Evaluation at the origin therefore induces an isomorphism

$$C^\infty(\mathbb{R})/(t) \cong \mathbb{R}.$$

For the second quotient, Taylor's formula with smooth remainder gives

$$f(t) = f(0) + f'(0)t + t^2 h(t).$$

Consequently, the map

$$C^\infty(\mathbb{R}) \longrightarrow \mathbb{R}[\varepsilon]/(\varepsilon^2), \qquad f \longmapsto f(0) + f'(0)\varepsilon$$

is a surjective homomorphism of commutative $\mathbb{R}$-algebras with kernel (t^2). It identifies

$$C^\infty(\mathbb{R})/(t^2) \cong \mathbb{R}[\varepsilon]/(\varepsilon^2).$$

The ring on the right is called the ring of *dual numbers*. Its C^∞-structure is not additional arbitrary data. It is the quotient structure from Proposition 2.8, and its operations are given by the first-order Taylor formula

$$f(a + b\varepsilon) = f(a) + b f'(a)\varepsilon.$$

To compare these quotients with their ordinary solution sets, we isolate the appropriate notion of point.

Definition 2.12 (Real point)**.** A *real point* of a C^∞-ring A is a C^∞-homomorphism $A \to \mathbb{R}$, where $\mathbb{R}$ carries its evident smooth functional calculus.

Proposition 2.13 (Real points of a quotient)**.** *For every ideal $I \subseteq C^\infty(\mathbb{R}^n)$, evaluation induces a bijection*

$$\mathrm{Hom}_{C^\infty\text{-Ring}}\big(C^\infty(\mathbb{R}^n)/I, \mathbb{R}\big) \cong \{p \in \mathbb{R}^n \mid f(p) = 0 \text{ for every } f \in I\}.$$

Proof. By Theorem 2.6, every homomorphism $C^\infty(\mathbb{R}^n) \to \mathbb{R}$ is evaluation at a unique point $p = (p_1, \ldots, p_n)$. It factors through the quotient precisely when every element of I vanishes at p. □

Both $C^\infty(\mathbb{R})/(t)$ and $C^\infty(\mathbb{R})/(t^2)$ therefore have exactly one real point. They are nevertheless not isomorphic: the class of t is zero in the first and is a nonzero square-zero element in the second. The quotient by (t^2) remembers first-order information which the zero set forgets.

These are ordinary smooth rings. They retain infinitesimal equations through nilpotents. Derived smooth rings will additionally retain higher homotopy, which is needed for the pullbacks in Chapter 5.

2.4 Recovering geometry from function rings

Different smooth rings can have the same real points, as the dual numbers illustrate. In the other direction, passing from a manifold to its function ring loses no information: every homomorphism between such rings comes from a unique smooth map. We prove this first for closed subsets of Euclidean spaces, then deduce the manifold case using a proper embedding.

Let $X \subseteq \mathbb{R}^n$ be an arbitrary closed subset. It need not be a manifold, so we specify what smoothness means in this setting.

Definition 2.14 (Smooth functions and maps on subsets)**.** A function $a\colon X \to \mathbb{R}$ is *smooth* if it extends to a smooth function on an open neighborhood of X in $\mathbb{R}^n$. We write $C^\infty(X)$ for the resulting C^∞-ring.

A map $u\colon X \to Y \subseteq \mathbb{R}^m$ is *smooth* if its component functions are smooth on X. Equivalently, it extends to a smooth map into $\mathbb{R}^m$ on an open neighborhood of X.

The equivalence in the second paragraph uses only that there are finitely many component functions: intersect their extension neighborhoods and combine the extensions.

Lemma 2.15 (Smooth extension from a closed subset)**.** *If $X \subseteq \mathbb{R}^n$ is closed, every element of $C^\infty(X)$ is the restriction of a globally defined smooth function on $\mathbb{R}^n$.*

Proof. Let $a \in C^\infty(X)$, and choose a smooth extension $\widetilde{a}$ on an open neighborhood U of X. A smooth partition of unity subordinate to the cover

$$\{U, \mathbb{R}^n \setminus X\}$$

supplies a smooth function $\chi\colon \mathbb{R}^n \to [0,1]$ whose support is contained in U and whose restriction to X is one. Define

$$\overline{a}(x) = \begin{cases} \chi(x)\widetilde{a}(x), & x \in U, \\ 0, & x \notin U. \end{cases}$$

Since the support of χ is contained in U, this formula defines a smooth function on all of $\mathbb{R}^n$. Its restriction to X is a. □

Let

$$\mathfrak{m}_X = \{h \in C^\infty(\mathbb{R}^n) \mid h|_X = 0\}.$$

The restriction homomorphism is surjective by Lemma 2.15, with kernel $\mathfrak{m}_X$. Hence

$$C^\infty(X) \cong C^\infty(\mathbb{R}^n)/\mathfrak{m}_X.$$

In particular, $C^\infty(X)$ is a finitely generated C^∞-ring, even when X is singular or has complicated topology.

The restricted coordinate functions generate $C^\infty(X)$. Their images under a homomorphism therefore determine a candidate smooth map. The relations in $\mathfrak{m}_X$ will force that map to land in X.

Theorem 2.16 (Full faithfulness for closed subsets)**.** *Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$ be closed subsets. Pullback induces a bijection*

$$\{\textit{smooth maps } Y \to X\} \longrightarrow \mathrm{Hom}_{C^\infty\text{-Ring}}\big(C^\infty(X), C^\infty(Y)\big).$$

Consequently, the functor

$$C^\infty(-)\colon (\textit{closed subsets of Euclidean spaces})^{\mathrm{op}} \longrightarrow C^\infty\text{-Ring}$$

is fully faithful.

Proof. Let

$$\Psi\colon C^\infty(X) \longrightarrow C^\infty(Y)$$

be a homomorphism. Write $x_1, \dots, x_n$ for the coordinate functions on $\mathbb{R}^n$, and define

$$g_i = \Psi(x_i|_X) \in C^\infty(Y), \qquad g = (g_1, \dots, g_n)\colon Y \longrightarrow \mathbb{R}^n.$$

The map g is smooth by Definition 2.14. For every $h \in C^\infty(\mathbb{R}^n)$, preservation of smooth functional calculus gives

$$\Psi(h|_X) = \Psi\big(\Phi_h(x_1|_X, \dots, x_n|_X)\big) = \Phi_h(g_1, \dots, g_n) = h \circ g.$$

We claim that g lands in X. If $g(y) = p \notin X$ for some $y \in Y$, choose a smooth bump function $h\colon \mathbb{R}^n \to \mathbb{R}$ supported in $\mathbb{R}^n \setminus X$ and satisfying $h(p) = 1$. Then $h|_X = 0$, whereas the displayed identity gives

$$0 = \Psi(h|_X)(y) = h(g(y)) = 1,$$

a contradiction.

It remains to identify Ψ. Given $a \in C^\infty(X)$, choose a global smooth extension $h \in C^\infty(\mathbb{R}^n)$ using Lemma 2.15. Then

$$\Psi(a) = \Psi(h|_X) = h \circ g = a \circ g.$$

Thus $\Psi = g^*$. The map g is unique because its component functions are the images of the restricted coordinates.

Conversely, pullback along a smooth map $g\colon Y \to X$ preserves every smooth operation pointwise, and hence defines a C^∞-homomorphism. The two constructions are inverse. □

The corresponding result for all locally closed subsets is Moerdijk and Reyes [MR91, Chapter I, Proposition 1.5].

For the usual class of finite-dimensional, second-countable smooth manifolds, Whitney's embedding theorem may be strengthened to give a proper Euclidean embedding [Lee02]. Its image is closed. The intrinsic smooth functions and smooth maps agree with those in Definition 2.14. We therefore obtain the case which will be used later.

Corollary 2.17 (Recovering manifolds)**.** *The contravariant functor*

$$C^\infty(-)\colon (\textit{smooth manifolds})^{\mathrm{op}} \longrightarrow C^\infty\text{-Ring}$$

is fully faithful.

The conclusion is stronger than a reconstruction of points. The entire smooth map $Y \to X$ is recovered from the homomorphism on function rings by looking at the images of finitely many ambient coordinate functions.

2.5 Further material: locally closed subsets

Full faithfulness extends from closed subsets to locally closed ones. The additional geometric step realizes an open subset as a closed hypersurface in one higher-dimensional Euclidean space. This is the construction in [MR91, Chapter I, Lemma 1.4 and Proposition 1.5].

Lemma 2.18 (Open subsets are smooth cozero loci)**.** *For every open subset $U \subseteq \mathbb{R}^n$, there is a smooth function $f\colon \mathbb{R}^n \to [0,\infty)$ such that*

$$U = \{x \in \mathbb{R}^n \mid f(x) \neq 0\}.$$

Proof. Choose a countable collection of open balls B_k which covers U and whose closures lie in U. For each k, choose a nonnegative smooth function ρ_k with compact support in U and strictly positive on B_k. Choose constants $\varepsilon_k > 0$ so small that

$$\varepsilon_k \sup_{x\in\mathbb{R}^n} |D^\alpha \rho_k(x)| \leq 2^{-k}$$

for every multi-index α with $|\alpha| \leq k$. Then

$$f = \sum_{k\geq 1} \varepsilon_k \rho_k$$

converges together with all its derivatives and hence defines a smooth function. It vanishes outside U. At every point of U, at least one ρ_k is positive, so f is positive there. □

Proposition 2.19 (Closed realization)**.** *Every locally closed subset $X \subseteq \mathbb{R}^n$ is diffeomorphic, in the sense of Definition 2.14, to a closed subset of $\mathbb{R}^{n+1}$.*

Proof. First let $U \subseteq \mathbb{R}^n$ be open, and choose f as in Lemma 2.18. The graph

$$H_f = \{(x,y) \in \mathbb{R}^n \times \mathbb{R} \mid yf(x) = 1\}$$

is closed, since it is the zero locus of the continuous function $(x,y) \mapsto yf(x) - 1$. The map

$$U \longrightarrow H_f, \qquad x \longmapsto \big(x, 1/f(x)\big)$$

is a diffeomorphism with inverse the first projection.

Now write the locally closed subset as $X = F \cap U$, with F closed in $\mathbb{R}^n$ and U open. Under the displayed diffeomorphism, X corresponds to

$$H_f \cap (F \times \mathbb{R}),$$

which is closed in $\mathbb{R}^{n+1}$. □

Corollary 2.20 (Moerdijk–Reyes)**.** *The contravariant functor from locally closed subsets of Euclidean spaces and smooth maps to finitely generated C^∞-rings is fully faithful.*

Proof. Replace the source and target by the closed realizations from Proposition 2.19, and apply Theorem 2.16. Their function rings are finitely generated because each closed realization is a quotient of a free C^∞-ring. □

2.6 Further material: infinite jets and Borel's theorem

This optional section extends the dual-number calculation to infinite Taylor expansions. It is not needed for the later chapters. For infinite jets, smooth functions have a striking flexibility: every formal Taylor series occurs. Let $\mathfrak{m}_0^\infty \subseteq C^\infty(\mathbb{R}^n)$ be the ideal of functions *flat at the origin*, meaning that every partial derivative of every order vanishes at the origin. The Taylor homomorphism is

$$T_0\colon C^\infty(\mathbb{R}^n) \longrightarrow \mathbb{R}[[x_1,\dots,x_n]], \qquad f \longmapsto \sum_{\alpha\in\mathbb{N}^n} \frac{D^\alpha f(0)}{\alpha!} x^\alpha.$$

Theorem 2.21 (Borel's theorem)**.** *The Taylor homomorphism T_0 is surjective and has kernel $\mathfrak{m}_0^\infty$. It therefore induces an isomorphism of commutative $\mathbb{R}$-algebras*

$$C^\infty(\mathbb{R}^n)/\mathfrak{m}_0^\infty \cong \mathbb{R}[[x_1, \ldots, x_n]].$$

In particular, the formal power series ring inherits a C^∞-ring structure from the quotient on the left.

This is the form of Borel's theorem recorded in Moerdijk and Reyes [MR91, Chapter I, Theorem 1.3].

Proof. The assertion about the kernel follows from the definition. Write a prescribed formal series as $\sum_{k\geq 0} P_k(x)$, with P_k homogeneous of degree k. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ equal to one near zero. We will construct

$$f(x) = \sum_{k\geq 0} P_k(x)\chi(\lambda_k x).$$

For a multi-index α, homogeneity gives

$$\sup_x |D^\alpha(P_k(x)\chi(\lambda x))| = \lambda^{|\alpha|-k} \sup_y |D^\alpha(P_k(y)\chi(y))|.$$

For each $k \geq 1$, choose λ_k sufficiently large that this quantity is at most 2^{-k} for every $|\alpha| < k$. Choose λ_0 arbitrarily positive.

For each fixed derivative order, the resulting derivative series has a uniformly summable tail; only finitely many initial terms are excluded from the estimate. It therefore converges to a smooth function and may be differentiated termwise to every order. The Taylor series at zero of the kth summand is exactly P_k, because its cutoff is constant near zero. Termwise evaluation of derivatives shows that f has the prescribed Taylor series. □

The quotient in Theorem 2.21 is generated as a C^∞-ring by the images of $x_1, \ldots, x_n$, but its underlying commutative algebra contains arbitrary formal power series. Generation under smooth operations allows every one of these formal series to be obtained from the coordinate generators. It is different from generation of the underlying commutative algebra.

Surjectivity of the Taylor map does not make Taylor series a complete description of smooth geometry. Its kernel has geometric consequences.

Example 2.22 (An equation invisible to all Taylor coefficients)**.** Define

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Every derivative away from zero is a polynomial in $1/x$ times e^{-1/x^2} and tends to zero as $x \to 0$. Successive differentiation therefore shows that f is smooth and flat at zero. The equations $f(x) = 0$ and $0 = 0$ have identical Taylor series at the origin, but their zero sets in every neighbourhood of the origin are respectively a singleton and the entire neighbourhood.

Borel's theorem says that every formal series occurs; this example shows what passage to that series can forget. Smooth function rings retain the flat functions as well as all finite and infinite Taylor data. The use of smooth rings in the seminar consequently retains more than the formal infinitesimal behaviour of an equation at one point.

The principal reference is [MR91, Chapter I, Section 1]. For a modern account of smooth rings and their spectra, see [Joy12, Sections 2–3].

3 Geometric constructions in C^∞-rings

The preceding chapter showed that a smooth manifold and all of its smooth maps are encoded by its ring of smooth functions. We now ask whether geometric constructions are encoded as well. Since the functor

$$C^\infty(-)\colon (\text{smooth manifolds})^{\mathrm{op}} \longrightarrow C^\infty\text{-Ring}$$

is contravariant, the first test is whether a pullback of manifolds becomes a pushout of C^∞-rings.

Question 3.1. When is the algebraic pushout of smooth function rings again the function ring of the geometric pullback?

Transversality is a sufficient condition. To understand why, we first show that independent smooth equations generate the full ideal of functions vanishing on their common zero locus. This lets us translate geometric constructions into presentations of smooth rings.

Two intermediate results prepare the general pullback theorem. Every manifold has a finitely presented function ring, and products and open restrictions have the expected algebraic descriptions. We then prove the transverse-pullback theorem locally and explain its consequences. The final proof section supplies the globalization argument. Together with the full faithfulness proved in the preceding chapter, these results identify manifolds inside smooth algebra and explain which intersections that embedding preserves.

3.1 Equations and zero loci

Return to the intersections in Figure 1.1. Both the transverse axes and the tangent line and parabola intersect in a single point. Their equations nevertheless give different algebraic pushouts. We begin by making this calculation precise, then ask when a quotient by equations is the function ring of their zero locus.

3.1.1 Coproducts impose simultaneous equations

We first fix the algebraic notation. The category of C^∞-rings has all small colimits. Following Moerdijk and Reyes, we write

$$B \otimes_\infty C$$

for the coproduct of two C^∞-rings. This notation records the categorical role of the construction. It is not, in general, the tensor product of the underlying commutative $\mathbb{R}$-algebras.

Proposition 3.2 (Coproducts of free rings and quotients)**.** *The following formulas hold.*

(1) There is a natural isomorphism

$$C^\infty(\mathbb{R}^n) \otimes_\infty C^\infty(\mathbb{R}^m) \cong C^\infty(\mathbb{R}^{n+m}).$$

(2) If $I \subseteq B$ and $J \subseteq C$ are ideals, then

$$(B/I) \otimes_\infty (C/J) \cong (B \otimes_\infty C)/(I, J),$$

where (I, J) denotes the ideal generated by the images of I and J.

(3) If $I, J \subseteq B$, then

$$(B/I) \amalg_B (B/J) \cong B/(I + J).$$

Proof. A homomorphism from $C^\infty(\mathbb{R}^n) \otimes_\infty C^\infty(\mathbb{R}^m)$ to a C^∞-ring D is a pair of homomorphisms from the two factors. By Theorem 2.6, this is the same as a pair consisting of an n-tuple and an m-tuple of elements of D, hence the same as an $(n + m)$-tuple. The free-ring theorem identifies the representing object as $C^\infty(\mathbb{R}^{n+m})$.

A homomorphism from the coproduct in the second formula to D is a pair of homomorphisms $B \to D$ and $C \to D$ which kill I and J, respectively. By the universal properties of the coproduct and the quotient, this is the same as a homomorphism from $(B \otimes_\infty C)/(I, J)$ to D. The third formula is proved in the same way: a compatible pair of maps from B/I and B/J is a map from B killing both ideals. □

Set

$$A = C^\infty(\mathbb{R}^2).$$

For the two coordinate axes, the function rings are $A/(x)$ and $A/(y)$. Imposing both equations gives

$$A/(x) \amalg_A A/(y) \cong A/(x, y) \cong \mathbb{R}.$$

This is the function ring of their intersection point. For the x-axis and the parabola $y = x^2$, the analogous calculation begins

$$A/(y) \amalg_A A/(y - x^2) \cong A/(y, y - x^2).$$

Eliminating y gives $C^\infty(\mathbb{R})/(x^2)$, the dual-number ring.

The first calculation recovers the function ring of the geometric intersection; the second retains a nonzero nilpotent element. The regular-equation theorem below explains why independent equations recover the function ring of the zero locus.

3.1.2 Regular equations

Let M be a smooth manifold and let $g_1, \ldots, g_q \in C^\infty(M)$. Write

$$Z(g_1, \ldots, g_q) = \{p \in M \mid g_1(p) = \cdots = g_q(p) = 0\}$$

and

$$\mathfrak{m}_Z = \{h \in C^\infty(M) \mid h|_Z = 0\}.$$

The inclusion $(g_1, \ldots, g_q) \subseteq \mathfrak{m}_Z$ is automatic. Equality is a geometric assertion about the equations.

We first isolate the globalization argument which will occur several times.

Lemma 3.3 (Local ideal membership)**.** *Let $h, g_1, \dots, g_q \in C^\infty(M)$. If every point of M has an open neighborhood U such that*

$$h|_U \in (g_1|_U, \dots, g_q|_U),$$

then $h \in (g_1, \dots, g_q)$ in $C^\infty(M)$.

Proof. Choose a locally finite open cover $\{U_\alpha\}$ on which

$$h|_{U_\alpha} = \sum_{i=1}^{q} a_{\alpha i} g_i|_{U_\alpha},$$

and choose a smooth partition of unity $\{\rho_\alpha\}$ subordinate to this cover. Since $\operatorname{supp}(\rho_\alpha) \subseteq U_\alpha$, the product $\rho_\alpha a_{\alpha i}$ extends by zero to a smooth function on M. Local finiteness then gives smooth functions

$$a_i = \sum_\alpha \rho_\alpha a_{\alpha i} \in C^\infty(M).$$

Since $\sum_\alpha \rho_\alpha = 1$, we obtain

$$h = \sum_\alpha \rho_\alpha h = \sum_{i=1}^{q} a_i g_i.$$

□

Theorem 3.4 (Regular equations)**.** *Let $g = (g_1, \dots, g_q) \colon M \to \mathbb{R}^q$, and let $Z = g^{-1}(0)$. Suppose that*

$$D_p g \colon T_p M \longrightarrow \mathbb{R}^q$$

is surjective for every $p \in Z$. Then

$$\mathfrak{m}_Z = (g_1, \dots, g_q).$$

This is Moerdijk and Reyes [MR91, Chapter I, Lemma 2.1].

Proof. Let $h \in \mathfrak{m}_Z$. We prove ideal membership locally and apply Lemma 3.3.

Suppose first that $p \notin Z$. Some $g_i(p)$ is nonzero, so after shrinking to a neighborhood U of p, the function $g_i|_U$ is invertible. Hence

$$h|_U = (h/g_i) g_i|_U.$$

Now suppose that $p \in Z$. The submersion theorem gives local coordinates

$$(x_1, \dots, x_q, y_1, \dots, y_{d-q})$$

centered at p in which $g_i = x_i$ for $1 \le i \le q$. After shrinking, we may take the coordinate neighborhood to be a product which is convex in the x-variables. In these coordinates, $h(0, y) = 0$. The parameterized form of Hadamard's lemma gives

$$h(x, y) = h(0, y) + \sum_{i=1}^{q} x_i \int_0^1 \frac{\partial h}{\partial x_i}(tx, y)\, dt = \sum_{i=1}^{q} x_i h_i(x, y)$$

for smooth functions h_i. Thus h belongs locally to the ideal generated by the g_i. The local ideal-membership lemma finishes the proof. □

The theorem says more than the regular-value theorem. The latter identifies Z as a submanifold; the former identifies the entire ideal of functions vanishing on Z.

Corollary 3.5 (Function ring of a regular zero locus)**.** *Under the hypotheses of Theorem 3.4, restriction induces an isomorphism*

$$C^\infty(M)/(g_1, \dots, g_q) \cong C^\infty(Z).$$

Proof. The zero locus Z is a closed embedded submanifold of M. Every smooth function on Z extends to M: extend it along a tubular-neighborhood retraction, multiply by a cutoff supported in that neighborhood, and extend by zero. Hence the restriction map $C^\infty(M) \to C^\infty(Z)$ is surjective. Its kernel is $\mathfrak{m}_Z$, which is $(g_1, \dots, g_q)$ by Theorem 3.4. □

The regularity hypothesis is essential. For $g(t) = t^2$, the zero locus is the origin, but

$$(t^2) \subsetneq (t) = \mathfrak{m}_{\{0\}}.$$

Consequently,

$$C^\infty(\mathbb{R})/(t^2) \not\cong C^\infty(\{0\}).$$

This is exactly the defect seen in the tangent line–parabola intersection.

3.2 Manifold function rings are finitely presented

Our first application is a finiteness theorem: every manifold function ring can be described by finitely many generators and smooth relations. Here finite presentation is understood in the sense of Definition 2.9.

Example 3.6 (The circle)**.** The circle has the finite presentation

$$C^\infty(S^1) \cong C^\infty(\mathbb{R}^2)/(x^2 + y^2 - 1).$$

The gradient $(2x, 2y)$ is nonzero on the circle, so this follows from the regular-equation theorem.

For a general manifold, a short global list of regular equations need not be available. We will instead use a tubular neighborhood and a retraction.

The first step is to present the function ring of an arbitrary open subset of Euclidean space. One extra generator and one relation suffice, regardless of the shape of the open subset.

Proposition 3.7 (One-relation presentation of an open subset)**.** *Let $U \subseteq \mathbb{R}^n$ be open. Choose a smooth function $a\colon \mathbb{R}^n \to [0, \infty)$ such that $U = \{a \neq 0\}$. Then*

$$C^\infty(U) \cong C^\infty(\mathbb{R}^{n+1})/(ya(x) - 1).$$

This is Moerdijk and Reyes [MR91, Chapter I, Corollary 2.2].

Proof. The map

$$U \longrightarrow \mathbb{R}^{n+1}, \qquad x \longmapsto \big(x, 1/a(x)\big)$$

is a diffeomorphism onto the hypersurface

$$H_a = \{(x, y) \mid ya(x) - 1 = 0\}.$$

Along H_a, the partial derivative of $ya(x) - 1$ with respect to y is $a(x)$, which is nonzero. Thus the defining equation is regular, and Corollary 3.5 gives

$$C^\infty(U) \cong C^\infty(H_a) \cong C^\infty(\mathbb{R}^{n+1})/(ya(x) - 1).$$

□

A properly embedded manifold is a retract of an open tubular neighborhood in Euclidean space. On function rings, it is therefore enough to know that finite presentation passes to retracts.

Lemma 3.8 (Retracts of finitely presented rings)**.** *A retract of a finitely presented C^∞-ring is finitely presented.*

Proof. Let A be a retract of

$$P = C^\infty(\mathbb{R}^n)/(r_1, \dots, r_k),$$

with maps $s\colon A \to P$ and $p\colon P \to A$ satisfying $p \circ s = \mathrm{id}_A$. The composite $e = s \circ p$ is an idempotent endomorphism of P whose image is $s(A)$. Choose smooth functions $e_i \in C^\infty(\mathbb{R}^n)$ representing the images under e of the coordinate generators x_i of P. Then A is the coequalizer of id_P and e, and hence

$$A \cong C^\infty(\mathbb{R}^n)/(r_1, \dots, r_k, x_1 - e_1, \dots, x_n - e_n).$$

Indeed, a homomorphism $u\colon P \to D$ factors through p precisely when $u = u \circ e$, and it is enough to check this equality on the coordinate generators. The displayed quotient therefore has the universal property of A. □

Theorem 3.9 (Finite presentation for manifolds)**.** *If M is a finite-dimensional, second-countable smooth manifold without boundary, then $C^\infty(M)$ is a finitely presented C^∞-ring.*

This is Moerdijk and Reyes [MR91, Chapter I, Theorem 2.3].

Proof. Choose a proper smooth embedding $i\colon M \hookrightarrow \mathbb{R}^N$. Its image is closed. The tubular-neighborhood theorem supplies an open neighborhood $U \subseteq \mathbb{R}^N$ of M and a smooth retraction $r\colon U \to M$. Since $r \circ i = \mathrm{id}_M$, the induced maps

$$C^\infty(M) \xrightarrow{r^*} C^\infty(U) \xrightarrow{i^*} C^\infty(M)$$

have composite the identity. By Proposition 3.7, the middle ring is finitely presented. The result follows from Lemma 3.8. □

The theorem does not say that the underlying commutative $\mathbb{R}$-algebra $C^\infty(M)$ is finitely presented. All finiteness statements take place in the algebraic theory whose operations are arbitrary smooth functions.

3.3 Products and open restrictions

The same passage from open subsets to manifolds also computes products. Once products are understood, the equation $ya = 1$ identifies restriction to the open subset $\{a \neq 0\}$ with adjoining an inverse to a. These two constructions will let us reduce transverse pullbacks to local equations and then compare the calculations on overlapping charts.

For manifolds M and P, pullback along the two projections determines a natural comparison homomorphism

$$\theta_{M,P} \colon C^\infty(M) \otimes_\infty C^\infty(P) \longrightarrow C^\infty(M \times P).$$

Lemma 3.10 (Products of open subsets)**.** *If $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ are open, then $\theta_{U,V}$ is an isomorphism.*

This is Moerdijk and Reyes [MR91, Chapter I, Lemma 2.4].

Proof. Choose smooth functions a and b with $U = \{a \neq 0\}$ and $V = \{b \neq 0\}$. By Propositions 3.2 and 3.7,

$$C^\infty(U) \otimes_\infty C^\infty(V) \cong \frac{C^\infty(\mathbb{R}^{n+m+2})}{(ya(x) - 1, zb(w) - 1)}.$$

The two displayed equations have independent differentials along their common zero locus, since the coefficients of dy and dz are $a(x)$ and $b(w)$ there. Their common zero locus is

$$H_a \times H_b \cong U \times V.$$

The regular-equation theorem therefore identifies the displayed quotient with $C^\infty(U \times V)$. Under these identifications the isomorphism is the natural comparison map $\theta_{U,V}$. □

Proposition 3.11 (Products of manifolds)**.** *For smooth manifolds M and P, the natural map*

$$\theta_{M,P} \colon C^\infty(M) \otimes_\infty C^\infty(P) \xrightarrow{\sim} C^\infty(M \times P)$$

is an isomorphism.

This is Moerdijk and Reyes [MR91, Chapter I, Proposition 2.5].

Proof. Choose proper embeddings of M and P, open tubular neighborhoods $U \supseteq M$ and $V \supseteq P$, and retractions $U \to M$ and $V \to P$. Thus (M, P) is a retract of (U, V). Naturality of θ gives a corresponding retraction diagram in the arrow category of C^∞-rings: $\theta_{M,P}$ is a retract of $\theta_{U,V}$. The latter is an isomorphism by Lemma 3.10. A retract of an isomorphism is an isomorphism, so the result follows. □

This result is the first complete answer to Question 3.1: a product is a pullback over a point, and its function ring is always the corresponding coproduct.

Every open subset of a manifold can be written as $\{a \neq 0\}$ for some smooth function a. We now extend the one-relation presentation of an open subset of Euclidean space to this setting.

Proposition 3.12 (Smooth localization)**.** *Let M be a manifold, let $a \in C^\infty(M)$, and put $U = \{a \neq 0\}$. Restriction exhibits $C^\infty(U)$ as the universal C^∞-ring under $C^\infty(M)$ in which a becomes invertible.*

Proof. By the product theorem, adjoining one free smooth generator y gives

$$C^\infty(M) \otimes_\infty C^\infty(\mathbb{R}) \cong C^\infty(M \times \mathbb{R}).$$

The quotient by $ya - 1$ therefore universally makes a invertible. The function $(m, y) \mapsto ya(m) - 1$ is regular along its zero set, since its derivative in y is the nonzero number $a(m)$. That zero manifold is the graph $m \mapsto (m, 1/a(m))$ over U. The regular-equation theorem identifies the quotient with $C^\infty(U)$, and its structure map with restriction. □

For $M = \mathbb{R}^n$, this is Moerdijk and Reyes [MR91, Chapter I, Proposition 1.6 and Corollary 2.2]. The argument extends it to manifolds using the product calculation in Proposition 3.11.

3.4 Transverse pullbacks

We now return to Question 3.1. The essential case is the inverse image of an embedded submanifold: transversality ensures that its local defining equations remain independent after pullback. General transverse fiber products reduce to this case by using the diagonal.

Theorem 3.13 (Transverse inverse images become pushouts)**.** *Let $f\colon M \to N$ be transverse to an embedded submanifold $i\colon Z \hookrightarrow N$, and put $P = f^{-1}(Z)$. Then*

$$\begin{array}{ccc} C^\infty(N) & \xrightarrow{f^*} & C^\infty(M) \\ {\scriptstyle i^*}\downarrow & \ulcorner & \downarrow \\ C^\infty(Z) & \longrightarrow & C^\infty(P) \end{array}$$

is a pushout of C^∞-rings.

This is Moerdijk and Reyes [MR91, Chapter I, Proposition 2.6].

The local calculation contains the geometric content. Suppose $U \subseteq N$ is an open set on which Z is defined by functions $g_1, \dots, g_q$ with independent differentials along $U \cap Z$. Set $V = f^{-1}(U)$. Transversality implies that

$$g_1 \circ f, \dots, g_q \circ f$$

have independent differentials along $V \cap P$. Hence Corollary 3.5 gives

$$C^\infty(U \cap Z) \cong C^\infty(U)/(g_1, \dots, g_q)$$

and

$$C^\infty(V \cap P) \cong C^\infty(V)/(g_1 \circ f, \dots, g_q \circ f).$$

The local function-ring square is therefore isomorphic to

$$\begin{array}{ccc} C^\infty(U) & \xrightarrow{\qquad f^* \qquad} & C^\infty(V) \\ \downarrow & \ulcorner & \downarrow \\ C^\infty(U)/(g_1, \dots, g_q) & \longrightarrow & C^\infty(V)/(g_1 \circ f, \dots, g_q \circ f). \end{array}$$

It is a pushout by Proposition 3.2: the lower-right ring is obtained by imposing on $C^\infty(V)$ the images of the same equations.

This calculation also identifies the exact use of transversality. It ensures that the pulled-back equations remain regular, so that their algebraic quotient is the function ring of the geometric inverse image.

The local calculation proves the theorem on coordinate neighbourhoods. Globalization uses partitions of unity and descent for functions modulo finitely many equations. We give that argument in Section 3.5; first we explain the consequence for arbitrary transverse fibre products.

We can now answer Question 3.1 for general fiber products. Let $f\colon M \to N$ and $g\colon P \to N$ be transverse. Then

$$f \times g\colon M \times P \longrightarrow N \times N$$

is transverse to the diagonal Δ_N, and

$$(f \times g)^{-1}(\Delta_N) = M \times_N P.$$

Theorem 3.14 (Transverse pullbacks become pushouts)**.** *For transverse smooth maps $f\colon M \to N$ and $g\colon P \to N$, the two squares*

$$\begin{array}{ccc} M \times_N P & \longrightarrow & P \\ \downarrow & \lrcorner & \downarrow{\scriptstyle g} \\ M & \xrightarrow[f]{} & N \end{array} \qquad \begin{array}{ccc} C^\infty(N) & \xrightarrow{g^*} & C^\infty(P) \\ {\scriptstyle f^*}\downarrow & \ulcorner & \downarrow \\ C^\infty(M) & \longrightarrow & C^\infty(M \times_N P) \end{array}$$

are respectively a pullback of manifolds and a pushout of C^∞-rings.

Proof. Apply Theorem 3.13 to $f \times g$ and $\Delta_N \subseteq N \times N$. Then use Proposition 3.11 to identify the function rings of $M \times P$ and $N \times N$ with the corresponding coproducts. The resulting universal property says precisely that a map out of $C^\infty(M \times_N P)$ is a pair of maps out of $C^\infty(M)$ and $C^\infty(P)$ whose restrictions along f^* and g^* agree. This is the pushout property of the square displayed above. □

Combining Corollary 2.17 and theorems 3.9 and 3.14 gives the classical structural theorem.

Theorem 3.15 (Manifolds inside smooth algebra)**.** *The contravariant functor*

$$C^\infty(-)\colon (\textit{smooth manifolds})^{\mathrm{op}} \longrightarrow C^\infty\text{-Ring}$$

is fully faithful, takes values in finitely presented C^∞-rings, and sends transverse pullback squares to pushout squares.

This is Moerdijk and Reyes [MR91, Chapter I, Theorem 2.8].

The tangency calculation in Section 3.1.1 shows why the transversality hypothesis cannot simply be dropped. The geometric intersection is a point, with function ring $\mathbb{R}$, while the pushout is $C^\infty(\mathbb{R})/(x^2) \cong \mathbb{R}[\varepsilon]/(\varepsilon^2)$. The comparison homomorphism kills ε: it forgets the infinitesimal information retained by the equations.

This pushout shows that ordinary smooth algebra already contains intersections which are not manifolds. It does not retain all intersection data, however: the ordinary pushout for the self-intersection $* \to \mathbb{R} \leftarrow *$ is just $\mathbb{R}$. The excess direction seen in Example 1.8 requires a derived construction. The final exercise asks us to compare these two descriptions. The transverse-pullback theorem proved here is the compatibility condition for both the ordinary and the derived universal constructions in the next chapter.

3.5 Further proofs: globalizing the pushout

We supply the globalization used in Theorem 3.13. The local regular-equation calculation is already established. The remaining task is to glue maps into a test ring, which may contain infinitesimal elements invisible on its real points.

3.5.1 Test rings and descent

To verify a pushout, we must test maps into arbitrary smooth rings. The following criterion will let us restrict to finitely presented test rings, where we can glue functions modulo a finite list of equations.

Proposition 3.16 (Filtered-colimit criterion)**.** *A C^∞-ring A is finitely presented if and only if*

$$\mathrm{Hom}_{C^\infty\text{-Ring}}(A, -)$$

preserves filtered colimits.

Proof. Suppose first that $A = C^\infty(\mathbb{R}^n)/(r_1, \dots, r_k)$. For every C^∞-ring B,

$$\mathrm{Hom}(A, B) \cong \big\{ b \in B^n \mid \Phi_{r_1}(b) = \cdots = \Phi_{r_k}(b) = 0 \big\}.$$

Filtered colimits of C^∞-rings are computed on underlying sets, and filtered colimits of sets commute with finite limits. The displayed functor therefore preserves filtered colimits.

Conversely, every C^∞-ring is a filtered colimit of finitely presented ones mapping to it. Concretely, finite sets of elements give maps from finitely generated free rings, finite sets of relations give finitely presented quotients, two such stages admit a common refinement by combining their generators and relations, and parallel maps become equal after adding finitely many relations on a finite set of generators. Apply this canonical filtered presentation to A. If $\mathrm{Hom}(A, -)$ preserves its colimit, then id_A factors through one finitely presented stage $P \to A$. Thus A is a retract of P, and Lemma 3.8 completes the proof. □

For a finitely presented test ring, the following descent statement lets us glue the local maps constructed by imposing regular equations.

Lemma 3.17 (Cech descent)**.** *Let*

$$B = C^\infty(\mathbb{R}^k)/(h_1, \dots, h_p), \qquad X = Z(h_1, \dots, h_p),$$

and let $\{W_\alpha\}$ be a cover of X by open subsets of $\mathbb{R}^k$. For a nonempty finite set S of indices, put

$$W_S = \bigcap_{\alpha \in S} W_\alpha, \qquad B_S = C^\infty(W_S)/(h_1|_{W_S}, \dots, h_p|_{W_S}).$$

Then restriction induces an isomorphism

$$B \xrightarrow{\sim} \varprojlim_S B_S,$$

where the limit is taken over the Cech diagram of nonempty finite intersections.

This is Moerdijk and Reyes [MR91, Chapter I, Lemma 2.7].

Proof. It is enough to prove the usual gluing statement. Suppose that we have $g_\alpha \in C^\infty(W_\alpha)$ such that on every pairwise intersection

$$g_\alpha - g_\beta \in (h_1, \dots, h_p).$$

We must construct a unique class $[g] \in B$ whose restriction is $[g_\alpha]$ for every α.

Add $\mathbb{R}^k \setminus X$ to the cover and choose a locally finite refinement $\{V_\lambda\}$ of the resulting cover of $\mathbb{R}^k$. If V_λ lies in some W_α, let $k_\lambda = g_\alpha|_{V_\lambda}$, choosing one such α. If V_λ lies in $\mathbb{R}^k \setminus X$, let $k_\lambda = 0$. The choices are compatible modulo $(h_1, \dots, h_p)$. Indeed, this follows from the original compatibility on overlaps of the W_α, while on an open set disjoint from X the function $h_1^2 + \cdots + h_p^2$ is invertible, so $(h_1, \dots, h_p)$ is the unit ideal.

Let $\{\rho_\lambda\}$ be a partition of unity subordinate to the refined cover and define

$$g = \sum_\lambda \rho_\lambda k_\lambda,$$

extending each summand by zero. This is a globally defined smooth function. Fix an original index β. Near any point of W_β, only finitely many ρ_λ are nonzero, so

$$g - g_\beta = \sum_\lambda \rho_\lambda (k_\lambda - g_\beta)$$

belongs locally to $(h_1, \dots, h_p)$. By Lemma 3.3, it belongs to that ideal on W_β. Thus $[g]$ restricts to $[g_\beta]$.

If g' is another choice, then $g - g'$ belongs locally to $(h_1, \dots, h_p)$ on a cover of $\mathbb{R}^k$. Another application of Lemma 3.3 shows that $g - g'$ belongs to the global ideal. Hence $[g] = [g']$ in B. The construction respects every smooth operation, since those operations are computed componentwise in the limit. □

We also record how a homomorphism into a finitely presented test ring restricts to an open cover. This is the bridge between the local geometric calculation and Lemma 3.17.

Lemma 3.18 (Restricting a test-ring map)**.** *Let $\eta\colon C^\infty(M) \to B$ be a homomorphism, where $B = C^\infty(\mathbb{R}^k)/(h_1, \dots, h_p)$, and let $\{V_\alpha\}$ be an open cover of M. There is an open cover $\{W_\alpha\}$ of $X = Z(h_1, \dots, h_p)$ such that η induces compatible homomorphisms*

$$\eta_S\colon C^\infty(V_S) \longrightarrow B_S$$

on every nonempty finite intersection.

Proof. Choose $a_\alpha \in C^\infty(M)$ with $V_\alpha = \{a_\alpha \neq 0\}$. Represent $\eta(a_\alpha)$ by a smooth function $\widetilde{a}_\alpha$ on $\mathbb{R}^k$, and set

$$W_\alpha = \{\widetilde{a}_\alpha \neq 0\}.$$

These open subsets of $\mathbb{R}^k$ cover X. Indeed, evaluation at $x \in X$ gives a real point $B \to \mathbb{R}$, and its composite with η is evaluation at a point $m_x \in M$. Since the V_α cover M, some $a_\alpha(m_x)$ is nonzero, and this value equals $\widetilde{a}_\alpha(x)$.

In B_S, each image of $\eta(a_\alpha)$ for $\alpha \in S$ is invertible. The universal property of adjoining smooth inverses in Proposition 3.12 extends η uniquely to $C^\infty(V_S) \to B_S$. Uniqueness makes these extensions compatible under restriction. □

3.5.2 Globalizing the local calculation

Proof of Theorem 3.13. We first reduce to a closed embedding. An embedded submanifold Z is locally closed in N, so it is closed in some open neighbourhood $N' \subseteq N$. The inclusions $N' \to N$ and $f^{-1}(N') \to M$ give smooth localizations. Their function-ring square is a pushout by the universal property of localization. Pasting with the square over N' therefore reduces the assertion to the case where Z is closed in N.

For a C^∞-ring B, consider the natural map

$$\mathrm{Hom}(C^\infty(P), B) \longrightarrow \mathrm{Hom}(C^\infty(M), B) \times_{\mathrm{Hom}(C^\infty(N), B)} \mathrm{Hom}(C^\infty(Z), B).$$

All four manifold function rings are finitely presented by Theorem 3.9. Both sides therefore preserve filtered colimits in B, by Proposition 3.16 and the fact that filtered colimits of sets commute with finite limits. Since every C^∞-ring is a filtered colimit of finitely presented ones, it is enough to prove that the displayed map is a bijection when

$$B = C^\infty(\mathbb{R}^k)/(h_1, \dots, h_p).$$

Let homomorphisms

$$\varphi \colon C^\infty(M) \to B, \qquad \psi \colon C^\infty(Z) \to B$$

agree after restriction from $C^\infty(N)$. Choose an open cover $\{U_\alpha\}$ of N such that $U_\alpha \cap Z$ is cut out in U_α by independent functions. Put $V_\alpha = f^{-1}(U_\alpha)$. The local calculation in Section 3.4 shows that the function-ring square over every nonempty finite intersection U_S is a pushout.

Choose $b_\alpha \in C^\infty(N)$ with $U_\alpha = \{b_\alpha \neq 0\}$. Then V_α is the cozero locus of $b_\alpha \circ f$, while $U_\alpha \cap Z$ is the cozero locus of $b_\alpha|_Z$. The compatibility of φ and ψ says that these two elements have the same image in B. Consequently, Lemma 3.18 produces one open cover $\{W_\alpha\}$ of $X = Z(h_1, \dots, h_p)$ on which both maps restrict. We obtain compatible local maps

$$C^\infty(V_S) \longrightarrow B_S, \qquad C^\infty(U_S \cap Z) \longrightarrow B_S.$$

The local pushout property gives a unique homomorphism

$$\chi_S \colon C^\infty(V_S \cap P) \longrightarrow B_S.$$

These homomorphisms are compatible on intersections, by their uniqueness.

For $a \in C^\infty(P)$, apply χ_S to the restriction of a to $V_S \cap P$. The resulting elements form a compatible family in the Cech diagram $\{B_S\}$. By Lemma 3.17, they determine a unique element $\chi(a) \in B$. This construction commutes with every smooth operation, so it defines a C^∞-homomorphism

$$\chi \colon C^\infty(P) \longrightarrow B.$$

Locally, χ has the required composites with the restriction maps from M and Z; descent makes these equalities global. The same descent argument shows uniqueness. Thus the displayed map of Hom-sets is a bijection. □

3.6 Further calculations

Example 3.19 (Smooth coproducts versus algebraic tensor products)**.** The difference between a smooth coproduct and an algebraic tensor product is visible in one function. The image of

$$C^\infty(\mathbb{R}) \otimes_{\mathbb{R}} C^\infty(\mathbb{R}) \longrightarrow C^\infty(\mathbb{R}^2)$$

consists of finite sums $\sum_j f_j(x) g_j(y)$. The function e^{xy} cannot be written in this form.

Indeed, if it were a sum of r such terms, choose $r+1$ distinct real numbers $x_0, \dots, x_r$. All functions $e^{x_i y}$ would lie in the span of $g_1, \dots, g_r$ and hence be linearly dependent. But differentiating a linear relation at $y = 0$ in orders $0, \dots, r$ gives a Vandermonde system with invertible matrix (x_i^j). The functions are therefore linearly independent, a contradiction. The smooth coproduct includes e^{xy} because it permits smooth substitution involving variables from both factors.

Exercise 3.20. Give a finite presentation of $C^\infty(\mathbb{R} \setminus \{0\})$ and use it to calculate its coproduct with $C^\infty(S^1)$.

Exercise 3.21. Show that the pushout for the self-intersection $* \to \mathbb{R} \leftarrow *$ in ordinary smooth rings is $\mathbb{R}$. Compare the result with the linearized matching complex $[0 \to \mathbb{R}]$ of Chapter 1.

The proofs follow [MR91, Chapter I, Section 2]. The differential-topological ingredients are standard tubular neighbourhood, partition-of-unity, and submersion results; see [Lee02]. The transverse compatibility becomes part of the universal property in [CS19].

4 Derived manifolds

The preceding chapter established the structural theorem

$$C^\infty(-)\colon \mathrm{Mfd}^{\mathrm{op}} \longrightarrow C^\infty\text{-}\mathrm{Ring}^{\mathrm{fp}}.$$

This functor is fully faithful, and it sends transverse pullback squares of manifolds to pushout squares of C^∞-rings. Thus smooth algebra already contains every manifold and every transverse intersection. It contains more: the nontransverse line–parabola intersection produced the finitely presented ring

$$C^\infty(\mathbb{R})/(x^2),$$

which is not the function ring of a manifold.

These examples suggest a universal enlargement of the category of manifolds.

Question 4.1. What is the universal way to enlarge manifolds so that all finite limits exist, while preserving transverse pullbacks and the terminal object?

The answer depends on the categories in which we ask the question. Ordinary smooth rings retain nilpotent thickenings, but their pushouts collapse the excess self-intersections seen in Example 1.8. To retain that information, we pass to ∞-categories, where morphisms carry identifications between them, identifications between those identifications, and so on. Pullbacks can then retain this higher information.

We first describe the universal construction for ordinary smooth rings and locate its limitation in a self-intersection calculation. We then introduce the pullback formalism in ∞-categories and extend smooth algebra to this setting. The final step relates the resulting animated smooth rings to a universal category of derived manifolds. We quote the algebraic and geometric extension theorems, explaining their content through presentations and examples. The next chapter will calculate the resulting derived intersections.

4.1 The ordinary finite-limit envelope

We begin with Euclidean spaces and freely adjoin finite limits while preserving their finite products. The result is the opposite category of finitely presented smooth rings: generators specify coordinates, and relations specify equations. Its universal property says that a finite-limit-preserving functor out of this category is determined by an interpretation of smooth operations in its target.

Recall that Euc consists of Euclidean spaces and smooth maps. In Proposition 2.5, an ordinary C^∞-ring was a finite-product-preserving functor $\mathrm{Euc} \to \mathrm{Set}$. The same definition works in any ordinary category with finite products.

Definition 4.2 (C^∞-ring object)**.** Let C be an ordinary category with finite products. A *C^∞-ring object in C* is a finite-product-preserving functor

$$R\colon \mathrm{Euc} \longrightarrow C.$$

We write $\mathrm{Alg}_{C^\infty}(C)$ for the category of C^∞-ring objects in C.

The object $R(\mathbb{R})$ carries an operation in C for every smooth map between Euclidean spaces. Since $R(\mathbb{R}^n) \cong R(\mathbb{R})^n$, the functor is determined by this one object together with its smooth operations. For $C = \mathrm{Set}$ we recover ordinary C^∞-rings.

A finite-limit-preserving functor transports such a ring object, because it preserves finite products and the identities between the smooth operations. We will use this to characterize an extension from Euclidean spaces by its value on the smooth line.

To describe the universal target for these operations, set

$$\mathrm{Aff}_{C^\infty}^{\mathrm{fp}} = (C^\infty\text{-}\mathrm{Ring}^{\mathrm{fp}})^{\mathrm{op}}.$$

An object of this opposite category is an affine smooth object described by its function ring. Its spectrum and structure sheaf will be constructed in Chapter 7.

There is a functor

$$j\colon \mathrm{Euc} \longrightarrow \mathrm{Aff}_{C^\infty}^{\mathrm{fp}}$$

which sends $\mathbb{R}^n$ to the free ring $C^\infty(\mathbb{R}^n)$, regarded as an object of the opposite category. A smooth map $f\colon \mathbb{R}^n \to \mathbb{R}^m$ induces

$$f^*\colon C^\infty(\mathbb{R}^m) \longrightarrow C^\infty(\mathbb{R}^n),$$

which becomes a morphism in the same direction as f after taking opposites. The functor j preserves finite products.

The category of formal affine smooth objects has the following universal property.

Theorem 4.3 (Ordinary finite-limit envelope)**.** *Let C be an ordinary category with finite limits. Restriction along j induces an equivalence of categories*

$$\mathrm{Fun}^{\mathrm{lex}}\big(\mathrm{Aff}_{C^\infty}^{\mathrm{fp}}, C\big) \simeq \mathrm{Alg}_{C^\infty}(C),$$

where the left side consists of finite-limit-preserving functors.

This is the 0-truncated case of the finite-limit-envelope theorem for an algebraic theory [CS19, Theorem 3.22 and Remark 3.26].

The universal property has an explicit consequence: a functor preserving finite limits must send a presentation by equations to the corresponding zero locus in its target. A finitely presented C^∞-ring can be written

$$A = \frac{C^\infty(\mathbb{R}^n)}{(f_1, \dots, f_k)}.$$

Equivalently, A is the coequalizer

$$C^\infty(\mathbb{R}^k) \underset{0^*}{\overset{(f_1,\dots,f_k)^*}{\rightrightarrows}} C^\infty(\mathbb{R}^n) \longrightarrow A,$$

where the ith coordinate function on $\mathbb{R}^k$ is sent respectively to f_i and to zero. In the opposite category, this becomes an equalizer diagram

$$A \longrightarrow C^\infty(\mathbb{R}^n) \rightrightarrows C^\infty(\mathbb{R}^k).$$

Consequently, if F preserves finite limits and

$$R = F \circ j \colon \mathrm{Euc} \longrightarrow C,$$

then $F(A)$ is forced to be the internal zero locus

$$F(A) \cong \mathrm{Eq}\left(R(\mathbb{R})^n \underset{0}{\overset{(f_1,\ldots,f_k)}{\rightrightarrows}} R(\mathbb{R})^k \right).$$

Thus the chosen C^∞-ring object determines the value of F on every finite presentation.

This argument explains uniqueness, but a theorem is still required. One must show that the construction is independent of the chosen presentation, functorial in A, and compatible with natural transformations. The general proof observes that finitely presented algebras are generated from finitely generated free algebras by finite colimits and retracts. Taking opposites turns those operations into finite limits and retracts. The same generation statements underlie the animated version of the theorem. Extending from Euclidean spaces to all manifolds requires a geometric theorem, which we will state in Section 4.5.

4.2 Why ordinary intersections are not enough

The ordinary envelope gives a formal zero locus for every smooth map, even when the geometric zero set is singular. Quotients retain the equations and can therefore distinguish nilpotent thickenings. They do not, however, distinguish a closed subobject from its self-intersection. This calculation will explain what needs to change in the universal problem.

Let M be a manifold and let

$$f = (f_1, \ldots, f_k) \colon M \longrightarrow \mathbb{R}^k$$

be smooth. The diagram of function rings is

$$\begin{array}{ccc} C^\infty(\mathbb{R}^k) & \xrightarrow{f^*} & C^\infty(M) \\ {\scriptstyle 0^*}\downarrow & \ulcorner & \downarrow \\ \mathbb{R} & \longrightarrow & A_f. \end{array}$$

By the quotient formula from Proposition 3.2,

$$A_f \cong \frac{C^\infty(M)}{(f_1, \ldots, f_k)}.$$

After taking opposites, the corresponding formal affine object is the pullback of

$$M \xrightarrow{f} \mathbb{R}^k \xleftarrow{0} *.$$

This construction makes sense without a spectrum or a structure sheaf.

Example 4.4 (A transverse zero)**.** The quotient $C^\infty(\mathbb{R})/(x) \cong \mathbb{R}$ says that the formal zero locus of x is the ordinary point, as required by transverse compatibility.

Example 4.5 (A singular zero)**.** The dual-number quotient $C^\infty(\mathbb{R})/(x^2)$ from the preceding chapters is the formal zero locus of x^2. Thus the ordinary envelope already retains its nilpotent thickening.

Now consider the self-intersection of a closed subobject. Let A be a C^∞-ring, let $I \subseteq A$ be an ideal, and put

$$B = A/I.$$

The self-intersection of the corresponding formal closed subobject is computed by

$$B \amalg_A B.$$

Proposition 4.6 (Ordinary self-intersections)**.** *There is a natural isomorphism*

$$(A/I) \amalg_A (A/I) \cong A/I.$$

Proof. Apply the quotient pushout formula:

$$(A/I) \amalg_A (A/I) \cong A/(I + I) = A/I.$$

□

For example, let $A = C^\infty(\mathbb{R}^2)$ and $I = (y)$. Then B represents the x-axis. Its ordinary self-intersection inside the plane is again represented by B. The quotient remembers that y vanishes, but it has no record that the relation $y = 0$ was imposed twice.

The collapse reflects how pullbacks work in an ordinary category. An ordinary pullback is characterized by an isomorphism of sets

$$\mathrm{Hom}_C(T, X \times_Z Y) \cong \mathrm{Hom}_C(T, X) \times_{\mathrm{Hom}_C(T,Z)} \mathrm{Hom}_C(T, Y).$$

An element on the right is a pair of maps whose composites to Z are equal. In a set, equality is a property. Once the equality holds, there is no further datum recording how it holds. In particular, imposing the same relation twice cannot create new equality data.

4.3 Pullbacks with higher identifications

We want a pullback to record how two composites are identified. For this, we first introduce animae, whose elements admit identifications and higher identifications. The based loops in a circle will illustrate what a self-pullback can retain. We then formulate pullbacks in ∞-categories before returning to smooth algebra.

Definition 4.7 (Anima, informal description)**.** An *anima* behaves like a set in which the identifications between any two elements themselves form an anima. Identifications can be composed and inverted, with the usual identities holding through coherent higher identifications.

For the purposes of this seminar, we take the notion of an anima as a foundational concept, much as we take the notion of a set in ordinary mathematics. It is possible to construct explicit models using, for example, simplicial sets. Such constructions require substantial setup and do not by themselves convey the intended intuition. We will instead work with the heuristic picture above: an anima behaves like a set, except that its identifications themselves form animae.

The following examples give different meanings to these identifications.

- Every set X determines an anima. For $x, y \in X$, the anima of identifications between x and y is a point if $x = y$, and empty otherwise. These are the 0-truncated animae.
- Every ordinary category C determines an anima $C^{\simeq}$, its *groupoid core*. Its points are the objects of C, and its identifications are isomorphisms in C. For example, there is an anima of manifolds whose identifications are diffeomorphisms.
- Every topological space X determines an anima $\Pi_\infty(X)$. Its points are points of X, and its identifications are paths in X. Here the anima of identifications can itself be interesting: identifications between paths are homotopies relative to their endpoints, and continuing in this way gives iterated homotopies.

An ordinary category can be described by a set of objects and a set of morphisms between each pair of objects. In an ∞-category C, we instead have an anima of objects and, for every two objects X, Y, an anima

$$\mathrm{Hom}_C(X, Y)$$

of morphisms. An identification in $\mathrm{Hom}_C(X, Y)$ is called a *homotopy* between morphisms. Composition is associative and unital with coherent higher identifications. Universal properties that are ordinarily phrased using equalities between morphisms are now phrased using homotopies between morphisms, together with their higher coherence.

Ordinary categories are the special case in which every mapping anima is a set, so there are no nontrivial homotopies between morphisms. When an ordinary category C is viewed in this way, its anima of objects is its core $C^{\simeq}$; isomorphisms between objects are still retained.

We can now write An for the ∞-category of animae. We will use products, pullbacks, functor categories, and equivalences in this setting without choosing a model.

The effect of retaining identifications is already visible in a self-pullback of animae. Let X be an anima and let $x\colon * \to X$ be a point. Form the pullback

$$\begin{array}{ccc} * \times_X * & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow{\scriptstyle x} \\ * & \xrightarrow[x]{} & X. \end{array}$$

An element of this pullback consists of the two chosen points together with an identification between their images in X. Both images are x. Writing $X(x, x)$ for the anima of identifications of x with itself, including all higher identifications, we obtain an isomorphism

$$* \times_X * \simeq X(x, x).$$

In sets, the right side is a singleton. In animae, it can be highly nontrivial. If

$$X = \Pi_\infty(S^1)$$

is the underlying anima of the circle with base point x, an identification of x with itself is represented by a continuous path

$$\gamma\colon [0,1] \longrightarrow S^1, \qquad \gamma(0) = \gamma(1) = x.$$

The endpoint condition is a strict equality of points, but the datum in the pullback is the entire path γ. It may travel around the circle any number of times. Paths between these choices are homotopies of loops that keep the base point fixed. Thus

$$X(x,x) \simeq \Omega\Pi_\infty(S^1) \simeq \mathbb{Z}.$$

The integers record the winding numbers of loops. Each winding-number component is contractible, which gives the last isomorphism with the discrete anima $\mathbb{Z}$.

The loop anima is a prototype for the information a homotopy pullback can retain. A geometric derived self-intersection uses the same pullback formalism in a different category.

The same description of pullbacks applies in any ∞-category. Let

$$X \xrightarrow{f} Z \xleftarrow{g} Y$$

be a diagram in an ∞-category C. Its pullback is characterized by natural isomorphisms of animae

$$\mathrm{Hom}_C(T, X \times_Z Y) \simeq \mathrm{Hom}_C(T,X) \times_{\mathrm{Hom}_C(T,Z)} \mathrm{Hom}_C(T,Y)$$

for all $T \in C$.

Informally, a generalized T-point consists of

$$a\colon T \to X, \qquad b\colon T \to Y, \qquad \alpha\colon f \circ a \cong g \circ b.$$

The equality α is part of the datum. Equalities between choices of α, and all subsequent coherence, are contained in the pullback of hom animae.

This is the only pullback notion needed in the sequel. In older sources the phrase *homotopy pullback* is often used to distinguish a construction in a chosen presentation from an ordinary strict pullback. We will work synthetically and simply say *pullback* in an ∞-category.

The passage from hom sets to hom animae now gives a precise reason to reconsider Question 4.1. We should ask for a universal finite-limit envelope among ∞-categories, not merely among ordinary categories.

4.4 Animated smooth rings and finite limits

We now apply the preceding pullback formalism to smooth algebra. The operations are still indexed by smooth maps between Euclidean spaces, but their values are animae. The finite-limit-envelope theorem has a corresponding animated form, with compactness replacing ordinary finite presentation and with retracts included explicitly.

The definition in Definition 4.2 extends to an ∞-category C with finite products: a smooth ring object is a finite-product-preserving functor $\mathrm{Euc} \to C$, with Euc regarded as an ∞-category. We retain the notation $\mathrm{Alg}_{C^\infty}(C)$. Taking $C = \mathrm{An}$ gives the following definition.

Definition 4.8 (Animated C^∞-ring)**.** An *animated C^∞-ring* is a finite-product-preserving functor

$$A\colon \mathrm{Euc} \longrightarrow \mathrm{An}\,.$$

We denote their ∞-category by

$$\mathrm{Alg}_{C^\infty}(\mathrm{An}) = \mathrm{Fun}^{\times}(\mathrm{Euc}, \mathrm{An}).$$

An ordinary C^∞-ring is the special case in which every value $A(\mathbb{R}^n)$ is a set, regarded as a 0-truncated anima. Animated rings allow the smooth operations themselves to interact with higher equality data.

For ordinary smooth rings, Proposition 3.16 expresses finite presentation by preservation of filtered colimits. We use the same criterion here, applied to the whole mapping anima.

Definition 4.9 (Homotopical finite presentation)**.** An animated C^∞-ring A is *homotopically finitely presented* if it is compact in $\mathrm{Alg}_{C^\infty}(\mathrm{An})$: the functor

$$\mathrm{Hom}_{\mathrm{Alg}_{C^\infty}(\mathrm{An})}(A, -)\colon\ \mathrm{Alg}_{C^\infty}(\mathrm{An}) \longrightarrow \mathrm{An}$$

preserves filtered colimits.

We use the subscript fp for the full subcategory of such objects. The word *homotopically* matters: finite presentation is now a statement about the entire hom anima, not only its set of connected components.

The next theorem identifies the universal target for Euclidean smooth operations when finite limits and retracts are required. It applies to every algebraic theory; we state only its C^∞ instance.

Theorem 4.10 (Animated finite-limit envelope)**.** *Let C be an idempotent-complete ∞-category with finite limits. Restriction along the finitely generated free animated C^∞-rings induces an equivalence of ∞-categories*

$$\mathrm{Fun}^{\mathrm{lex}}\left(\mathrm{Alg}_{C^\infty}(\mathrm{An})^{\mathrm{op}}_{\mathrm{fp}}, C\right) \simeq \mathrm{Alg}_{C^\infty}(C).$$

This is Carchedi and Steffens [CS19, Theorem 3.22]. The ordinary Theorem 4.3 is its 0-truncated counterpart.

Recall that an ∞-category is idempotent-complete if every retract diagram which exists formally is represented by an actual object. Ordinary categories with finite limits are automatically idempotent-complete. This is not automatic for ∞-categories, which is why the condition appears in the theorem.

We sketch the ingredients of the quoted theorem. The starting point is compact generation by free algebras.

(1) The free C^∞-ring on n generators is compact. Mapping out of it is evaluation on an n-tuple, and evaluation commutes with filtered colimits.

(2) Every animated C^∞-ring is assembled as a colimit of free rings. This is the homotopical form of presenting an algebra by generators and relations.

(3) A compact animated C^∞-ring is a retract of a finite colimit of finitely generated free rings.

(4) After taking opposites, the preceding statement says that every object of $\mathrm{Alg}_{C^\infty}(\mathrm{An})^{\mathrm{op}}_{\mathrm{fp}}$ is a retract of a finite limit of Euclidean objects. This describes the values forced on any finite-limit-preserving extension from Euc.

The complete argument constructs the extension by right Kan extension and checks that it takes values in the chosen target C. The compactness and idempotent-completeness hypotheses are precisely what make that last step work. Finite colimits and retracts of free algebras explain the form of the universal property; the Kan-extension argument supplies existence and compatibility with all presentations.

Replacing sets by animae changes both the limits in the universal problem and the finiteness condition on its algebraic objects. The new pullbacks retain homotopies and their coherence. Compactness must likewise control the whole mapping anima.

Warning 4.11. An ordinary finitely presented C^∞-ring need not remain compact when it is regarded as a discrete animated C^∞-ring. Thus the ordinary envelope is not simply a full subcategory of the animated envelope obtained by declaring every ordinary object discrete.

The analogous warning for an arbitrary algebraic theory is recorded in Carchedi and Steffens [CS19, Warning 3.28]. The truncation functor π_0 does send homotopically finitely presented animated rings to finitely presented ordinary rings, but the reverse passage can require retaining higher relation data.

In particular, the animated pushout associated with a self-intersection can retain higher homotopy groups beyond its ordinary quotient.

4.5 The universal property of derived manifolds

The animated envelope was constructed from Euclidean spaces. To answer the chapter's question, we must relate its universal property to functors on all manifolds that preserve transverse pullbacks. A geometric extension theorem provides this link. We first state that theorem, then define the universal category of derived manifolds and identify it with the opposite category of compact animated smooth rings.

Chapter 3 showed that the function-ring functor preserves transverse pullbacks. We need a stronger statement: any interpretation of Euclidean smooth operations extends uniquely to manifolds while preserving transverse pullbacks and the terminal object. We quote this geometric extension theorem.

Let

$$q\colon \mathrm{Euc} \longrightarrow \mathrm{Mfd}$$

be the inclusion. For every idempotent-complete ∞-category C with finite limits, restriction along q induces an equivalence

$$\mathrm{Fun}^{\pitchfork}(\mathrm{Mfd}, C) \simeq \mathrm{Fun}^{\times}(\mathrm{Euc}, C) = \mathrm{Alg}_{C^\infty}(C).$$

Here $\mathrm{Fun}^{\pitchfork}$ denotes the functors preserving transverse pullbacks and the terminal object. This is Carchedi and Steffens [CS19, Lemma 5.1 and Corollary 5.2].

The geometric idea comes from the presentations in Chapter 3. An open Euclidean subset is a transverse zero locus of $ya(x) - 1$, and every manifold is a retract of an open tubular neighborhood

in Euclidean space. A product-preserving functor on Euc can therefore assign an object to each such presentation using finite limits and retracts. This explains the candidate extension. Proving its independence of choices, functoriality, and preservation of transverse pullbacks is the substantive part of the theorem; the proof uses right Kan extension.

The earlier transverse-pullback theorem concerned the actual function-ring functor. This extension theorem applies to any internal C^∞-ring: it determines a unique transverse-pullback-preserving functor on manifolds.

For an ordinary category C with finite limits, combining this extension theorem with Theorem 4.3 gives

$$\mathrm{Fun}^{\mathrm{lex}}(\mathrm{Aff}^{\mathrm{fp}}_{C^\infty}, C) \simeq \mathrm{Fun}^{\pitchfork}(\mathrm{Mfd}, C).$$

Thus ordinary finitely presented smooth rings solve the ordinary version of the chapter's universal problem. We now formulate the version for ∞-categories.

Definition 4.12 (Universal ∞-category of derived manifolds)**.** The ∞-category DMfd of *derived manifolds* is equipped with a functor

$$i\colon \mathrm{Mfd} \longrightarrow \mathrm{DMfd}$$

and is characterized by the following properties.

(1) The ∞-category DMfd has finite limits and is idempotent-complete.

(2) The functor i preserves transverse pullbacks and the terminal object.

(3) For every idempotent-complete ∞-category C with finite limits, restriction along i induces an equivalence

$$\mathrm{Fun}^{\mathrm{lex}}(\mathrm{DMfd}, C) \simeq \mathrm{Fun}^{\pitchfork}(\mathrm{Mfd}, C),$$

where the right side consists of functors preserving transverse pullbacks and the terminal object.

This is Carchedi and Steffens [CS19, Definition 2.1]. The universal property determines DMfd uniquely. It fixes the existing transverse pullbacks, adjoins all finite homotopy limits, and includes their retracts.

The geometric extension theorem turns the defining universal property under manifolds into a universal property in terms of smooth ring objects. Comparing this with the animated envelope will both establish existence and give an algebraic description of DMfd.

Theorem 4.13 (Universal property through C^∞-rings)**.** *For every idempotent-complete ∞-category C with finite limits, there is a natural equivalence*

$$\mathrm{Fun}^{\mathrm{lex}}(\mathrm{DMfd}, C) \simeq \mathrm{Alg}_{C^\infty}(C).$$

Proof. By the universal property under manifolds,

$$\mathrm{Fun}^{\mathrm{lex}}(\mathrm{DMfd}, C) \simeq \mathrm{Fun}^{\pitchfork}(\mathrm{Mfd}, C).$$

Restriction to Euclidean spaces gives

$$\mathrm{Fun}^{\pitchfork}(\mathrm{Mfd}, C) \simeq \mathrm{Fun}^{\times}(\mathrm{Euc}, C) = \mathrm{Alg}_{C^\infty}(C).$$

Composing these equivalences proves the claim. □

This is Carchedi and Steffens [CS19, Theorem 5.3].

The animated finite-limit envelope has exactly the same universal property. We therefore obtain the promised algebraic presentation.

Corollary 4.14 (Derived manifolds as formal animated affines)**.** *There is an equivalence of* ∞*-categories*

$$\mathrm{DMfd} \simeq \mathrm{Alg}_{C^\infty}(\mathrm{An})^{\mathrm{op}}_{\mathrm{fp}}.$$

Proof. By Theorems 4.10 and 4.13, both sides are idempotent-complete ∞-categories with finite limits which universally carry a C^∞-ring object. Universal objects are unique, so the two ∞-categories are equivalent. □

This is Carchedi and Steffens [CS19, Corollary 5.4]. This identifies the universal category with an algebraic construction. In particular, manifolds embed fully faithfully, transverse pullbacks are preserved, and all finite pullbacks exist.

For a smooth map $f\colon U \to \mathbb{R}^r$, the function ring of its derived zero locus is consequently the pushout

$$C^\infty(U) \otimes^{\mathbb{L}}_{\infty, C^\infty(\mathbb{R}^r)} \mathbb{R}$$

in animated smooth rings. Chapter 5 computes it by a differential graded resolution. The category of derived schemes introduced in Chapter 7 allows these affine objects to be described locally and glued.

4.6 Further details: retracts and coherence

The universal properties above require idempotent-complete targets. We explain more precisely what this means in an ∞-category and why finite limits alone do not supply it. The exercises revisit this point and the self-pullback example.

Informally, an idempotent looks like a morphism $e\colon X \to X$ together with a homotopy from $e \circ e$ to e. Formally, an idempotent in an ∞-category is a coherent idempotent diagram: the displayed homotopy is part of the data, together with its higher compatibilities. Splitting it means finding an object Y and maps

$$Y \xrightarrow{i} X \xrightarrow{r} Y$$

together with coherent identifications of $r \circ i$ with id_Y and of $i \circ r$ with the given idempotent. In an ordinary category with finite limits, an idempotent $e\colon X \to X$ splits by taking the equalizer $i\colon Y \to X$ of e and the identity. The map e factors as ir, and $ri = \mathrm{id}_Y$ follows from the equalizer property. For ∞-categories this is not forced by finite limits alone. The compact algebraic objects are closed under retracts, so their opposite category is naturally idempotent-complete. The universal property of derived manifolds must record this closure explicitly.

Exercise 4.15. Prove the equalizer construction of a splitting just described. Compare it with the explicit finite presentation for a retract ring in Chapter 3.

Exercise 4.16. Use the universal property of a homotopy pullback to identify $* \times_{BG} *$ with a discrete group G, where BG is its classifying anima and both maps select the base point. Compare this with the self-pullback over a set.

The primary source is [CS19, Definition 2.1, Theorem 3.22, Remark 3.26, and Section 5]. It treats both universal properties, their geometric comparison, and the relation with Spivak's construction.

5 Calculating derived intersections

A smooth map $f\colon U \to \mathbb{R}^r$ determines a derived manifold

$$Z^{\mathrm{der}}(f) = U \times_{\mathbb{R}^r} \{0\}.$$

The preceding chapter guarantees that this pullback exists. We now ask what its functions remember about the equations defining it. To calculate them, we need an algebraic presentation of the pullback. The resulting algebra has one generator for each equation, together with a differential recording that equation. Products of the generators keep track of relations among the equations.

Write $f = (f_1, \dots, f_r)$. Under the contravariant description of derived manifolds by animated C^∞-rings, the defining pullback becomes the pushout

$$\begin{array}{ccc} C^\infty(\mathbb{R}^r) & \xrightarrow{f^*} & C^\infty(U) \\ {\scriptstyle 0^*}\downarrow & \ulcorner & \downarrow \\ \mathbb{R} & \longrightarrow & O(Z^{\mathrm{der}}(f)). \end{array}$$

Thus

$$O(Z^{\mathrm{der}}(f)) \cong C^\infty(U) \otimes^{\mathbb{L}}_{\infty, C^\infty(\mathbb{R}^r)} \mathbb{R}. \tag{5.1}$$

Here the tensor notation denotes a pushout of animated smooth rings. The corresponding ordinary pushout is $C^\infty(U)/(f_1, \dots, f_r)$. Computing that quotient first would discard the higher homotopy of the animated pushout.

There are already two different classical objects in this discussion. The ring $C^\infty(U)/(f_1, \dots, f_r)$ records the equations, whereas the set $f^{-1}(0)$ only records their common real solutions. For example, x and x^2 have the same zero set but give different quotients. The derived construction adds a third level: homotopy groups beyond π_0.

We first introduce smooth differential graded algebras as a tool for computing animated smooth rings, then construct the Koszul model of a zero locus. Comparing a transverse equation, a multiple zero, and a repeated equation will make these three levels visible. The same construction will then apply to families and to sections of vector bundles.

5.1 Smooth differential graded algebras

Smooth differential graded algebras make the higher homotopy of an animated smooth ring accessible through homology. We use homological grading, as in Steffens: the differential lowers degree by one. Our connective dg-algebra models are concentrated in nonnegative degrees. After describing their smooth structure, we state the comparison theorem that justifies using them to compute derived pushouts.

Definition 5.1 (Smooth dg-algebra)**.** A *connective C^∞-dg-algebra* is a commutative differential graded $\mathbb{R}$-algebra

$$\cdots \longrightarrow A_2 \xrightarrow{d} A_1 \xrightarrow{d} A_0$$

together with a C^∞-ring structure on A_0 whose underlying $\mathbb{R}$-algebra structure is the given one. A morphism preserves the differential, the multiplication, and the smooth operations on degree zero.

For homogeneous elements, commutativity and the differential rule mean

$$ab = (-1)^{|a||b|}ba, \qquad d(ab) = d(a)b + (-1)^{|a|}a\,d(b).$$

In particular, an odd-degree generator has square zero over $\mathbb{R}$. The smooth operations are supplied in degree zero. There is no operation $f(a_1, \dots, a_n)$ for arbitrary nonhomogeneous elements in this definition.

This is the convention of the foundations paper [Ste23, Definition 3.3.1]. Because d vanishes on A_0, there is no additional chain-rule condition to impose on its degree-zero elements.

Example 5.2. An ordinary C^∞-ring is a smooth dg-algebra concentrated in degree zero. More generally, for an ordinary smooth ring A and elements $f_1, \dots, f_r \in A$, form

$$A[\varepsilon_1, \dots, \varepsilon_r], \qquad |\varepsilon_i| = 1, \qquad d\varepsilon_i = f_i.$$

The notation denotes a free graded-commutative extension: as a graded A-module it is $A \otimes_{\mathbb{R}} \Lambda^\bullet(\mathbb{R}^r)$, with Λ^q placed in degree q. Its differential is determined by the displayed equations and the Leibniz rule. In particular,

$$d(\varepsilon_i\varepsilon_j) = f_i\varepsilon_j - f_j\varepsilon_i.$$

The equality $d^2 = 0$ follows on generators and hence on the whole algebra.

The image of $d\colon A_1 \to A_0$ is an ideal: for $a \in A_0$ and $b \in A_1$, $a\,d(b) = d(ab)$. Therefore $H_0(A)$ is an ordinary C^∞-ring. To see that all smooth operations descend, apply Hadamard's lemma to a smooth function $\varphi\colon \mathbb{R}^n \to \mathbb{R}$:

$$\varphi(x) - \varphi(y) = \sum_i (x_i - y_i)\psi_i(x, y).$$

Replacing each x_i by an element congruent to y_i modulo the ideal changes the output by an element of the same ideal. The groups $H_i(A)$ are modules over $H_0(A)$, and their products form a graded algebra.

The passage from dg-algebras to animated smooth rings identifies maps that induce isomorphisms on homology.

Definition 5.3. A morphism of smooth dg-algebras is a *quasi-isomorphism* if it induces an isomorphism on every homology group.

Theorem 5.4 (Smooth Dold–Kan correspondence)**.** *The localization of connective C^∞-dg-algebras at quasi-isomorphisms, taken as an ∞-categorical localization, is equivalent to the ∞-category of animated C^∞-rings. If A_{an} corresponds to a dg-algebra A, then*

$$\pi_i(A_{\mathrm{an}}) \cong H_i(A), \qquad i \geq 0.$$

We use this as a comparison theorem [Ste23, Theorem 3.3.12]. It includes multiplicative and smooth structure. The ordinary Dold–Kan correspondence for vector spaces alone does not supply that structure.

One way to describe the animated object attached to A is through its values on free smooth rings. In the localized category D of smooth dg-algebras, set

$$A_{\mathrm{an}}(\mathbb{R}^n) = \mathrm{Hom}_D(C^\infty(\mathbb{R}^n), A).$$

The functor is covariant in Euclidean spaces because smooth functions are contravariant. Free smooth rings turn finite products of Euclidean spaces into coproducts, so this functor preserves finite products. It is therefore an animated C^∞-ring. The mapping objects in this formula are animae; using only strict dg-algebra maps would lose the higher homotopies.

The theorem allows computations using dg-algebras provided we compute their *derived* pushouts. An arbitrary strict pushout of dg-algebras need not have the required universal property. In the zero-locus calculation, we will replace evaluation at the origin by a free resolution before taking the strict pushout.

5.2 The Koszul model of a zero locus

We compute the pushout in Equation (5.1) in two stages. First we resolve evaluation at the origin of $\mathbb{R}^r$. We then substitute the functions defining our zero locus into this resolution.

Start with one coordinate t and put $B = C^\infty(\mathbb{R})$. Consider the map

$$B[\varepsilon] \longrightarrow \mathbb{R}, \qquad |\varepsilon| = 1, \quad d\varepsilon = t,$$

sending t and ε to zero. Its underlying complex is

$$B \xrightarrow{t} B$$

in degrees $1, 0$. Multiplication by t is injective: a smooth function annihilated by t vanishes away from the origin, hence everywhere. Its cokernel is $\mathbb{R}$, since a smooth function vanishing at zero is divisible by t. Thus this map is a quasi-isomorphism.

For $B = C^\infty(\mathbb{R}^r)$, the same construction gives

$$K_B(t_1, \ldots, t_r) = \big(B[\varepsilon_1, \ldots, \varepsilon_r], d\varepsilon_i = t_i\big) \longrightarrow \mathbb{R}.$$

The coordinate functions form a regular sequence. Indeed, quotienting by the first j coordinates gives the smooth functions on the remaining coordinate subspace, and the next coordinate is a non-zero-divisor there. Successively applying the one-variable calculation proves that the augmentation is a quasi-isomorphism.

The map $B \to K_B(t_1, \ldots, t_r)$ is built by adjoining generators in positive degree with prescribed differential. Together with the augmentation to $\mathbb{R}$, it is a relative free resolution of evaluation. Taking its strict pushout computes derived base change. The criterion used here is Steffens [Ste23, Corollary 3.3.15]: this map is an isomorphism in degree zero, is surjective on H_0, and is a relative cell extension of underlying commutative dg-algebras. The resulting zero-locus calculation is given in Steffens [Ste23, Example 3.3.16].

Now base change along $f^*\colon B \to A = C^\infty(U)$. Each t_i becomes f_i, so the pushout is

$$K_A(f) = \big(A[\varepsilon_1, \ldots, \varepsilon_r], d\varepsilon_i = f_i\big). \tag{5.2}$$

Theorem 5.5 (Koszul model)**.** *The smooth dg-algebra $K_A(f)$ presents the animated function ring of $Z^{\mathrm{der}}(f)$. Consequently,*

$$\pi_i O(Z^{\mathrm{der}}(f)) \cong H_i(K_A(f)).$$

Proof. Replace the evaluation map $B \to \mathbb{R}$ in Equation (5.1) by the relative free resolution $B \to K_B(t)$. Its strict pushout along $B \to A$ computes the derived pushout and is exactly Equation (5.2). Apply the smooth Dold–Kan correspondence. □

The construction resolves the coordinates on $\mathbb{R}^r$, which are a regular sequence, before substituting the possibly singular functions f_i. No regular-sequence assumption on the f_i is needed. The positive-degree homology can retain additional relations among them.

In particular,

$$H_0(K_A(f)) = A/(f_1, \ldots, f_r).$$

In degree 1, a cycle $\sum_i a_i \varepsilon_i$ is a relation $\sum_i a_i f_i = 0$. Boundaries include the relations $f_i \varepsilon_j - f_j \varepsilon_i$. Thus higher homology measures relations that survive after these elementary ones have been identified. The algebra also retains the multiplication between such classes.

5.3 Equations with the same zero set

The preceding chapters computed ordinary quotient rings. We now return to those equations to ask what survives in the positive degrees of the Koszul model. In particular, a multiple zero need not have higher homotopy of functions, whereas an excess equation can produce it.

Example 5.6 (A transverse equation)**.** For $f(x) = x$, the model is

$$C^\infty(\mathbb{R})[\varepsilon], \qquad d\varepsilon = x.$$

The resolution calculation shows that it is quasi-isomorphic to $\mathbb{R}$. Thus the derived zero locus is the ordinary point.

More generally, if $f\colon U \to \mathbb{R}^r$ is transverse to zero, its components can be completed locally to coordinates along the zero set. The same Koszul calculation shows that the derived zero locus is the ordinary manifold $f^{-1}(0)$. Away from the zero set, one of the functions is invertible, and the Koszul complex is contractible. These local statements recover the transverse-pullback property of derived manifolds.

Example 5.7 (A multiple zero)**.** For $f(x) = x^2$, multiplication by x^2 is still injective on $C^\infty(\mathbb{R})$. Hence the Koszul algebra has no positive homology, but

$$H_0(K(x^2)) = C^\infty(\mathbb{R})/(x^2) \cong \mathbb{R}[\delta]/(\delta^2).$$

The last identification follows from Taylor's formula:

$$g(x) = g(0) + xg'(0) + x^2 r(x),$$

with r smooth. The smooth operations on the quotient are given by the first-order Taylor formula. For instance, $\varphi(a + b\delta) = \varphi(a) + \varphi'(a)b\delta$.

Thus $Z^{\mathrm{der}}(x^2)$ is not the ordinary point, even though its animated ring is discrete. Nilpotents in π_0 can already distinguish it. The same argument gives $C^\infty(\mathbb{R})/(x^m) \cong \mathbb{R}[\delta]/(\delta^m)$ for $m \geq 1$.

Example 5.8 (A zero equation)**.** For the zero function on $\mathbb{R}$, the differential vanishes:

$$K(0) = C^\infty(\mathbb{R})[\varepsilon], \qquad d\varepsilon = 0.$$

Its degree-zero homology is $C^\infty(\mathbb{R})$, and its degree 1 homology is another copy of $C^\infty(\mathbb{R})$. The classical truncation is the line, while the derived object remembers that an equation was imposed which vanishes identically.

Similarly, the self-intersection of $\{0\} \subset \mathbb{R}$ has function algebra $\mathbb{R}[\varepsilon]$ with $d\varepsilon = 0$. Its underlying topological space is a point and its classical function ring is $\mathbb{R}$, but $\pi_1 = \mathbb{R}$. This is a local instance of the excess-intersection phenomena in Spivak [Spi10, Example 2.7].

Example 5.9 (A repeated equation)**.** The maps $x \mapsto x$ and $x \mapsto (x, x)$ have the same zero set and the same ordinary quotient ring. For the second map the Koszul algebra is

$$C^\infty(\mathbb{R})[\varepsilon_1, \varepsilon_2], \qquad d\varepsilon_1 = d\varepsilon_2 = x.$$

Set $\eta = \varepsilon_2 - \varepsilon_1$. Then $d\eta = 0$, and the algebra becomes $K(x) \otimes_{\mathbb{R}} \Lambda(\eta)$. It is quasi-isomorphic to $\mathbb{R}[\eta]$ with zero differential. The repeated equation produces a nonzero π_1.

These examples separate three comparisons. A point and a multiple point differ in their classical function rings. An ordinary point and a derived self-intersection differ in higher homotopy. Different lists generating the same ordinary ideal can define different derived zero loci, because the derived construction remembers the chosen map to the space of equations.

5.4 Families of equations

The model also lets us follow an equation as its parameters vary. Even when the total zero locus is a manifold, its fibres can acquire multiplicity or higher homotopy.

Let S be a parameter manifold and let $f\colon S \times U \to \mathbb{R}^r$ be smooth. The relative equation defines a derived object over S:

$$X = (S \times U) \times_{S \times \mathbb{R}^r} (S \times \{0\}).$$

Its Koszul model has coefficients in $C^\infty(S \times U)$ and differential $d\varepsilon_i = f_i(s, x)$.

For a smooth map $g\colon T \to S$, pullback gives the same construction with $f(g(t), x)$. This follows by pasting the defining pullback squares. In particular, the fibre at $s \in S$ is the derived zero locus of f_s. The fibre is a derived pullback even when the total object is an ordinary manifold.

For example, consider $f(s, x) = x^2 - s$. The total zero locus is the ordinary parabola $s = x^2$, since the derivative in the s-direction is -1. Over $s > 0$ its fibre consists of two points; over $s < 0$ it is empty. The derived fibre at $s = 0$ has function ring $\mathbb{R}[\delta]/(\delta^2)$. The parameter map from the parabola to S has a critical point there.

A second family, $f(s, x) = sx$, illustrates the appearance of higher homotopy in a fibre. For $s \neq 0$, the fibre is an ordinary point. For $s = 0$, its Koszul model is $C^\infty(\mathbb{R})[\varepsilon]$ with zero differential. Although multiplication by sx on $C^\infty(\mathbb{R}^2)$ is injective, substitution $s = 0$ destroys injectivity. Derived base change retains this change in the equation.

5.5 Zero loci of vector bundle sections

The same model works without a chosen list of global equations. Let $p\colon E \to U$ be a vector bundle of rank r, and let s be a section. Its derived zero locus is the pullback of s and the zero section as maps from U to E. Define a sheaf of smooth dg-algebras on U by

$$\mathcal{K}(s)_q = \Gamma(-, \Lambda^q E^\vee), \qquad d = \iota_s.$$

Here ι_s contracts an alternating form with s. In a local frame, this is the Koszul algebra already calculated. Changes of frame respect contraction, so the local algebras glue.

Proposition 5.10. *The sheaf $\mathcal{K}(s)$ presents the functions of the derived zero locus of s, supported on its ordinary zero set. In particular, for the zero section its differential vanishes and its homotopy sheaves are $\Lambda^q E^\vee$ in degree q.*

Proof. Trivialize E on an open cover and apply the Koszul model theorem. On the complement of the zero set, contraction has a local contracting homotopy: choose $\lambda \in E^\vee$ with $\lambda(s) = 1$ and use exterior multiplication by λ. The frame-independent contraction formula identifies the models on overlaps, giving the stated sheaf. □

This is the bundle form of the calculation in [Ste23, Example 3.3.17]. It explains how the local excess directions of a self-intersection fit into a vector bundle rather than a fixed vector space.

5.6 Further reading

The following observations explain the role of regular sequences in the examples and the effect of allowing generators in higher degrees. The exercises give further comparisons between presentations of zero loci.

A regular sequence $f_1, \dots, f_r$ in an ordinary smooth ring has Koszul homology concentrated in degree zero. One proves this inductively: adjoining the last generator forms the cone of multiplication by f_r on the preceding Koszul complex. If the preceding homology is $A/(f_1, \dots, f_{r-1})$ and f_r is a non-zero-divisor there, the cone has only the desired quotient as its degree-zero homology. This explains both the transverse case and the multiple-zero example. Concentration in degree zero is weaker than smoothness of the resulting ordinary ring.

The smooth dg-algebra model also permits generators in degrees above 1. For example, a free generator z in degree 2 with $dz = 0$ gives $\mathbb{R}[z]$. Since z has even degree, its powers need not vanish: $H_{2j}(\mathbb{R}[z]) = \mathbb{R}z^j$ for every $j \geq 0$. A finite list of dg-generators therefore does not imply that the function algebra has only finitely many nonzero homotopy groups. The boundedness of a tangent complex is a separate issue, treated next.

Exercise 5.11. Compute the Koszul homology of $(x, 0)\colon \mathbb{R} \to \mathbb{R}^2$ and compare it with the repeated-equation example.

Exercise 5.12. For $f = (x_1, \dots, x_q, 0, \dots, 0)\colon \mathbb{R}^n \to \mathbb{R}^r$, show that the Koszul algebra is quasi-isomorphic to $C^\infty(\mathbb{R}^{n-q})[\eta_1, \dots, \eta_{r-q}]$ with zero differential. Interpret the surviving generators as excess equations.

Exercise 5.13. Show directly that adjoining a new coordinate y and the equation $y = 0$ does not change the derived zero locus. Compare this with adjoining the identically zero equation.

6 Tangent and cotangent complexes

Let x be a zero of a smooth map $f\colon U \to \mathbb{R}^r$. A first-order variation $v \in T_x U$ preserves the equation precisely when $d_x f(v) = 0$. If the equation itself is varied by a vector $e \in \mathbb{R}^r$, the linearized problem becomes $d_x f(v) = e$. Its solvability is measured by the class of e in $\operatorname{coker}(d_x f)$. Both questions are encoded by

$$[T_x U \xrightarrow{d_x f} \mathbb{R}^r]$$

in degrees 0 and -1.

The Koszul model of the preceding chapter lets us differentiate a chosen list of equations. We want a construction that depends on the derived zero locus itself. The cotangent complex supplies this intrinsic linearization by representing smooth derivations.

We begin with ordinary differentials and explain why their relations must be retained in a complex. Fibres and cofibres then provide the linear algebra needed to define and calculate the cotangent complex. Its base-change and transitivity properties lead back to the displayed two-term tangent complex, whose homology describes infinitesimal deformations and obstructions. We quote the existence and module comparison theorems; the zero-locus calculation is the main conclusion.

6.1 Smooth derivations and differentials

Let A be an ordinary C^∞-ring and let M be an A-module. A *smooth derivation* is a map $D\colon A \to M$ satisfying

$$D\big(\varphi(a_1, \dots, a_n)\big) = \sum_i (\partial_i \varphi)(a_1, \dots, a_n) D(a_i) \tag{6.1}$$

for every smooth function $\varphi\colon \mathbb{R}^n \to \mathbb{R}$. Addition, multiplication, and constants among the smooth operations show that D is $\mathbb{R}$-linear and satisfies the Leibniz rule. The displayed condition also requires compatibility with the nonpolynomial smooth operations.

Definition 6.1 (Smooth Kähler differentials)**.** The module Ω^1_A of smooth Kähler differentials is equipped with a derivation $d\colon A \to \Omega^1_A$ such that

$$\operatorname{Hom}_A(\Omega^1_A, M) \cong \operatorname{Der}_{C^\infty}(A, M)$$

naturally in M.

One construction takes the free A-module on symbols da, for $a \in A$, and imposes all instances of the smooth chain rule. The universal property follows immediately from the imposed relations.

Proposition 6.2. *For $A = C^\infty(\mathbb{R}^n)$, there is an isomorphism*

$$\Omega^1_A \cong \bigoplus_{i=1}^{n} A\, dx_i, \qquad dg = \sum_i \partial_i g\, dx_i.$$

Proof. A derivation is determined by its values $m_i = D(x_i)$, because the chain rule forces $D(g) = \sum_i \partial_i g \, m_i$. Conversely, any such choice defines a smooth derivation by the usual chain rule. □

For a manifold U, the corresponding formula is

$$\Omega^1_{C^\infty(U)} \cong \Gamma(U, T^*U).$$

It follows by localizing the coordinate calculation and gluing. After tensoring with $\mathbb{R}$ along evaluation at x, its dual is $T_x U$.

For a quotient $A = B/I$, differentiating the equations gives a right exact sequence

$$I/I^2 \longrightarrow \Omega^1_B \otimes_B A \longrightarrow \Omega^1_A \longrightarrow 0.$$

Taking the cokernel discards the kernel of the first map. This matters already for a hypersurface. For $B = C^\infty(\mathbb{R})$ and $I = (x^2)$, the differentiated relation is $2x\,dx$; at the origin it vanishes. The ordinary cotangent space has dimension one, but the equation still contributes a relation direction. A complex can retain both.

The same issue appears in families. The maps $\mathbb{R} \xrightarrow{t} \mathbb{R}$ vary smoothly with t, whereas both kernel and cokernel jump in dimension at zero. Keeping the differential allows a single linear description throughout the family.

6.2 Complexes and the derived category

To retain both solutions and relations, we work with complexes rather than individual modules. We first describe their maps and homotopies, then the shifts and cones that will be used to linearize a pullback.

A chain complex C consists of real vector spaces and maps

$$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots$$

with $d_{n-1} d_n = 0$. Its homology is $H_n(C) = \ker(d_n)/\mathrm{im}(d_{n+1})$. The underlying complexes of the preceding Koszul algebras are examples, concentrated in nonnegative degrees. Here we allow all degrees.

A morphism of complexes commutes with the differentials. It is a *quasi-isomorphism* if it induces isomorphisms on all homology groups. For example, the acyclic complex $[\mathbb{R} \xrightarrow{1} \mathbb{R}]$ is quasi-isomorphic to zero. Adding this complex as a direct summand should not change a linear invariant.

Definition 6.3 (Derived ∞-category)**.** The derived ∞-category $D(\mathbb{R})$ is the ∞-categorical localization of chain complexes of real vector spaces at quasi-isomorphisms.

Objects can be represented by complexes, and quasi-isomorphic complexes represent isomorphic objects. Mapping objects are animae: they retain homotopies between maps and their higher homotopies.

For vector spaces these can be calculated explicitly. A chain homotopy from f to g is a degree 1 map h with $f - g = dh + hd$. The mapping complex has terms

$$\underline{\mathrm{Hom}}(C, D)_n = \prod_p \mathrm{Hom}_{\mathbb{R}}(C_p, D_{p+n}), \qquad d(h) = d_D h - (-1)^n h\, d_C.$$

Its degree-zero cycles are chain maps, and its degree-zero homology is chain maps modulo chain homotopy. Over a field this complex computes the derived mapping object; positive homology gives the higher homotopy groups of the mapping anima.

To move between degrees, we use shifts. The shift of C is defined by

$$C[1]_n = C_{n-1}, \qquad d_{C[1]} = -d_C.$$

A vector space in degree zero moves to degree 1 under $[1]$. The inverse shift $[-1]$ moves it to degree -1. Thus $H_n(C[m]) = H_{n-m}(C)$. In particular, a Koszul generator contributes in the degree of $A[1]$; dualizing puts the corresponding tangent direction in degree -1.

As a simple example of higher homotopy, the mapping anima from $\mathbb{R}$ to $\mathbb{R}[1]$ has one component and fundamental group $\mathbb{R}$. Every map is nullhomotopic, but its nullhomotopy is additional data. Such data is part of a homotopy fibre.

Fibres and cofibers can be calculated using cones. For a chain map $f\colon C \to D$, the cone is

$$\mathrm{Cone}(f)_n = D_n \oplus C_{n-1}, \qquad d(y, x) = (d_D y + f(x), -d_C x).$$

It represents the cofiber of f in $D(\mathbb{R})$. Its shift $\mathrm{Cone}(f)[-1]$ represents the fibre. These constructions replace cokernels and kernels by operations invariant under quasi-isomorphism.

If C, D are vector spaces in degree zero, the cone is $[C \xrightarrow{f} D]$ in degrees $1, 0$, so

$$H_1(\mathrm{Cone}(f)) = \ker f, \qquad H_0(\mathrm{Cone}(f)) = \operatorname{coker} f.$$

For $f = 0$, the cone is $D \oplus C[1]$. For an injective f, the cone maps quasi-isomorphically to $\operatorname{coker} f$. An ordinary short exact sequence $0 \to C \to D \to E \to 0$ therefore gives a cofiber sequence $C \to D \to E$.

Every morphism fits into an exact triangle

$$C \xrightarrow{f} D \longrightarrow \mathrm{Cone}(f) \longrightarrow C[1].$$

Its homology forms a long exact sequence

$$\cdots \longrightarrow H_n(C) \longrightarrow H_n(D) \longrightarrow H_n(\mathrm{Cone}(f)) \longrightarrow H_{n-1}(C) \longrightarrow \cdots.$$

The connecting map relates the neighbouring homology groups. An exact triangle includes these maps and their compatibility, rather than merely three objects.

Short exact sequences of complexes can have nonzero connecting maps even over a field. Put $K = [\mathbb{R} \xrightarrow{1} \mathbb{R}]$ in degrees $1, 0$. Its degree-zero term is a subcomplex, giving a short exact sequence

$$0 \longrightarrow \mathbb{R} \longrightarrow K \longrightarrow \mathbb{R}[1] \longrightarrow 0.$$

The middle complex is acyclic. In the derived category, its triangle therefore becomes

$$\mathbb{R} \longrightarrow 0 \longrightarrow \mathbb{R}[1] \xrightarrow{\sim} \mathbb{R}[1].$$

The nonzero connecting map is essential to this triangle, even though its middle object is zero. This example also identifies the shift $\mathbb{R}[1]$ as the cofiber of $\mathbb{R} \to 0$.

The cone calculations describe categorical constructions in $D(\mathbb{R})$. The zero complex is both initial and terminal. The fibre and cofiber of $f\colon A \to B$ fit into the squares

$$\begin{array}{ccc} \mathrm{fib}(f) & \longrightarrow & A \\ \downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ 0 & \longrightarrow & B \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & \llcorner & \downarrow \\ 0 & \longrightarrow & \mathrm{cofib}(f). \end{array}$$

Each square includes its specified commutativity homotopy. In $D(\mathbb{R})$, a square is a pullback if and only if it is a pushout. This can be checked using the cone descriptions above. Consequently, $A \to B \to C$, with a specified nullhomotopy of its composite, is a fibre sequence exactly when it is a cofiber sequence.

The fibre and cofiber of the *same map* differ by a shift:

$$\mathrm{cofib}(f) \cong \mathrm{fib}(f)[1].$$

In particular, $A[1] = 0 \amalg_A 0$ and $A[-1] = 0 \times_A 0$. For vector spaces concentrated in degree zero, a fibre-cofiber sequence is precisely a short exact sequence. For general complexes, its homology instead forms a long exact sequence, and no injectivity or surjectivity of the first map is required. Finite products and coproducts in $D(\mathbb{R})$ agree and are given by direct sums of complexes.

6.3 Modules and derived smooth derivations

We now apply this linear algebra to an animated smooth ring. Modules provide the coefficients for infinitesimal variations, square-zero extensions turn those variations into maps of smooth rings, and the cotangent complex represents the resulting derivations.

The tensor product of real complexes gives $D(\mathbb{R})$ a symmetric monoidal structure with unit $\mathbb{R}$. Forgetting the nonpolynomial smooth operations of an animated C^∞-ring A gives a connective commutative algebra object A^{alg} in $D(\mathbb{R})$. We define its modules in this ∞-category.

Definition 6.4. The ∞-category Mod_A is the category of modules over the commutative algebra object A^{alg} in $D(\mathbb{R})$.

Informally, such a module is an object M of $D(\mathbb{R})$ equipped with an action $A^{\mathrm{alg}} \otimes_{\mathbb{R}} M \to M$. Associativity and the unit law are supplied by homotopies, together with their higher compatibilities. Module morphisms likewise respect the action with coherent homotopies, and the morphisms between two modules form an anima. This is the higher-categorical notion of a module; the coherence data is part of its structure.

For an explicit model, choose a C^∞-dg-algebra presenting A. Then Mod_A is modeled by dg-modules over its underlying commutative dg-algebra, with quasi-isomorphisms inverted in the ∞-categorical sense. This description presents the module category just defined. Modules may have homology in all degrees. For an ordinary smooth ring, Mod_A is the derived ∞-category of ordinary A-modules; in particular, $\mathrm{Mod}_{\mathbb{R}} = D(\mathbb{R})$.

Shifts, fibres, and cofibres work as they do for real complexes, and a fibre sequence is also a cofiber sequence. In a dg-module model they are calculated by shifts and cones of dg-modules. Extension

of scalars along $A \to B$ is written $M \otimes_A B$. All tensor products over A here are derived tensor products over the underlying commutative algebras.

The smooth operations re-enter in square-zero extensions. For ordinary A, M, the ring $A \oplus M$ has multiplication

$$(a, m)(b, n) = (ab, an + bm).$$

The ideal M has square zero. The smooth operations are

$$\varphi_{A\oplus M}((a_1, m_1), \ldots, (a_n, m_n)) = \left(\varphi_A(a), \sum_i (\partial_i \varphi)_A(a) m_i\right). \tag{6.2}$$

A section $a \mapsto (a, D(a))$ of $A \oplus M \to A$ preserves these operations precisely when D is a smooth derivation.

For animated A and connective M, there is similarly a split square-zero extension $A \oplus M$. In connective dg-models its underlying complex is the direct sum, with the displayed multiplication and smooth operations in degree zero. This construction respects the derived module category. For a general module, $A \oplus M$ uses its connective cover, since animated rings are connective. In our homological convention, connectivity means that homology vanishes in negative degrees.

As in the ordinary case, we define the cotangent object by its maps into every coefficient module. The sets of derivations are now replaced by animae of derivations.

Definition 6.5 (Derived derivations and the cotangent complex)**.** The anima of smooth derivations is

$$\mathrm{Der}_{C^\infty}(A, M) = \mathrm{Hom}_{\mathrm{Alg}_{C^\infty}(\mathrm{An})_{/A}}(A, A \oplus M).$$

The cotangent complex $\mathbb{L}_A$ represents this functor:

$$\mathrm{Hom}_{\mathrm{Mod}_A}(\mathbb{L}_A, M) \cong \mathrm{Der}_{C^\infty}(A, M). \tag{6.3}$$

For $A \to B$, the relative cotangent complex $\mathbb{L}_{B/A}$ represents derivations of B vanishing on A: the maps $B \to B \oplus M$ are taken over B and under A.

Existence is part of the general cotangent-complex construction. The complex is connective, so maps from it depend only on the connective cover of the target module, as required by the square-zero convention. For an ordinary ring A, $H_0(\mathbb{L}_A) \cong \Omega^1_A$. The other degrees retain relations in a presentation of A. They may be present even when A itself has no higher homotopy groups.

Proposition 6.6 (Smooth manifolds)**.** *For a manifold U, $\mathbb{L}_{C^\infty(U)} \cong \Gamma(U, T^*U)$ in degree zero. Its derived fibre at x is $T^*_x U$.*

For Euclidean space the proof uses universal derivations of a free animated smooth ring. Smooth localization has vanishing relative cotangent complex, so the calculation extends to coordinate opens and then to manifolds. We use that localization result here; the geometric construction appears in the next chapter.

6.4 Base change and transitivity

The universal definition makes the cotangent complex independent of a presentation. The following two properties let us calculate it from one. Base change transports equations, and transitivity compares successive stages of imposing them.

Theorem 6.7 (Base change)**.** *For a pushout of animated smooth rings*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & \ulcorner & \downarrow \\ A' & \longrightarrow & B', \end{array}$$

there is a natural isomorphism $\mathbb{L}_{B/A} \otimes_B B' \cong \mathbb{L}_{B'/A'}$.

Theorem 6.8 (Transitivity)**.** *For* $A \to B \to C$*, there is a natural cofiber sequence*

$$\mathbb{L}_{B/A} \otimes_B C \longrightarrow \mathbb{L}_{C/A} \longrightarrow \mathbb{L}_{C/B}.$$

For base change, an A'-linear derivation of B' is the same as an A-linear derivation of B with the indicated coefficients. For transitivity, restriction gives a fibre sequence

$$\mathrm{Der}_B(C, M) \longrightarrow \mathrm{Der}_A(C, M) \longrightarrow \mathrm{Der}_A(B, M).$$

The fibre includes the nullhomotopy expressing vanishing on B. Representing these functors gives the cofiber sequence. These arguments explain the formal content of the two theorems; their construction for animated smooth rings is included in the cotangent-complex theory cited below.

Absolute cotangent complexes are relative to the initial ring $\mathbb{R}$. For $D = B \otimes^{\mathbb{L}}_{\infty, A} C$, base change and transitivity imply

$$\mathbb{L}_A \otimes_A D \xrightarrow{(df, -dg)} (\mathbb{L}_B \otimes_B D) \oplus (\mathbb{L}_C \otimes_C D) \longrightarrow \mathbb{L}_D. \tag{6.4}$$

This is a cofiber sequence. The signs identify the two copies of a differential coming from A.

6.5 The tangent complex of a zero locus

We can now recover the complex from the chapter opening without choosing a dg-algebra resolution. We first linearize a pullback, specialize to a vector bundle section, and interpret the two homology groups.

For a real point x of a derived manifold X, write $\mathbb{L}_X|_x$ for the derived fibre of its cotangent complex.

Definition 6.9. The tangent complex is $T_x X = \mathrm{RHom}_{\mathbb{R}}(\mathbb{L}_X|_x, \mathbb{R})$.

For $X = U \times_W V$ a derived pullback of ordinary manifolds, Equation (6.4) at (u, v) gives

$$T^*_w W \longrightarrow T^*_u U \oplus T^*_v V \longrightarrow \mathbb{L}_X|_{(u,v)}.$$

Dualizing identifies

$$T_{(u,v)} X \cong [T_u U \oplus T_v V \xrightarrow{df - dg} T_w W]$$

in degrees $0, -1$. Its degree-zero homology is the ordinary tangent space of the equations.

Let s be a section of $p\colon E \to U$, and let $s(x) = 0_x$. At the zero section there is a canonical splitting

$$T_{0_x} E \cong T_x U \oplus E_x.$$

It uses dp, the derivative of the zero section, and the vertical tangent of the vector bundle. Since $p \circ s$ is the identity, $d_x s(v) = (v, D_x s(v))$ for a map $D_x s\colon T_x U \to E_x$. This vertical linearization is canonical at a zero; no connection is needed.

The pullback of s and the zero section has tangent complex

$$[T_xU \oplus T_xU \longrightarrow T_xU \oplus E_x], \qquad (v, w) \longmapsto (v - w, D_x s(v)).$$

Changing source coordinates to $(v, v - w)$ separates the acyclic identity complex on T_xU.

Theorem 6.10. *There are isomorphisms*

$$T_x Z^{\mathrm{der}}(s) \cong [T_xU \xrightarrow{D_x s} E_x], \qquad \mathbb{L}_{Z^{\mathrm{der}}(s)}|_x \cong [E_x^\vee \xrightarrow{(D_x s)^\vee} T_x^* U],$$

in degrees $0, -1$ *and* $1, 0$*, respectively.*

The Koszul model gives the same answer. Its derived differentials have generators dx_j in degree zero and $d\varepsilon_i$ in degree 1, with differential

$$d_{\mathrm{complex}}(d\varepsilon_i) = d_{\mathrm{dR}} f_i = \sum_j \frac{\partial f_i}{\partial x_j}\, dx_j.$$

Tensoring with the residue field gives the transpose of the Jacobian. The relative free resolution used in the Koszul model is what makes this a derived differential calculation.

The homology groups are

$$H_0(T_x Z^{\mathrm{der}}(s)) = \ker D_x s, \qquad H_{-1}(T_x Z^{\mathrm{der}}(s)) = \operatorname{coker} D_x s.$$

The first consists of infinitesimal solutions. In coordinates, $x + \varepsilon v$ over $\mathbb{R}[\varepsilon]/(\varepsilon^2)$ satisfies the equation exactly when $D_x s(v) = 0$.

The second is the target for obstruction classes. Perturbing the equation by εe gives $D_x s(v) = -e$, whose obstruction is $[e]$ in the cokernel. Extending a first-order solution illustrates the same phenomenon. In local coordinates and a bundle frame, write

$$x(\varepsilon) = x + \varepsilon v + \varepsilon^2 w.$$

Modulo ε^3, Taylor expansion gives

$$D_x s(v) = 0, \qquad D_x s(w) + \tfrac{1}{2} D_x^2 s(v, v) = 0.$$

Thus the class of $\frac{1}{2} D_x^2 s(v, v)$ obstructs extension to second order. The representative uses coordinates; the obstruction has an intrinsic cotangent-complex interpretation. A nonzero obstruction space permits nonzero obstruction classes, but does not imply that every deformation is obstructed.

For $f(x) = x$ the tangent complex is acyclic. For x^2 at zero it is $[\mathbb{R} \xrightarrow{0} \mathbb{R}]$: every v solves the first-order equation, but second-order extension requires $v^2 = 0$. For x^3 the tangent complex is identical, and the first obstruction occurs one order later. A tangent complex therefore does not determine the nonlinear derived object.

For the zero section of E, the tangent complex is $[T_xU \xrightarrow{0} E_x]$. In all these cases,

$$\dim H_0(T_x X) - \dim H_{-1}(T_x X) = \dim U - \operatorname{rank} E.$$

This is the virtual dimension. Its constancy is compatible with jumps in the individual homology dimensions.

6.6 Further reading

The following observations distinguish smooth from algebraic differentials and clarify how the finiteness of a tangent complex differs from that of the homotopy of functions.

Modules have an algebraic description, while their extensions carry smooth operations. Thus the smooth cotangent complex represents smooth derivations. In general it cannot be replaced by the algebraic cotangent complex of the underlying $\mathbb{R}$-algebra. For example, $C^\infty(\mathbb{R})$ is free on one generator as a smooth ring, but is not the polynomial $\mathbb{R}$-algebra on that generator.

A finitely presented animated smooth ring has a perfect cotangent complex. After taking its fibre at a real point, a perfect complex is represented by a bounded complex of finite-dimensional vector spaces. The tangent complex at that point consequently has finite-dimensional homology in finitely many degrees. There is no universal two-term bound for all derived manifolds. The two-term form above reflects the particular presentation by a section of an ordinary vector bundle. The next chapter characterizes the objects that have such local presentations.

For example, the free smooth dg-algebra $\mathbb{R}[z]$ with $|z| = 2$ and $dz = 0$ is a finite cell algebra. Its derived differentials are freely generated by $d_{\mathrm{dR}}z$ in degree 2. At its real point, the cotangent complex is $\mathbb{R}[2]$ and the tangent complex is $\mathbb{R}[-2]$, concentrated in degree -2. Its function algebra has nonzero homotopy in every nonnegative even degree. Bounded tangent information and bounded homotopy of functions are therefore different finiteness conditions.

Exercise 6.11. For a linear map $f\colon V \to W$, write down its fibre complex and verify that its cofiber is the shift of its fibre. Check zero maps and identity maps.

Exercise 6.12. Calculate the tangent complexes of the derived zero loci of $(x, 0)$ and (x, x) as maps $\mathbb{R} \to \mathbb{R}^2$. Compare the Koszul models.

Exercise 6.13. For a smooth family of equations and zeros, show that the Euler characteristic of the tangent complex is locally constant even where its homology dimensions jump.

The smooth module comparison and cotangent construction are developed in [Ste22, Section 4.3 and Construction 5.1.0.1, Definition 5.1.0.2, Remark 5.1.0.4]. For a complementary account, see [Pas25, Sections 3.2–3.3 and Appendix D.3]. Spivak's geometric cotangent calculation is in [Spi10, Section 7].

7 Quasi-smooth derived geometry and Kuranishi charts

The preceding chapters described the functions and tangent complexes of derived zero loci. We now ask when a derived object admits such a presentation locally. The answer is detected by its cotangent complex: the quasi-smooth objects are precisely those locally presented as zero loci of sections of finite-rank vector bundles.

To make this local statement precise, we first pass from animated smooth rings to their spectra and structure sheaves. Smooth localization supplies the restriction maps, and gluing affine objects leads to derived C^∞-schemes. We then state the local zero-locus theorem and study its presentations, called Kuranishi charts. Stabilization and changes of bundle frame illustrate how different equations can describe the same derived object. These comparisons will be used later in the finite-dimensional reduction of elliptic equations.

7.1 Smooth localization

For a smooth function a on a manifold M, put $D(a) = \{x \in M : a(x) \neq 0\}$. The smooth localization satisfies

$$C^\infty(M)[a^{-1}] \cong C^\infty(D(a)).$$

One way to see this is to identify $D(a)$ with the graph of $1/a$ in $M \times \mathbb{R}$. This graph is the transverse zero locus of $ya(x) - 1$, so adjoining a smooth inverse to a gives precisely its function ring.

Smooth localization is larger than algebraic localization of the underlying ring. For instance, $C^\infty(\mathbb{R})[x^{-1}]$ contains the smooth function $\exp(1/x^2)$ on the punctured line. It is not a fraction $g(x)/x^n$ with g smooth on all of $\mathbb{R}$: multiplication by any power of x leaves its divergence at zero. Smooth functional calculus is part of the localization.

Let now A be an animated C^∞-ring and $a \in \pi_0 A$.

Definition 7.1. The smooth localization $A[a^{-1}]$ is initial among animated smooth rings under A in which the image of a is invertible in π_0.

Choose a representative of a, hence a map $C^\infty(\mathbb{R}) \to A$. The localization is the pushout

$$\begin{array}{ccc} C^\infty(\mathbb{R}) & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow \\ C^\infty(\mathbb{R} \setminus \{0\}) & \longrightarrow & A[a^{-1}]. \end{array}$$

The universal property makes the result independent of this choice.

Theorem 7.2 (Flatness of smooth localization)**.** *There are natural isomorphisms*

$$\pi_0(A[a^{-1}]) \cong (\pi_0 A)[a^{-1}], \qquad \pi_i(A[a^{-1}]) \cong \pi_i(A) \otimes_{\pi_0 A} (\pi_0 A)[a^{-1}].$$

The smooth localization of the ordinary ring $\pi_0 A$ is flat as a module over its underlying ring.

We use this result without proof [Ste22, Proposition 4.1.3.13]. The tensor product in its second formula is an ordinary module tensor product. Flatness explains why restricting functions creates no additional higher homotopy from a derived tensor product. It restricts each existing homotopy module.

The corresponding relative cotangent complex vanishes:

$$\mathbb{L}_{A[a^{-1}]/A} \cong 0.$$

By transitivity,

$$\mathbb{L}_{A[a^{-1}]} \cong \mathbb{L}_A \otimes_A A[a^{-1}]. \tag{7.1}$$

This is the compatibility needed to regard cotangent complexes as local geometric objects.

7.2 The spectrum and its structure sheaf

Assume A is finitely presented as an animated smooth ring. Its spectrum has underlying set

$$|\operatorname{Spec} A| = \operatorname{Hom}_{C^\infty\text{-rings}}(\pi_0 A, \mathbb{R}).$$

These are real points, not arbitrary prime ideals of an underlying commutative ring. A point determines a maximal ideal with residue field $\mathbb{R}$, namely the kernel of its evaluation map. The evaluation map is the useful datum in smooth geometry.

For $a \in \pi_0 A$, the subsets

$$D(a) = \{x : x(a) \neq 0\}$$

form a basis for a topology, since $D(a) \cap D(b) = D(ab)$. If $\pi_0 A = C^\infty(\mathbb{R}^n)/I$, evaluation identifies this topological space with the common zero set $Z(I) \subseteq \mathbb{R}^n$, with its subspace topology. To check the topology, bump functions give basic neighbourhoods inside any prescribed Euclidean open set.

Construction 7.3 (Derived spectrum)**.** The derived spectrum is

$$\operatorname{Spec} A = (|\operatorname{Spec} A|, \mathcal{O}_{\operatorname{Spec} A}),$$

where the sheaf of animated smooth rings is obtained from smooth localizations. In the finite-presentation setting,

$$\mathcal{O}_{\operatorname{Spec} A}(D(a)) \cong A[a^{-1}].$$

The local ring at x is the filtered colimit of these localizations over basic neighbourhoods of x.

The assertion that these assignments have the required sheaf and reconstruction properties is a theorem about smooth rings. It is part of the affine comparison [Ste22, Section 4.1.3, especially Corollary 4.1.3.34]. For unrestricted smooth rings one must take care with sheafification and reconstruction; finite presentation is the class used here.

The term *locally animated C^∞-ringed space* means that π_0 of every stalk is a local smooth ring. At x its residue map is evaluation in $\mathbb{R}$. The higher homotopy modules remain in the stalk. A stalk belongs to a point; the values on basic opens are the localizations before taking that colimit.

For an ordinary manifold, the construction recovers its familiar ringed space:

$$\operatorname{Spec} C^\infty(M) \cong (M, C^\infty_M).$$

Real homomorphisms are evaluations at points, the basic opens give the manifold topology, and the localization formula identifies the structure sheaf with smooth functions.

The examples from Chapter 5 now have an explicit topological meaning. The spectra of $\mathbb{R}$, $\mathbb{R}[\delta]/(\delta^2)$, and $\mathbb{R}[\varepsilon]$ with $|\varepsilon| = 1$, $d = 0$, all have one point. Their structure sheaves differ. The second has nilpotents in degree zero; the third has a nonzero homotopy group in degree one.

7.3 Derived schemes and quasi-smoothness

The spectrum construction supplies affine objects and their open subsets. We can therefore define locally affine objects, then use the cotangent complex to single out those that should admit local presentations by vector bundle sections.

An open subset $V \subseteq |X|$ of a locally ringed object defines an open subobject $(V, \mathcal{O}_X|_V)$. In particular,

$$\operatorname{Spec} A[a^{-1}] \cong (D(a), \mathcal{O}_{\operatorname{Spec} A}|_{D(a)}).$$

Thus algebraic localization becomes geometric open restriction.

Definition 7.4. A derived C^∞-scheme locally of finite presentation is a locally animated C^∞-ringed space with an open cover by spectra of finitely presented animated C^∞-rings.

The affine objects in this definition are exactly the derived manifolds of the universal-property convention in Chapter 4:

$$\mathrm{DMfd} \simeq \left(\mathrm{Alg}_{C^\infty}(\mathrm{An})^{\mathrm{fp}}\right)^{\mathrm{op}}.$$

In particular, an ordinary manifold is affine in this sense. Affineness does not mean that it is a single Euclidean coordinate chart. For example, $S^1 = \operatorname{Spec} C^\infty(S^1)$ is affine. Allowing locally affine objects supplies the category in which open gluing can be performed.

The classical truncation is

$$t_0 X = (|X|, \pi_0 \mathcal{O}_X).$$

It keeps the topology and removes higher homotopy from the structure sheaf. It can still contain nilpotents. Thus classical truncation and the underlying zero set are different operations.

Cotangent complexes restrict by Equation (7.1), so they glue over affine opens. A module over an animated ring A is *perfect* if it is a retract of a finite construction by cofibers from shifts of finite sums of A. For an ordinary ring this is equivalent to having a bounded complex of finitely generated projective modules as a presentation. Perfection on a derived scheme is checked on its affine opens. A module has Tor-amplitude in $[a, b]$ if tensoring with any discrete module produces homology only in degrees $a, \ldots, b$. This condition concerns derived tensor products; it is stronger than merely locating the homology of the original complex.

Definition 7.5 (Quasi-smoothness)**.** A derived C^∞-scheme locally of finite presentation is *quasi-smooth* if its cotangent complex is perfect of Tor-amplitude contained in $[0, 1]$. For a morphism, the same definition uses the relative cotangent complex.

Perfection and Tor-amplitude can be checked locally. A two-term complex of finite-rank vector bundles in degrees $1, 0$ has this amplitude. The cotangent calculation of Chapter 6 therefore proves that every derived zero locus of a section of a finite-rank bundle is quasi-smooth.

For a perfect complex the amplitude condition can also be tested on residue fields. Consequently, at each real point of a quasi-smooth object its tangent complex has homology only in degrees $0, -1$. The bundle presentation of the cotangent complex, rather than just a dimension count at individual points, is the local structure behind this assertion.

7.4 Local presentations by Kuranishi charts

The cotangent calculation already shows that a vector bundle zero locus is quasi-smooth. The local normal-form theorem supplies the converse. We first specify what data is included in a chart.

Definition 7.6 (Kuranishi chart)**.** A Kuranishi presentation is a triple (U, E, s) consisting of a finite-dimensional manifold, a finite-rank vector bundle on it, and a smooth section. Its associated derived object is $Z^{\mathrm{der}}(s)$. A Kuranishi chart for X also includes an identification of this derived zero locus with an open subobject of X.

This usage concerns charts without isotropy, boundary, or corners. It is the local presentation needed in the representability proof. A choice of charts alone does not specify any of the more elaborate global notions called Kuranishi structures in the literature.

Theorem 7.7 (Local zero-locus form)**.** *A derived C^∞-scheme locally of finite presentation is quasi-smooth if and only if it admits an open cover by Kuranishi charts.*

The forward implication is the substantive normal-form theorem. The affine statement is proved in Steffens [Ste22, Proposition 5.1.1.11 and Corollary 5.1.1.13]. In that statement, a finitely presented quasi-smooth affine object can be expressed as the zero locus of a bundle section over an open subset of a Euclidean space. Applying it on affine opens proves the displayed local form. The reverse implication is the cotangent calculation.

The proof begins by choosing generators for the ring, giving an ambient smooth affine object. The relative cotangent complex describes its equations. The amplitude assumption implies that the shifted relative cotangent complex is a finite projective module, which can be extended to a vector bundle on an ambient neighbourhood. Lifting the relations then produces a section of that bundle. A cotangent criterion identifies its derived zero locus with the original object. This explains the role of the amplitude assumption; the extension and comparison theorems in this argument are quoted.

At a point of a chart, the tangent complex is $[T_x U \to E_x]$, and the virtual dimension is $\dim U - \operatorname{rank} E$. Quasi-smoothness permits both singular classical truncations and nonzero higher homotopy. The zero locus of x^2 illustrates the first, and a zero equation illustrates the second.

7.5 Changing a presentation

The quotient rings, Koszul algebras, and tangent complexes calculated earlier distinguish different geometric objects. We now use those calculations to recognize changes of equations that preserve an object.

Let $f\colon U \to \mathbb{R}^r$, and define

$$f'\colon U \times \mathbb{R}^q \longrightarrow \mathbb{R}^r \times \mathbb{R}^q, \qquad f'(x,y) = (f(x), y).$$

The additional equation forces the additional variable to zero.

Proposition 7.8 (Stabilization)**.** *Projection onto U induces an isomorphism $Z^{\mathrm{der}}(f') \cong Z^{\mathrm{der}}(f)$.*

Proof. The second factor is the derived zero locus of the identity of $\mathbb{R}^q$. That map is transverse to zero, so its zero locus is the ordinary point. Pasting pullbacks gives the stated isomorphism. Equivalently, the new Koszul variables satisfy $d\eta_j = y_j$. Evaluation at $y = \eta = 0$ removes an acyclic Koszul resolution. □

For example, x^2 and (x^2, y) give the same derived object. Their tangent presentations differ by $[\mathbb{R} \xrightarrow{1} \mathbb{R}]$, an acyclic summand. Both ambient dimension and the number of equations increase by one, leaving virtual dimension unchanged.

Changing a bundle frame likewise changes the list of equations without changing the object. If $g(x)$ is an invertible matrix of smooth functions, then f and gf present isomorphic derived zero loci. Explicitly, the map from the Koszul algebra for gf to that for f sends its generator ε'_i to $\sum_j g_{ij}\varepsilon_j$. The inverse matrix gives the inverse dg-algebra map. The intrinsic bundle model with differential ι_s makes this independence immediate.

In contrast, x and $(x, 0)$ have the same ordinary zero set but different derived zero loci. The extra zero equation contributes a degree-minus-one tangent direction and a degree-one homotopy class of functions. Nor does equality of tangent complexes suffice: x^2 and x^3 have the same tangent complex at zero but distinct classical function rings. Comparisons of charts must respect their equations.

7.6 Further reading

Two refinements of the chart description will recur later: removing transverse directions makes a chart minimal at a point, while a nontrivial obstruction bundle shows what local frames can conceal.

A useful refinement makes a chart minimal at a chosen point. Suppose $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}^r$ vanishes at x and $d_x f$ has rank q. After linear changes of coordinates, q components of f have independent differentials. The inverse function theorem allows these components to be used as coordinates $y \in \mathbb{R}^q$. Writing the remaining coordinates as z, the map becomes

$$(y, z) \longmapsto (y, g(y, z)).$$

Hadamard's lemma gives a smooth matrix H with $g(y,z) = g(0,z) + H(y,z)y$. An invertible change of equation generators subtracts $H(y,z)y$, giving $(y, g(0,z))$. Stabilization now removes the coordinates y.

The resulting local chart is the zero locus of

$$\kappa\colon \mathbb{R}^{n-q} \supseteq V \longrightarrow \mathbb{R}^{r-q}, \qquad \kappa(z) = g(0,z),$$

with $d\kappa$ zero at the chosen point. Its source tangent and target identify with $\ker d_x f$ and $\operatorname{coker} d_x f$. These identifications use choices; the derived object they present is the original open neighbourhood. The analytic reduction of an elliptic equation will produce a chart of this kind from an initially infinite-dimensional problem.

Example 7.9 (An obstruction bundle with no global frame)**.** Let $L \to S^1$ be the Möbius line bundle and form $X = S^1 \times_L S^1$ using two zero sections. The intrinsic Koszul algebra is the exterior algebra of $L^\vee$ with zero differential. Therefore

$$\pi_0 \mathcal{O}_X = C^\infty_{S^1}, \qquad \pi_1 \mathcal{O}_X = \Gamma(-, L^\vee).$$

Each trivializing open gives a single zero equation. On overlaps the generator changes by the transition function of $L^\vee$. There is no globally nonvanishing generator, although the derived self-intersection is defined globally.

After restriction to the classical circle, the tangent complex is $[TS^1 \xrightarrow{0} L]$. Its obstruction sheaf remembers the nontrivial line bundle. This shows why vector bundles, rather than only fixed vector spaces of equations, naturally occur in Kuranishi charts.

Exercise 7.10. Show that the derived zero locus of $(x, xy)\colon \mathbb{R}^2 \to \mathbb{R}^2$ is locally, and in this case globally, the product of a line and the derived zero locus of the zero map from a point to $\mathbb{R}$.

Exercise 7.11. For $f(x,y) = x^2 + y^2$, calculate the tangent complex at the origin and compare its virtual dimension with the dimension of its zero set. Explain why the zero set alone cannot carry this information.

For ordinary smooth spectra and their structure sheaves, see [Joy12, Sections 2.2 and 3.1]. The localization and normal-form statements used here are in [Ste22, Sections 4.1.3 and 5.1.1]. They provide the passage from the affine universal theory to the local charts used in geometric moduli problems.

8 Derived bordism and fundamental classes

A quasi-smooth derived manifold is locally the zero locus of a section of a vector bundle. Its tangent complex records both the infinitesimal solutions and the failure of transversality. A transverse perturbation usually changes the derived manifold, and can change even its underlying topological space. It nevertheless preserves a useful invariant: the bordism class.

We begin with a homotopy of sections, whose derived zero locus gives a cobordism between the endpoint zero loci. Using the global presentation theorem, which we quote, we extend this construction to compact quasi-smooth derived manifolds and prove the comparison with classical bordism. Derived intersections then represent intersection products without a transversality hypothesis, and the self-intersection of a zero section represents an Euler class. Finally, we explain how bordism classes give homology classes and where orientations enter.

Throughout this chapter, derived manifolds are quasi-smooth and have the Hausdorff, second-countable underlying spaces of our affine setting. Spivak's bordism theorem applies to these objects. His local models agree with the quasi-smooth local models used here; the comparison with the universal framework is discussed in [CS19, Section 5.2, especially Theorem 5.10].

8.1 From perturbations to derived cobordisms

A family joining a section to a transverse perturbation is the basic example of a derived cobordism. We construct this family first, so that properness and the collar conditions in the definition have a concrete geometric purpose.

Let M be a compact manifold, let $q\colon E \to M$ be a rank-r vector bundle, and let $s\colon M \to E$ be a section. Recall that its derived zero locus is defined by the pullback square

$$\begin{array}{ccc} Z^{\mathrm{der}}(s) & \longrightarrow & M \\ \downarrow & \lrcorner & \downarrow{\scriptstyle 0} \\ M & \xrightarrow[s]{} & E. \end{array}$$

Its underlying set is the ordinary zero set of s, but its structure sheaf retains the equations defining that set.

Choose another section $s'\colon M \to E$. We first assume that s' is transverse to the zero section. Choose a smooth function

$$\beta\colon \mathbb{R} \longrightarrow [0,1]$$

which is identically zero on a neighborhood of 0 and identically one on a neighborhood of 1. Fiberwise affine combination defines a section

$$H\colon \mathbb{R}\times M \longrightarrow \mathrm{pr}_M^{*}E, \qquad H(u,m) = (1-\beta(u))s(m) + \beta(u)s'(m).$$

Its derived zero locus is taken over $\mathbb{R} \times M$:

$$\begin{array}{ccc} W & \longrightarrow & \mathbb{R} \times M \\ \downarrow & \lrcorner & \downarrow 0 \\ \mathbb{R} \times M & \xrightarrow[H]{} & \mathrm{pr}_M^* E. \end{array}$$

Projection to the first factor induces a map

$$p \colon W = Z^{\mathrm{der}}(H) \longrightarrow \mathbb{R}.$$

Both endpoint zero loci occur as fibers of W.

Proposition 8.1 (A homotopy of sections gives a cobordism)**.** *With notation as above, the map $p \colon W \to \mathbb{R}$ is proper and is a product near its fibers over* 0 *and* 1*. Moreover,*

$$W_{p=0} \simeq Z^{\mathrm{der}}(s), \qquad W_{p=1} \simeq Z(s').$$

Proof. Pulling the defining square of W back along $\{0\} \to \mathbb{R}$ replaces H by s, so

$$W_{p=0} \simeq Z^{\mathrm{der}}(s).$$

The same argument at 1 gives $W_{p=1} \simeq Z^{\mathrm{der}}(s')$. Since s' is transverse to the zero section, its derived zero locus is the ordinary zero manifold $Z(s')$.

Choose $\varepsilon > 0$ so that β is zero on $(-\varepsilon, \varepsilon)$ and one on $(1-\varepsilon, 1+\varepsilon)$. Over these two intervals the section H is constant in the parameter. Consequently,

$$W|_{(-\varepsilon,\varepsilon)} \simeq Z^{\mathrm{der}}(s) \times (-\varepsilon, \varepsilon)$$

and

$$W|_{(1-\varepsilon,1+\varepsilon)} \simeq Z(s') \times (1-\varepsilon, 1+\varepsilon).$$

It remains to check properness. Let $K \subset \mathbb{R}$ be compact. The underlying space of $p^{-1}(K)$ is the ordinary zero set of H inside $K \times M$. It is closed because the zero section of the vector bundle is closed. Since $K \times M$ is compact, the space underlying $p^{-1}(K)$ is compact. Thus p is proper. □

The proof separates the two roles of the hypotheses. The constancy of the homotopy near 0 and 1 supplies product neighborhoods of the endpoint fibers. Compactness of M prevents zeros from escaping to infinity while the section is perturbed.

The preceding construction contains every clause in the general definition.

Definition 8.2 (Collared map)**.** Let $p \colon W \to \mathbb{R}$ be a map of derived manifolds and let $i \in \mathbb{R}$. We say that p is *i-collared* if there is an $\varepsilon > 0$ and an isomorphism over $(i-\varepsilon, i+\varepsilon)$

$$W|_{|p-i|<\varepsilon} \simeq W_{p=i} \times (-\varepsilon, \varepsilon).$$

On the right, the interval is translated so that its origin maps to i.

Definition 8.3 (Derived cobordism)**.** Let X_0 and X_1 be compact quasi-smooth derived manifolds. A *derived cobordism* from X_0 to X_1 consists of a quasi-smooth derived manifold W and a proper map

$$p \colon W \longrightarrow \mathbb{R}$$

such that:

(1) The map p is 0-collared and 1-collared.

(2) There are isomorphisms

$$W_{p=0} \simeq X_0, \qquad W_{p=1} \simeq X_1.$$

If the objects carry maps to a manifold T, a derived cobordism *over* T also carries a continuous map from the underlying space of W to T which restricts to the endpoint maps.

This is Spivak's boundaryless formulation of cobordism [Spi10, Definitions 3.5–3.6]. The portion over $[0, 1]$ is compact, and its collars allow it to be joined to another cobordism along an endpoint.

Corollary 8.4 (Homotopic sections)**.** *Let M be compact and let s_0, s_1 be sections of the same vector bundle over M. Then their derived zero loci are derived cobordant. If s_1 is transverse, then $Z^{\mathrm{der}}(s_0)$ is derived cobordant to the ordinary manifold $Z(s_1)$.*

Proof. Apply the construction above to a homotopy of sections which is constant near its endpoints. Transversality is needed only to identify the second endpoint with an ordinary manifold. □

8.2 Classical and derived bordism

The homotopy construction applies directly to a single bundle section. To compare bordism theories, we need global equations for an arbitrary compact quasi-smooth derived manifold. After stating the comparison, we explain this geometric input and use perturbations to prove both surjectivity and injectivity.

Let T be a manifold. Write $\Omega_d(T)$ for the ordinary unoriented bordism group of continuous maps from compact d-manifolds without boundary to T, with compact bordisms over T. Equivalently, $\Omega_*(T)$ is the homology theory represented by MO. Define $\Omega_d^{\mathrm{der}}(T)$ in the same way using compact quasi-smooth derived manifolds of virtual dimension d and derived cobordisms over T. Disjoint union gives the group operation.

The inclusion of ordinary manifolds into derived manifolds induces a map

$$i_* \colon \Omega_*(T) \longrightarrow \Omega_*^{\mathrm{der}}(T).$$

At first sight the target might be larger. Derived manifolds provide many new representatives, including singular zero loci and nontransverse intersections. The comparison theorem says that these representatives do not produce new bordism classes.

Theorem 8.5 (Spivak's bordism comparison)**.** *For every manifold T, inclusion induces an isomorphism*

$$i_* \colon \Omega_*(T) \xrightarrow{\sim} \Omega_*^{\mathrm{der}}(T).$$

The absolute and relative assertions are, respectively, Spivak [Spi10, Theorem 3.12 and Corollary 3.13].

The proof reduces the general case to the construction from the first section. Two geometric inputs make this possible.

Theorem 8.6 (Embedding and global zero-locus presentation)**.** *Let X be a compact quasi-smooth derived manifold in Spivak's framework. Then there are:*

(1) An embedding $j\colon X \to \mathbb{R}^N$.

(2) An open neighborhood $U \subseteq \mathbb{R}^N$ of the underlying space of X.

(3) A finite-rank vector bundle $E \to U$.

(4) A section $s\colon U \to E$.

With this data, there is an isomorphism over U

$$X \simeq Z^{\mathrm{der}}(s).$$

Explanation of the geometric input. The embedding is supplied by Spivak [Spi10, Proposition 3.3], and the bundle and section by his normal-bundle axiom [Spi10, Definition 2.1(7)].

In our affine description, the starting point is particularly simple. Write $X = \mathrm{Spec}(A)$ and choose finitely many generators for the finitely presented smooth ring $\pi_0 A$. They give a map $C^\infty(\mathbb{R}^N) \to A$ which is surjective on π_0, hence a map $X \to \mathbb{R}^N$ whose underlying space is a closed subset. The quasi-smooth normal-form theorem upgrades this algebraic presentation to a vector-bundle zero locus on an open neighborhood of that subset. This is the global form of the theorem used in the preceding chapter [Ste22, Proposition 5.1.1.11 and Corollary 5.1.1.13].

Geometrically, Spivak obtains the embedding by taking finitely many local zero-locus charts and extending their coordinate functions with cutoffs. The cutoffs and coordinates separate points; near each point, one chart still supplies its original coordinates. The resulting map is locally a derived embedding. The normal-bundle theorem then collects the local equations into a section of one vector bundle. □

There is a small but essential difference from the opening construction. The ambient manifold U need not be compact. Compactness of X nevertheless allows the perturbation to be supported near its zero locus.

Choose a compact subset $K \subset U$ whose interior contains the underlying space of X. Since s has no zeros outside the interior of K, relative transversality gives a transverse section s' which agrees with s outside K. The straight-line homotopy, made constant near its endpoints, is also fixed outside K. Every zero of every section in the homotopy therefore lies in K, which restores the properness argument.

Proof of Theorem 8.5. We first prove surjectivity in the absolute case. Let X be a compact quasi-smooth derived manifold. Choose a global zero-locus presentation

$$X \simeq Z^{\mathrm{der}}(s)$$

as in Theorem 8.6. Perturb s to a transverse section s' while keeping the perturbation fixed outside a compact neighborhood of the zero locus. The homotopy-zero-locus construction of Proposition 8.1 gives a derived cobordism

$$X \sim_{\mathrm{bord}} Z(s').$$

The endpoint $Z(s')$ is an ordinary compact manifold. Hence every derived bordism class is in the image of i_*.

For injectivity, suppose that ordinary compact manifolds M_0 and M_1 are derived cobordant. Let

$$p\colon W \longrightarrow \mathbb{R}$$

be a derived cobordism between them. Restrict to the part of W over an open interval containing $[0,1]$. Spivak's relative embedding theorem embeds this restricted cobordism over the parameter into $\mathbb{R}^N \times \mathbb{R}$ [Spi10, Corollary 3.4]. The normal-bundle axiom presents it as the derived zero locus of a section. The collared regions of W are ordinary manifolds $M_i \times I$. Their tangent complexes have no degree-minus-one homology, so the defining section is already transverse there. Relative transversality permits a perturbation on a compact neighbourhood of the middle portion, keeping the collars fixed and the zeros confined to that neighbourhood. The perturbed zero locus between the collars is an ordinary compact cobordism from M_0 to M_1. Thus two ordinary manifolds which become derived cobordant were already classically cobordant, and i_* is injective.

For the relative theorem over T, one performs the same constructions while retaining the map to T. The only additional point is that a map from the compact zero locus to the manifold T extends to a neighborhood in the ambient Euclidean space. To see this, embed T in a Euclidean space with a tubular-neighborhood retraction. Tietze extension extends the coordinate functions of the given continuous map. Near the zero locus its image stays in the tubular neighborhood, so composing with the retraction gives the required extension. The perturbation and cobordism can then be carried out over T. □

Example 8.7 (A singular fiber of a Morse function)**.** Let $f\colon T^2 \to \mathbb{R}$ be a Morse function with exactly four critical points and distinct critical values, and let c be its lower saddle value. The underlying space of the derived fiber

$$T^2 \times^{\mathrm{der}}_{\mathbb{R}} \{c\}$$

is a figure eight. It is not the underlying space of a one-dimensional manifold. Moving c slightly to either side produces a regular fiber, either one circle or two circles depending on the side. The parameter interval gives derived cobordisms from the singular fiber to both regular fibers. Thus the two regular fibers are classically cobordant, while the singular derived fiber remains a distinct geometric object. This is one of Spivak's examples [Spi10, Example 2.7].

8.3 Intersection products and Euler classes

The comparison theorem lets us compute a classical bordism class using a derived representative. We apply it first to intersections, then to the self-intersection of a zero section. The trivial and Möbius line bundles illustrate how the resulting class detects global bundle data.

Let M be a closed manifold, and let $A, B \subseteq M$ be closed submanifolds. We use cohomological notation for their Poincare-dual unoriented bordism classes:

$$[A] \in MO^{\mathrm{codim}(A)}(M), \qquad [B] \in MO^{\mathrm{codim}(B)}(M).$$

Bordism Poincare duality identifies these cohomology classes with the homology classes of the compact manifolds A and B over M.

If A and B are transverse, their ordinary intersection represents the cup product. Derived geometry removes the transversality condition from the geometric representative.

Theorem 8.8 (General cup-product formula)**.** *Let A and B be closed submanifolds of a closed manifold M. Then*

$$[A] \smile [B] = [A \times_M^{\mathrm{der}} B]$$

in the unoriented bordism cohomology of M.

Proof. Perturb the inclusion $A \to M$ through smooth maps until it is transverse to $B \to M$. Denote the perturbed map by $A' \to M$. Pulling the homotopy back along $B \to M$ gives a derived cobordism over M from

$$A \times_M^{\mathrm{der}} B$$

to the ordinary transverse fiber product $A' \pitchfork B$. Hence the two objects determine the same class over M. The classical transverse intersection formula gives

$$[A] \smile [B] = [A' \pitchfork B].$$

Combining the two equalities proves the result. □

This is Spivak [Spi10, Corollary 3.13].

Let M be a closed m-manifold and let $q\colon E \to M$ be a rank-r vector bundle. Write $z\colon M \to E$ for the zero section. Its derived self-intersection is

$$X_E = M \times_E^{\mathrm{der}} M.$$

The two maps to E in this expression are both z. The underlying space of X_E is all of M, but the derived object remembers that the same normal equation has been imposed twice.

Choose a transverse section $s'\colon M \to E$. Its graph is a perturbation of one copy of the zero section. The transverse intersection of $z(M)$ and $s'(M)$ is precisely the zero manifold $Z(s')$. The homotopy from z to s' therefore gives a derived cobordism over M

$$X_E \sim_{\mathrm{bord}} Z(s').$$

The bordism Euler class

$$e_{MO}(E) \in MO^r(M)$$

is represented by the zero locus of a transverse section. We have proved the following self-intersection formula.

Corollary 8.9 (Zero-section self-intersection)**.** *The cohomological bordism class represented by the derived self-intersection of the zero section is the Euler class:*

$$[M \times_E^{\mathrm{der}} M] = e_{MO}(E).$$

In homological notation, the corresponding class over M is

$$e_{MO}(E) \cap [M] = [Z(s') \longrightarrow M] \in MO_{m-r}(M).$$

The derived self-intersection is canonical, whereas the transverse section s' is a choice. Any two choices are joined by a homotopy of sections, so their zero manifolds are cobordant. Derived geometry packages the class before the choice of perturbation is made.

Example 8.10 (The trivial line bundle)**.** Let $E = M \times \mathbb{R}$ be the trivial line bundle. Its derived zero-section self-intersection is

$$X_{\underline{\mathbb{R}}} = M \times^{\mathrm{der}}_{M \times \mathbb{R}} M.$$

Locally, its algebra of functions is the Koszul algebra

$$\big(C^\infty(U) \otimes \Lambda(\eta), d\eta = 0\big).$$

Thus its underlying space and classical truncation are both M, while it has H_1 locally generated by η, or equivalently a nonzero π_1.

On the other hand, the zero section can be perturbed to the nowhere-zero constant section 1. Its zero locus is empty. Hence

$$[X_{\underline{\mathbb{R}}}] = 0$$

in bordism. The derived object is visibly nonempty and nonclassical, but its bordism class vanishes. Its tangent complex is $[TM \to \underline{\mathbb{R}}]$ with zero differential, so its obstruction bundle remains visible even though its bordism class vanishes.

Example 8.11 (The Möbius line bundle)**.** Return to the Möbius obstruction bundle from Chapter 7. Let $L \to S^1$ be this line bundle. A section can be described by a smooth function $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(t + 1) = -f(t).$$

The function $f(t) = \sin(\pi t)$ defines a transverse section. Modulo the identification $t \sim t + 1$, it has one zero. Therefore the derived self-intersection of the zero section is derived cobordant to one point over S^1. Its mod-2 Euler number is nonzero.

This complements the trivial-line example. For the trivial line, a nowhere-zero section makes the self-intersection class vanish. For the Möbius line, every section has a zero and the transverse zero set detects the nontrivial unoriented Euler class.

8.4 Fundamental classes

A compact derived manifold now determines a bordism class. Passing to a chosen homology theory gives a fundamental class over a manifold target. We describe the unoriented construction first, then the orientation data needed for an integral class.

Let X be a compact quasi-smooth derived manifold of virtual dimension d. The comparison theorem supplies a class

$$[X] \in \Omega_d^{\mathrm{der}} \cong \Omega_d = MO_d.$$

A continuous map from this compact X to a manifold T supplies a class in $MO_d(T)$.

Suppose that a generalized homology theory $\mathbb{E}_*$ is equipped with a morphism

$$MO \longrightarrow \mathbb{E}$$

of spectra representing the two theories. This induces a natural transformation

$$MO_*(T) \longrightarrow \mathbb{E}_*(T).$$

Definition 8.12 (Generalized fundamental class)**.** The image of $[X \to T]$ under this natural transformation is the $\mathbb{E}$-homology fundamental class of X over T:

$$[X \to T]_{\mathbb{E}} \in \mathbb{E}_d(T).$$

For example, the Thom orientation $MO \to H\mathbf{F}_2$ gives a mod-2 homology class. A transverse perturbation computes this class by the usual fundamental class of its zero manifold.

For an integral fundamental class, one needs an orientation. Spivak formulates an orientation using a normal bundle. A compact derived manifold is embedded in some $\mathbb{R}^N$, and the normal-bundle axiom supplies a normal vector bundle. After adding trivial summands, this normal bundle is stable under changing the embedding. An orientation of the stable normal bundle is the input for oriented derived bordism [Spi10, Remark 3.8].

In the oriented case, MO is replaced by the appropriate Thom spectrum, for example MSO for oriented bordism. A morphism from that spectrum to $\mathbb{E}$ transports the oriented bordism class to $\mathbb{E}$-homology. For a zero locus $X = Z^{\mathrm{der}}(s)$ of a section of $E \to U$, the virtual tangent bundle is $[TU|_X] - [E|_X]$. Its determinant line is

$$\det(TU)|_X \otimes \det(E)^\vee|_X.$$

Orienting this line gives the corresponding orientation of the virtual tangent bundle. It is the determinant of the intrinsic tangent complex, so it is independent of the presentation. For example, stabilizing a chart by a new variable together with that variable as a new equation contributes an identity complex, whose determinant has its canonical trivialization.

For a moduli problem, this construction applies once its representing quasi-smooth derived manifold has been shown to be compact and supplied with the desired orientation. Compactifications and objects with automorphisms require further geometric input.

8.5 Further reading

The remaining details explain why cobordisms compose and how the comparison theorem relates to the Pontryagin–Thom construction. References at the end give the geometric inputs used in the proof.

Proposition 8.13. *Derived cobordism is an equivalence relation on compact quasi-smooth derived manifolds.*

Proof. Reflexivity is represented by the projection

$$X \times \mathbb{R} \longrightarrow \mathbb{R}.$$

This map is proper when X is compact, is collared everywhere, and has fiber X over both 0 and 1.

Symmetry follows by reversing the parameter. If $p\colon W \to \mathbb{R}$ is a cobordism from X_0 to X_1, then $1 - p$ is a cobordism from X_1 to X_0.

For transitivity, write $p_{01}\colon W_{01} \to \mathbb{R}$ and $p_{12}\colon W_{12} \to \mathbb{R}$ for two cobordisms. Translate the second parameter by 1, so their common fiber X_1 lies over 1. Choose $\varepsilon > 0$ within both collars. Retain the part $p_{01} < 1 + \varepsilon/2$ of the first cobordism and the part $p_{12} > -\varepsilon/2$ of the second. Their overlaps

are both $X_1 \times (1 - \varepsilon/2, 1 + \varepsilon/2)$, and can be glued by the collar identifications. This produces a Hausdorff derived manifold with a map to $\mathbb{R}$. The map is proper: away from the seam this follows from the original maps, and a closed interval in the seam has compact preimage because X_1 is compact. The remaining endpoints lie over 0 and 2; dividing the parameter by 2 gives the required cobordism. □

This is Spivak [Spi10, Proposition 3.10]. The collar condition is exactly what makes the gluing step formal.

For an ordinary compact manifold N embedded in $\mathbb{R}^k$, the Pontryagin–Thom collapse produces a stable map from a sphere to the Thom space of the normal bundle. Stabilizing over all embeddings yields the class of N in MO_*. Stabilization removes dependence on the chosen ambient Euclidean space.

For a compact quasi-smooth derived manifold X, the embedding and normal-bundle theorem again supplies stable normal data. A transverse perturbation of a defining section produces an ordinary manifold N derived cobordant to X. Define the Pontryagin–Thom class of X to be the class of N. If a second perturbation produces N', a homotopy of perturbations gives a derived cobordism from N to N'. Injectivity in Theorem 8.5 turns this into an ordinary cobordism, so the stable class is independent of the perturbation.

The definitions of collared derived cobordism, the embedding theorem, the comparison with ordinary bordism, and the general cup-product formula are all from Spivak's *Derived smooth manifolds* [Spi10, Sections 1–3]. The comparison which places Spivak's model inside the universal modern framework is Carchedi and Steffens [CS19, Section 5.2].

9 Stacks and local representability

A moduli problem assigns families to parameter objects. For example, a family of zeros of $f\colon U \to \mathbb{R}^r$ parametrized by T is a map $T \to U$ satisfying the equation. In derived geometry the equation includes a specified homotopy to zero. Pulling the family back along $T' \to T$ gives a family over T'.

This viewpoint defines a moduli problem before a geometric representing object has been constructed. Descent expresses that families and their isomorphisms can be glued. The main result of this chapter gives a way to recognize when the resulting stack is a derived scheme: construct an open cover by representable stacks.

This returns to the coherence problem posed in Section 1.4. Starting with a globally defined stack supplies the overlaps and their higher compatibilities; the work is to construct open charts of that functor. We develop descent and atlases with the vector bundle zero locus as a running example, then formulate representability relative to a parameter stack. The distinction between gluing families and gluing charts will be essential throughout.

9.1 The open topology and descent

Descent concerns a cover of a parameter object: it reconstructs an anima of global families from coherent local families. We specify the test objects and covers first, then express this reconstruction as a limit.

We use the category DMfd of derived manifolds as test objects. By the affine comparison, these are spectra of finitely presented animated smooth rings. An open embedding is an inclusion of an open subobject with its restricted structure sheaf. An *open covering family* $\{T_i \to T\}$ is a family of open embeddings whose underlying images cover $|T|$. Smooth localization describes these embeddings algebraically. They are preserved by pullback.

Definition 9.1 (Prestack)**.** A derived smooth prestack is a functor $\mathcal{F}\colon \mathrm{DMfd}^{\mathrm{op}} \to \mathrm{An}$.

The anima $\mathcal{F}(T)$ contains the T-families, their isomorphisms, and higher identifications between those isomorphisms. Replacing it by its set of components loses automorphisms as well as higher coherence. Pullback along a map of parameter objects is part of the functor.

For an open cover, write $T_{i_0\cdots i_n}$ for the iterated intersection. Restriction gives a cosimplicial diagram with degree-n term

$$C^n = \prod_{i_0,\dots,i_n} \mathcal{F}(T_{i_0\cdots i_n}).$$

Its face maps restrict to smaller intersections, and its degeneracy maps repeat indices. The following formula for its limit avoids any need for an infinite disjoint union to be a test object.

Definition 9.2 (Stack)**.** A prestack is a stack if, for every open covering family, the natural map

$$\mathcal{F}(T) \longrightarrow \lim_{[n]\in\Delta} \prod_{i_0,\dots,i_n} \mathcal{F}(T_{i_0\cdots i_n}) \tag{9.1}$$

is an isomorphism of animae.

For a set-valued functor this reduces to the usual sheaf condition: local elements must agree on double overlaps. For groupoid-valued functors one instead chooses isomorphisms on double overlaps and imposes the cocycle condition on triple overlaps. For general animae the cocycle condition itself is supplied by a homotopy, followed by higher compatibility data. The limit in Equation (9.1) packages all of it.

Descent applies to maps between families too. It says that the anima of global families is the anima of coherent local data, rather than merely that some global family exists.

The resulting ∞-category of stacks is denoted $\mathrm{dSt}^{\mathrm{fp}}_{C^\infty}$ here. The superscript records our site of finitely presented test objects. Steffens embeds this category fully faithfully into his category using all affine derived smooth schemes, and calls its objects stacks locally of finite presentation [Ste24, Proposition 2.1.40 and Definition 2.1.41]. Our constructions take place in this subcategory.

9.2 Represented stacks and the zero-locus example

Representability asks whether all families arise as maps to a geometric object. The universal property of a derived pullback answers this question for the zero-locus functor and gives a model for the later elliptic moduli problem.

For a derived scheme X locally of finite presentation, its functor of points is

$$h_X(T) = \mathrm{Hom}(T, X).$$

It satisfies descent because morphisms of locally ringed objects can be glued over an open cover of their source. More explicitly, the continuous maps on the underlying spaces glue uniquely. For the resulting map $f\colon |T| \to |X|$, the maps $f^{-1}\mathcal{O}_X \to \mathcal{O}_T$ glue as morphisms of sheaves, including all homotopies between them; this is the mapping assertion of Lemma 9.7 below. The condition that the maps on stalks are local can be checked on the same cover. This proves descent for the entire mapping anima and hence the subcanonical property of the open topology. Moreover, the functor $X \mapsto h_X$ is fully faithful.

Definition 9.3 (Representability)**.** A stack $\mathcal{F}$ is represented by a derived C^∞-scheme X if there is a natural isomorphism $\mathcal{F}(T) \cong \mathrm{Hom}(T, X)$ for every test object T.

The representing object is unique. If X is not affine, the notation $\mathcal{F}(X)$ means the value obtained by descent from an affine open cover. Under representability, the identity of X corresponds to a universal family in $\mathcal{F}(X)$. Every other family is its pullback, by the Yoneda lemma.

Let $f\colon U \to \mathbb{R}^r$ be smooth. Define

$$\mathcal{Z}_f(T) = \mathrm{Hom}(T, U) \times_{\mathrm{Hom}(T,\mathbb{R}^r)} \mathrm{Hom}(T, *), \tag{9.2}$$

with pullback in animae. A point is a map $u\colon T \to U$ together with a homotopy from $f \circ u$ to zero. Identifications in this pullback give the isomorphisms between such data and their higher compatibilities. Limits of stacks are computed on each test object, so $\mathcal{Z}_f$ is a stack.

By the universal property of a pullback,

$$\mathcal{Z}_f(T) \cong \mathrm{Hom}(T, U \times^{\mathrm{der}}_{\mathbb{R}^r} *).$$

Thus it is represented by the derived zero locus computed in Chapter 5. This is a simple instance of the representability question that will later arise for differential equations.

For a section $s\colon U \to E$ of a vector bundle, the analogous formula is most conveniently written in the category over U:

$$\mathcal{Z}_s = U \times^{\mathrm{der}}_{E} U,$$

using s and the zero section. Restricting to a trivializing open U_i gives the Koszul presentation for its list of local components. These local representatives will form an open atlas of $\mathcal{Z}_s$.

9.3 Smooth stacks as derived stacks

A smooth stack is a sheaf of animae on ordinary manifolds with their open-cover topology. The inclusion of manifolds into derived manifolds induces a fully faithful functor

$$j_!\colon \mathrm{Shv}(\mathrm{Mfd}) \longrightarrow \mathrm{dSt}^{\mathrm{fp}}_{C^\infty}.$$

It is obtained by left Kan extension and sheafification, and sends an ordinary manifold to its derived functor of points. We can therefore regard smooth stacks as particular derived stacks.

The derived values of $j_!\mathcal{F}$ are not obtained just by evaluating $\mathcal{F}$ on an underlying point set. They are determined by this extension construction. Conversely, restriction to ordinary test manifolds does not recover every derived stack. For example, a multiple point and an ordinary point have the same families over ordinary manifolds, while maps from infinitesimal derived tests distinguish them.

There is also an essential distinction between the two categories of pullbacks. In smooth stacks,

$$* \times_{\mathbb{R}} * = *$$

for the two origins. After passing to derived stacks, the pullback is the derived self-intersection, with algebra $\mathbb{R}[\varepsilon]$, where $|\varepsilon| = 1$ and $d\varepsilon = 0$. Thus $j_!$ does not preserve arbitrary pullbacks. The pullbacks needed in the analytic argument will be compared using submersiveness in Chapter 12.

9.4 Open atlases and local representability

We now turn from gluing families over a parameter cover to presenting the moduli stack itself by charts. Here a Čech diagram reconstructs the stack as a colimit. Effectiveness expresses that the charts cover the functor; openness makes it possible to glue their representing objects into a derived scheme. We use two sheaf-theoretic inputs below, with references to Steffens and heuristic explanations, before giving the geometric proof of open local representability.

Let $p\colon \mathcal{U} \to \mathcal{F}$ be a morphism of stacks. Its Čech nerve is the simplicial stack

$$\mathcal{U}_n = \underbrace{\mathcal{U} \times_{\mathcal{F}} \cdots \times_{\mathcal{F}} \mathcal{U}}_{n+1 \text{ factors}}.$$

The augmentation maps its geometric realization to $\mathcal{F}$. Here geometric realization means a colimit in stacks, computed by taking the presheaf colimit and then sheafifying.

Definition 9.4 (Effective epimorphism)**.** The map p is an effective epimorphism if $|\mathcal{U}_\bullet| \to \mathcal{F}$ is an isomorphism.

Lemma 9.5 (Local lifting criterion)**.** *A map $p\colon \mathcal{U} \to \mathcal{F}$ is an effective epimorphism if and only if, for every test object T and every $x \in \mathcal{F}(T)$, there is an open cover $\{T_a \to T\}$ on which x lifts to $\mathcal{U}$. A lift includes an identification of its image with $x|_{T_a}$. Effective epimorphisms are preserved by pullback.*

We take this standard sheaf-theoretic criterion as an input. For the reduction to sheaves on underlying topological spaces, see Steffens [Ste24, Lemma 2.1.38]; the local lifting criterion is used explicitly in the proof of Corollary 2.1.49 there.

Heuristic explanation. For a map of animae $A \to B$, the realization of its Čech nerve is the union of the components of B met by A. Indeed, its fibre over $b \in B$ is the realization of the Čech nerve of $A_b \to *$. This is empty if A_b is empty. Otherwise, choosing a point of A_b contracts the augmented nerve by inserting that point, so its realization is a point.

Apply this observation on each test object. The presheaf realization of $\mathcal{U}_\bullet$ is the subpresheaf $\mathcal{I} \subseteq \mathcal{F}$ whose value on T consists of the components containing an image of $\mathcal{U}(T)$. Its sheafification is the substack of families which locally belong to $\mathcal{I}$: this is a local condition, and all identifications between such families are inherited from $\mathcal{F}$. A map from $\mathcal{I}$ to any stack extends uniquely to this substack by descent on the covers witnessing local membership, which verifies the universal property of sheafification. This substack is all of $\mathcal{F}$ precisely when the stated local lifts exist. Since realization in stacks is presheaf realization followed by sheafification, this proves the criterion.

If $\mathcal{G} \to \mathcal{F}$ is any map, a family in $\mathcal{G}(T)$ can locally be lifted to $\mathcal{U}$ together with an identification in $\mathcal{F}(T)$. These are exactly the data of a local lift to $\mathcal{U} \times_{\mathcal{F}} \mathcal{G}$. Thus the criterion also proves stability under pullback. □

For a family $\{\mathcal{U}_i \to \mathcal{F}\}$, the criterion says that each family in $\mathcal{F}(T)$ locally lifts to one of the $\mathcal{U}_i$. A map into a coproduct of stacks locally lands in one summand, since that coproduct is the sheafification of the pointwise coproduct. This accounts for the possible refinement of the parameter cover when applying the criterion to a family.

Starting with a globally defined stack supplies the overlap data. If $U_i \to \mathcal{F}$ and $U_j \to \mathcal{F}$ are two charts, their overlap is $U_i \times_{\mathcal{F}} U_j$. Its associativity and higher compatibilities come from pullbacks in the ambient ∞-category. The following condition makes these overlaps geometric open subsets.

Definition 9.6 (Open atlas)**.** A morphism $\mathcal{V} \to \mathcal{F}$ is an open inclusion if every pullback along $T \to \mathcal{F}$ is represented by an open subobject of T. An open atlas is a small family of open inclusions $h_{U_i} \to \mathcal{F}$ from derived schemes such that $\coprod_i h_{U_i} \to \mathcal{F}$ is an effective epimorphism.

The coproduct here is in stacks. Each map is a condition on families: after choosing a family over T, belonging to the indicated chart cuts out an open subset of T.

The word *open* is crucial for representing a stack by a scheme. For example, a nontrivial finite group G has a classifying stack BG. The map $* \to BG$ is an effective epimorphism, expressing local triviality of principal G-bundles. It is not an open inclusion: its pullback along itself is $G \to *$. The automorphisms of the trivial bundle also show that BG cannot be represented by a derived scheme, whose real points form a set. More general notions of atlas lead to orbifolds

and other geometric stacks, but an arbitrary effective epimorphism is insufficient for scheme representability.

To reconstruct a derived scheme, we must glue both its topology and its animated structure sheaf. Our second input concerns this sheaf construction. In the following lemma, coherent identifications mean isomorphisms on double intersections, homotopies between their composites on triple intersections, and all higher compatibilities.

Lemma 9.7 (Gluing structure sheaves)**.** *Let Y be a topological space with a small open cover $\{V_i\}$. Suppose that each V_i carries a sheaf $\mathcal{A}_i$ of animated C^∞-rings, with coherent identifications of their restrictions to the intersections. There is a sheaf $\mathcal{A}$ on Y whose restrictions, with their identifications, recover this datum. It is unique, and maps between such sheaves are likewise determined by their coherent restrictions to the cover.*

We also take this descent statement as an input. It is the sheaf gluing mechanism in the proof of Steffens [Ste24, Proposition 2.1.18]; Remark 2.1.19(c) there describes the structure sheaf by the limit formula explained below.

Heuristic explanation. Write $V_{i_0\cdots i_n} = V_{i_0} \cap \cdots \cap V_{i_n}$. For an open subset $W \subseteq Y$, define

$$\mathcal{A}(W) = \lim_{[n]\in\Delta} \prod_{i_0,\ldots,i_n} \mathcal{A}_{i_0}(W \cap V_{i_0\cdots i_n}). \tag{9.3}$$

The transition identifications supply the maps which change the first index, and their coherences make this a cosimplicial diagram. All limits are taken in the ∞-category of animated smooth rings. Each functor of W occurring on the right is a sheaf on Y: an open cover of W restricts to an open cover of its intersection with $V_{i_0\cdots i_n}$. Limits of sheaves are computed on opens, because limits commute with the limits expressing descent. Thus the formula defines a sheaf.

If $W \subseteq V_k$, the transition identifications identify the diagram with the descent diagram of $\mathcal{A}_k$ for the cover $\{W \cap V_i\}$ of W. Hence the sheaf condition gives $\mathcal{A}(W) \cong \mathcal{A}_k(W)$. These isomorphisms recover the specified transition data, including its higher coherences.

Conversely, a sheaf on Y is recovered from its restrictions by the same formula, using descent for $\{W \cap V_i\}$. The construction is functorial in the coherent restriction data, so gluing and restriction are inverse functors. This proves both uniqueness and the assertion about maps, including homotopies between them. □

Using these two inputs, we can now prove that an open atlas contains all the data required for representability.

Theorem 9.8 (Open local representability)**.** *A derived smooth stack is represented by a derived C^∞-scheme locally of finite presentation if and only if it has an open atlas by affine derived schemes of finite presentation. One may equivalently allow derived schemes locally of finite presentation as the atlas objects.*

This is Steffens [Ste24, Proposition 2.1.48].

Proof. Suppose first that $\mathcal{F} = h_X$, and choose an affine open cover $\{U_i \to X\}$. Each $h_{U_i} \to h_X$ is an open inclusion: its pullback along $T \to X$ is the open subscheme $T \times_X U_i$. These pullbacks

cover T. Refining them by affine opens gives local lifts of every $T \to X$ to the charts. By Lemma 9.5, the chart family is an effective epimorphism and hence an open atlas.

Conversely, let $p_i \colon h_{U_i} \to \mathcal{F}$ be an affine open atlas. An open inclusion is a monomorphism: on each test object its homotopy fibres are either empty or a point, since a factorization through an open subscheme, when it exists, is unique. In particular, the diagonal gives $h_{U_i} \cong h_{U_i} \times_{\mathcal{F}} h_{U_i}$. For every pair of indices, write

$$h_{U_{ij}} = h_{U_i} \times_{\mathcal{F}} h_{U_j}.$$

By openness, this is represented by an open subscheme of U_i, and its other projection identifies it with an open subscheme of U_j. Write $W_{ij} \subseteq |U_i|$ for the first image and $\varphi_{ij} \colon W_{ij} \to W_{ji}$ for the resulting homeomorphism.

The diagonal, symmetry, and composition in the Čech nerve give $\varphi_{ii} = \mathrm{id}$, $\varphi_{ji} = \varphi_{ij}^{-1}$, and $\varphi_{jk}\varphi_{ij} = \varphi_{ik}$ wherever the composite is defined. For the last assertion, a pair of composable overlap points comes from $U_{ij} \times_{U_j} U_{jk}$, which represents $h_{U_i} \times_{\mathcal{F}} h_{U_j} \times_{\mathcal{F}} h_{U_k}$; the remaining projection gives the corresponding point of U_{ik}. Passing to underlying topological spaces turns the specified homotopies into equalities.

Define a topological space

$$|X| = \left(\coprod_i |U_i| \right) / \sim, \qquad x \sim \varphi_{ij}(x) \quad (x \in W_{ij}),$$

with the quotient topology. The preceding identities show that the displayed relation is already an equivalence relation. It identifies no two distinct points of a single $|U_i|$, because $W_{ii} = |U_i|$ and $\varphi_{ii} = \mathrm{id}$. Each induced map $q_i \colon |U_i| \to |X|$ is therefore injective. It is also open: for $O \subseteq |U_i|$ open, the inverse image of $q_i(O)$ in $|U_j|$ is $\varphi_{ij}(O \cap W_{ij})$, which is open. The definition of the quotient topology then makes $q_i(O)$ open in $|X|$. Thus q_i is a homeomorphism onto an open subset V_i, the V_i cover $|X|$, and $q_i^{-1}(V_i \cap V_j) = W_{ij}$.

Transport $\mathcal{O}_{U_i}$ to V_i along q_i. The two open embeddings from U_{ij} identify the restrictions of $\mathcal{O}_{U_i}$ and $\mathcal{O}_{U_j}$ on $V_i \cap V_j$. All the required coherences are supplied by the higher terms of the original Čech nerve. Apply Lemma 9.7 to obtain $\mathcal{O}_X$ on $|X|$. At a point in V_i, its stalk is the corresponding stalk of $\mathcal{O}_{U_i}$, so its degree-zero ring is local with the same residue map to $\mathbb{R}$. Consequently $X = (|X|, \mathcal{O}_X)$ is a locally animated smooth-ringed space, and $(V_i, \mathcal{O}_X|_{V_i}) \cong U_i$. Since the U_i are affine of finite presentation, this is a derived C^∞-scheme locally of finite presentation by Definition 7.4. No global Hausdorff or second-countability condition is imposed on a derived scheme; imposing either would require an additional hypothesis on this gluing datum.

It remains to identify its functor of points with $\mathcal{F}$. The open subscheme $U_i \times_X U_j$ has underlying open subset $W_{ij} \subseteq |U_i|$ and structure sheaf $\mathcal{O}_{U_i}|_{W_{ij}}$. Hence its two maps to the charts identify it with U_{ij}. The same argument on iterated intersections identifies all higher overlaps, with the compatibilities used in constructing $\mathcal{O}_X$.

Set $\mathcal{U} = \coprod_i h_{U_i}$, with coproduct in stacks. The chart embeddings give a map $q \colon \mathcal{U} \to h_X$. For every test map $T \to X$, the open subschemes $T \times_X U_i$ cover T, so the local lifting criterion shows that q is an effective epimorphism. Its Čech nerve in degree n is

$$\coprod_{i_0, \dots, i_n} h_{U_{i_0} \times_X \cdots \times_X U_{i_n}}.$$

Here pullbacks distribute over coproducts of stacks: locally on a test object a map into a coproduct chooses a summand, so the two sides have the same local families and identifications. Also,

functors of points preserve the displayed pullbacks by their defining universal property. The overlap identifications above therefore identify this simplicial stack with the Čech nerve of $p\colon \mathcal{U} \to \mathcal{F}$. Since both augmentations are effective epimorphisms, realization gives

$$h_X \cong |\check{C}(q)_\bullet| \cong |\check{C}(p)_\bullet| \cong \mathcal{F}.$$

This is an isomorphism of stacks, and thus identifies the animae of families on every derived test object.

Finally, suppose the atlas objects are only locally of finite presentation. Choose an affine open cover of each U_i. The composites of these open inclusions with p_i are open. A family in $\mathcal{F}(T)$ first locally lifts to some U_i, and then, after pulling back its affine cover, locally lifts to one of these affine charts. The local lifting criterion shows that the refined family is again an effective epimorphism. The affine case just proved applies. □

In the bundle-section example, the charts are the zero loci over a trivializing cover of U. Their overlaps are the zero loci over the corresponding intersections. The transition maps of the bundle identify the Koszul models, and the theorem recovers the global derived zero locus.

The same argument preserves a local property such as quasi-smoothness. If the charts are quasi-smooth, the cotangent amplitude condition holds on the resulting scheme.

9.5 Relative representability and base change

Moduli problems often have a parameter stack S. For example, the metric, the domain manifold, or a coefficient in an equation may vary. The correct representability statement is then relative to S.

Definition 9.9. A map $\mathcal{F} \to S$ is representable by derived C^∞-schemes locally of finite presentation if, for each affine derived test $T \to S$, the pullback $\mathcal{F} \times_S T$ is represented by such a scheme. It is relatively quasi-smooth if these representing morphisms to T are quasi-smooth.

This asserts representability of every pulled-back family. It does not assert that the total stack $\mathcal{F}$ is itself a scheme when S has automorphisms.

Corollary 9.10 (Locality on the base)**.** *Representability by derived schemes locally of finite presentation can be checked after an effective epimorphic covering family of the base. The same holds with the additional requirement of relative quasi-smoothness.*

Proof. Let $\{S_i \to S\}$ be a family whose coproduct is an effective epimorphism, and suppose each $\mathcal{F}_i = \mathcal{F} \times_S S_i \to S_i$ is representable. Fix an affine derived test $T \to S$. By Lemma 9.5, there is an open cover $\{T_a \to T\}$ and, for each a, a lift $T_a \to S_{i(a)}$ with a specified identification over S. Refine the cover so that each T_a is affine of finite presentation. The pullback identity

$$\mathcal{F} \times_S T_a \cong \mathcal{F}_{i(a)} \times_{S_{i(a)}} T_a$$

and the assumed relative representability give a representing scheme X_a locally of finite presentation.

Put $\mathcal{G} = \mathcal{F} \times_S T$. Each $h_{X_a} \to \mathcal{G}$ is the pullback of the open inclusion $T_a \to T$, so is open. Their coproduct is an effective epimorphism: for any test map $R \to \mathcal{G}$, the opens $R \times_T T_a$ cover R

and give local lifts to the h_{X_a}. Theorem 9.8, including its non-affine atlas formulation, therefore represents $\mathcal{G}$ by a scheme X locally of finite presentation. This works for every $T \to S$ and proves the required relative representability.

If every $\mathcal{F}_i \to S_i$ is relatively quasi-smooth, then $X_a \to T_a$ is quasi-smooth by its definition on affine tests. Moreover $X_a = X \times_T T_a$. Base change for relative cotangent complexes identifies $\mathbb{L}_{X/T}|_{X_a}$ with $\mathbb{L}_{X_a/T_a}$. Perfection and Tor-amplitude in $[0, 1]$ can be checked on this open cover of X, so $X \to T$ is quasi-smooth. Conversely, representability and relative quasi-smoothness are preserved by base change, so the asserted conditions are necessary as well. □

The source formulation is [Ste24, Definition 2.1.42 and Corollary 2.1.49]. This permits a proof over a smooth stack to be carried out on manifold charts of the base, provided the moduli construction is compatible with pullback.

9.6 Further reading

The local representability criterion does not make the construction of a chart automatic. One must identify the entire derived functor on an open neighbourhood, including its values on derived tests. An identification of real solution sets, or even of their tangent complexes, proves less. The later Kuranishi argument is designed to establish precisely this functorial identification.

Exercise 9.11. Let $f : U \to \mathbb{R}^r$ and let $\{U_i\}$ cover U. Use the functor-of-points formula to verify that the corresponding zero loci form an effective open atlas of $\mathcal{Z}_f$.

Exercise 9.12. For a nontrivial finite group G, calculate $* \times_{BG} *$ and explain both the effectiveness of $* \to BG$ and its failure to be an open inclusion.

All the descent and representability results used in this chapter are contained in [Ste24, Section 2.1].

10 Section stacks and derived solution stacks

A differential operator assigns to a function, section, or map an expression in its derivatives. Its solutions often describe a geometric condition: harmonicity, stationarity of an energy, or compatibility with a complex structure. The equation can be defined for smooth families of unknowns. The resulting functor of solutions is a stack, whose geometric representability is a separate question.

Our running example is the nonlinear elliptic equation

$$P(u) = \Delta u + u^2 = 0$$

on a closed connected Riemannian manifold. Its ordinary solution set is a single point, but its infinitesimal solutions and obstructions are both nontrivial. The eventual conclusion is concrete: near zero, its derived solution object is the multiple point defined by $c^2 = 0$. Chapter 13 will prove this by eliminating the mean-zero part of u, leaving its constant part c as the remaining variable.

This chapter sets up the functor that the later analytic argument will represent. Geometric examples lead to section stacks and their derived fibres. Jets then express the finite-order condition, and linearization gives the definition of ellipticity. Each construction is formulated for families, so it can be used over a parameter stack.

10.1 Differential equations from geometry

The examples below identify the unknowns, their equations, and the bundles in which the equations take values. We begin with scalar functions and then consider maps, where the output bundle depends on the unknown.

Example 10.1 (Harmonic functions)**.** For a Riemannian manifold N, the Laplacian is an operator

$$\Delta\colon C^\infty(N) \longrightarrow C^\infty(N).$$

We use the nonnegative convention $\Delta = d^*d$. The equation $\Delta u = 0$ says that u is harmonic. It occurs in potential theory and in extending boundary data into a domain. On a closed connected manifold, integration by parts shows that harmonic functions are constant:

$$0 = \int_N u\Delta u = \int_N |du|^2.$$

Boundary-value problems require additional analytic hypotheses; the representability theorem considered here concerns closed fibres.

Both source and target are sections of the trivial real line bundle $N \times \mathbb{R} \to N$. The operator takes an entire section as its input and returns another section.

Example 10.2 (Critical points of energies)**.** Let N be a closed Riemannian manifold and let $V\colon \mathbb{R} \to \mathbb{R}$ be smooth. Consider

$$\mathcal{E}(u) = \int_N \left(\tfrac{1}{2}|du|^2 + V(u) \right) \, d\mathrm{vol}.$$

Differentiating in the direction v and integrating by parts gives

$$d\mathcal{E}_u(v) = \int_N \big(\Delta u + V'(u) \big) v \, d\mathrm{vol}.$$

Therefore the critical points are the solutions of $P(u) = 0$, where

$$P\colon C^\infty(N) \longrightarrow C^\infty(N), \qquad P(u) = \Delta u + V'(u).$$

The energy is a real-valued functional; its Euler–Lagrange operator P has a function as its output. The metric and volume form identify the first variation with this function.

Taking $V(u) = u^3/3$ gives the running example $P(u) = \Delta u + u^2$.

Example 10.3 (Pseudo-holomorphic maps)**.** Let (Σ, j) be a fixed Riemann surface and (Z, J) an almost complex manifold. A smooth map $u\colon \Sigma \to Z$ is pseudo-holomorphic when

$$\bar\partial_J u = \tfrac{1}{2}(du + J(u) \circ du \circ j) = 0.$$

The unknown is a section of $\Sigma \times Z \to \Sigma$. Its output lies in

$$\Gamma\big(\Sigma, \mathrm{Hom}^{0,1}_{\mathbb{R}}(T\Sigma, u^*TZ)\big),$$

where the superscript denotes complex-antilinear real linear maps. The output bundle depends on u.

Globally, this is a section of a bundle over the space of maps. Near a fixed map, exponential coordinates on Z and identifications of nearby output bundles turn it into an operator

$$Q \subseteq \Gamma(\Sigma, F) \longrightarrow \Gamma(\Sigma, E)$$

for fixed finite-rank bundles F, E. Chapter 11 explains the coordinate principle, and Chapter 14 carries out the Cauchy–Riemann application. Keeping track of these source and target bundles is necessary before applying an ellipticity theorem.

10.2 Section stacks

To apply the functor-of-points viewpoint, we retain smooth families of sections and their pullbacks. We first fix the domain manifold, then allow it to vary over a base.

For a submersion $p\colon X \to N$, the ordinary section set is

$$\Gamma(N; X) = \{u\colon N \to X : \ p \circ u = \mathrm{id}_N\}.$$

If X is a vector bundle, this is a vector space. A family parametrized by an ordinary manifold T is a smooth map $T \times N \to X$ over N. Smoothness is joint smoothness in the parameter and in the point of N.

The same formula makes sense with derived test objects. For T derived, it uses maps of derived stacks, and so retains infinitesimal and higher data in the parameter.

Definition 10.4 (Section stack)**.** The section stack of $X \to N$ is

$$\mathrm{Sec}_N(X)(T) = \mathrm{Hom}_{/N}(T \times N, X).$$

This is the mapping stack often written $\mathrm{Map}_N(N, X)$. The notation $\Gamma(N; X)$ will be reserved for ordinary smooth sections, and, for a vector bundle on compact N, their Fréchet manifold. The section stack is a functor on parameter objects.

A family of manifolds over a smooth stack S is a map $q\colon M \to S$ whose pullback along every ordinary manifold $T \to S$ is a submersion of ordinary manifolds. The family is proper if each such pullback is proper. For a manifold base this is the usual notion of a submersion, or a proper submersion, respectively. Suppose $q\colon M \to S$ is such a family and $X \to M$ is a family of submersions.

Definition 10.5 (Relative section stack)**.** The relative section stack $\mathrm{Sec}_{M/S}(X) \to S$ is characterized by

$$\mathrm{Hom}_{/S}(T, \mathrm{Sec}_{M/S}(X)) \cong \mathrm{Hom}_{/M}(T \times_S M, X).$$

It is the Weil restriction $\mathrm{Res}_{M/S}(X)$.

The terminology describes the right adjoint to pullback along $M \to S$. Its existence here follows from the locally cartesian closed ∞-category of stacks [Ste24, Definition 3.2.1]. The formula also proves base-change compatibility:

$$\mathrm{Sec}_{M/S}(X) \times_S T \cong \mathrm{Sec}_{M_T/T}(X_T), \qquad M_T = M \times_S T.$$

Over $s \in S$, its fibre is the section stack on M_s.

10.3 Solution stacks as derived fibres

Let $P\colon \mathrm{Sec}_N(X) \to \mathrm{Sec}_N(Y)$ be any morphism of stacks, and let b be a section of Y. Define

$$\mathrm{Sol}(P; b) = \mathrm{Sec}_N(X) \times^{\mathrm{der}}_{\mathrm{Sec}_N(Y)} *, \tag{10.1}$$

where the second map classifies b. The adjective derived emphasizes the category in which the pullback is taken; all pullbacks in derived stacks are already homotopy pullbacks.

On a derived parameter T, a solution is a family u together with a homotopy from $P(u)$ to the constant family b_T. Thus a solution stack can be defined without any finite-order assumption. One does need a morphism of stacks, not merely a map between sets of smooth sections.

For closed N, a smooth map between the corresponding convenient section manifolds supplies such a morphism. This follows from the comparison between section stacks and convenient manifolds explained in Chapter 12. For example, on S^1 the map

$$u \longmapsto \left(x \longmapsto \int_{S^1} u \, d\theta \right)$$

is a smooth linear map of section manifolds and defines a solution stack. It is not a finite-order differential operator: changing u away from x can change the output at x while leaving every derivative there fixed.

For a parameter base S, a right-hand side is a section $b\colon S \to \mathrm{Sec}_{M/S}(Y)$, and the same pullback is formed over S. All these constructions commute with base change. The finite-order differential condition introduced next provides a geometric way to define P and the analytic structure needed for representability.

10.4 Jets and finite-order operators

Jets encode the dependence of an operator on finitely many derivatives. Their relative construction also makes the induced map of section stacks available on derived parameters.

A k-jet of a local section of $X \to N$ at $x \in N$ records its value and its derivatives through order k at x. Two sections define the same jet when those derivatives agree in local coordinates. The chain rule makes this independent of the coordinates. These jets form the bundle $J^k_N(X) \to N$.

For example, for $X = N \times \mathbb{R}$, a first jet records a scalar and a covector. A second jet records the second derivatives as well; a connection identifies that last datum with a symmetric tensor. The jet prolongation sends a section u to $j^k u$.

Definition 10.6 (Finite-order differential operator)**.** A nonlinear differential operator of order at most k from X to Y over N is a smooth map

$$p \colon J^k_N(X) \longrightarrow Y$$

over N. Its induced map on sections is $P(u) = p \circ j^k u$.

Using the lowercase p distinguishes the finite-dimensional map on jets from the induced map P on sections. The latter is determined at x by finitely many derivatives of u at that same point. A general map on sections need not have this locality.

For $\Delta u + u^2$, $k = 2$. In local coordinates the Laplacian depends on derivatives of u of order at most two, while u^2 uses only its value. For the Cauchy–Riemann equation only first jets are needed.

For $M \to S$, relative jets $J^k_{M/S}(X)$ record derivatives in the directions tangent to the fibres M_s. No differentiation in the parameter s is involved. The resulting map P is therefore an S-family of operators on the fibre manifolds.

The derived jet stack has a coordinate-independent construction. Let $(M/S)^{(k)}$ be the order-k infinitesimal neighbourhood of the relative diagonal, with projections π_1, π_2 to M. Then

$$J^k_{M/S}(X) = \mathrm{Res}_{(M/S)^{(k)}/M}\big(X \times_{M,\pi_1} (M/S)^{(k)}\big),$$

where the Weil restriction uses π_2. It parametrizes sections on infinitesimal neighbourhoods. For an ordinary submersion this recovers the classical jet bundle. The construction, the prolongation map, and base-change compatibility are proved in [Ste24, Definition 3.3.7 and Proposition 3.3.9].

In particular, a map $p \colon J^k_{M/S}(X) \to Y$ induces a morphism of relative section stacks. This makes the preceding solution construction available over all derived test objects.

10.5 Differential moduli problems and ellipticity

We now collect the geometric data of an equation into a differential moduli problem. Its linearization gives the deformation-obstruction complex, and its principal symbol determines whether elliptic analysis can be applied.

A useful common formulation asks two operators to have the same output:

$$P_1(u_1) = P_2(u_2).$$

It includes $P(u) = b$ by taking the second unknown bundle to be $N \to N$, whose only section is the identity, and taking its operator to be b.

Definition 10.7. Over a family $M \to S$, a differential moduli problem consists of submersions $Y_1, Y_2, X \to M$ and maps

$$p_i \colon J^k_{M/S}(Y_i) \longrightarrow X.$$

Its solution stack is

$$\mathrm{Sol}(\mathcal{E}) = \mathrm{Sec}_{M/S}(Y_1) \times_{\mathrm{Sec}_{M/S}(X)} \mathrm{Sec}_{M/S}(Y_2),$$

with maps induced by p_1, p_2.

This is [Ste24, Definitions 3.4.1–3.4.2]. The order k is a common order: an operator of lower order can be composed with a jet truncation map. Ellipticity below is then a condition on the symbol for that chosen common order.

When S is a smooth stack, these data are specified on manifold families over S and are compatible with pullback. Relative jets and section stacks commute with that pullback, so the equations and solution stacks descend. This includes families with symmetries encoded in the base stack.

Fix first $S = *$ and a solution u of $P(u) = b$. Variations of u are sections of $F = u^* T_{X/N}$; variations of the output at b are sections of $E = b^* T_{Y/N}$. Differentiating the finite jet formula gives a linear differential operator

$$D_u P \colon \Gamma(N; F) \longrightarrow \Gamma(N; E).$$

In coordinates it has the form

$$D_u P(v) = \sum_{|\alpha| \le k} A_\alpha(x) \partial^\alpha v.$$

Its coefficients depend on u and its derivatives, but the expression is linear in v.

For a matching problem and solution (u_1, u_2) with common output w, put

$$F = u_1^* T_{Y_1/N} \oplus u_2^* T_{Y_2/N}, \qquad E = w^* T_{X/N}.$$

The linearized matching equation is

$$D(v_1, v_2) = D_{u_1} P_1(v_1) - D_{u_2} P_2(v_2). \tag{10.2}$$

Its deformation-obstruction complex is $[\Gamma(N; F) \xrightarrow{D} \Gamma(N; E)]$ in degrees $0, -1$. The kernel is the infinitesimal solution space. The cokernel receives obstructions to solving inhomogeneous correction equations, as in the finite-dimensional zero-locus calculation. The representability proof will identify this complex with the finite-dimensional tangent complex of a local derived chart.

Ellipticity is a condition on the highest-order part of the linearized operator. In particular, it depends on the source and target bundles of the full matching equation.

Definition 10.8 (Principal symbol and ellipticity)**.** For a linear operator D of order k, its principal symbol is

$$\sigma_D(x, \xi) = \sum_{|\alpha| = k} A_\alpha(x) \xi^\alpha \colon F_x \longrightarrow E_x.$$

It is elliptic if this map is invertible for every nonzero covector ξ. A differential moduli problem is elliptic if its linearized matching operator is elliptic at every solution on every fibre M_s.

This real symbol convention omits the optional factors of i used in Fourier analysis; the invertibility condition is unchanged. For the nonnegative Laplacian the real symbol with this convention is $-|\xi|^2$, while the Fourier convention gives $|\xi|^2$. Both are invertible away from the zero covector.

Ellipticity concerns the highest-order part. On a compact manifold it leads to finite-dimensional kernel and cokernel after Sobolev completion. It does not assert surjectivity. The latter failure is exactly why a derived zero locus can be needed.

10.6 Solutions and linearization of the running example

Return to $P(u) = \Delta u + u^2$ on a closed connected manifold N. If $P(u) = 0$, integration gives

$$0 = \int_N (\Delta u + u^2) = \int_N u^2,$$

so $u = 0$. Its ordinary solution set is one point.

Differentiating gives

$$D_u P(v) = \Delta v + 2uv.$$

The zeroth-order term does not change the principal symbol, so this operator is elliptic. At zero it is Δ. The standard solvability theorem for the Laplacian on a closed connected manifold says that its image consists of the functions of integral zero. Consequently,

$$\ker \Delta \cong \mathbb{R}, \qquad \operatorname{coker} \Delta \cong \mathbb{R}.$$

Constants give representatives in both cases.

This explains the usefulness of the example. Its real zero set is particularly simple, while its infinitesimal solutions and obstructions are both nontrivial. Chapter 13 will obtain its local Kuranishi equation explicitly, recovering a multiple point.

10.7 Further reading

The jet tower makes the highest-order part of an operator explicit. We describe it for vector bundles, then illustrate parameter dependence with a family of metrics and give two further calculations.

For a vector bundle $F \to N$, the jet tower has exact sequences

$$0 \longrightarrow \operatorname{Sym}^k(T^*N) \otimes F \longrightarrow J^k_N(F) \longrightarrow J^{k-1}_N(F) \longrightarrow 0.$$

The kernel consists of the highest derivatives when all lower derivatives vanish. Connections can split these sequences, but the exact sequences themselves are canonical. This is also the part of the jet data used by the principal symbol.

A smooth family of metrics g_s gives operators $P_s(u) = \Delta_{g_s} u + u^2$ on the fibres. For a test family $T \to S$, the equation is evaluated using the pulled-back metric. The parameter affects the coefficients, while all derivatives of u remain in the fibre directions. The relative solution stack is defined before any assertion about its projection to S being a submersion or being representable.

Exercise 10.9. Compute the Euler–Lagrange operator for $\int_N (\frac{1}{2}|du|^2 + \frac{1}{4}u^4 - \frac{1}{2}u^2)$ and its linearization at each constant solution. Check ellipticity.

Exercise 10.10. Let P_1, P_2 be first-order linear operators with a common target. Write the principal symbol of the matching equation. Explain why ellipticity of either operator individually does not imply ellipticity of their difference on the direct sum of the sources.

The section and jet constructions are in [Ste24, Sections 3.2–3.3 and Definitions 3.4.1–3.4.2]. The remainder of the seminar proves the representability statement for the elliptic problems defined here.

11 Elliptic representability I: local reductions

We now begin the proof that an elliptic equation has a finite-dimensional derived moduli object. The preceding chapter constructed its solution stack; we must now find open neighbourhoods represented by derived zero loci. The proof occupies three chapters, beginning here with a coordinate reduction of the nonlinear equation.

Near a chosen solution, relative exponential coordinates replace the unknowns and outputs by sections of fixed vector bundles. These are coordinates on the full nonlinear problem, not an approximation by its linearization. We show that the resulting zero locus is an open substack of the original solution stack, then extend the construction to families. The problem remains infinite-dimensional at this stage; the next two chapters carry out its finite-dimensional reduction.

11.1 The representability theorem and proof strategy

We state the relative theorem first, since parameters must be retained throughout the proof. The terms in its statement are those introduced in the preceding two chapters.

Theorem 11.1 (Relative elliptic representability)**.** *Let S be a smooth stack, let $M \to S$ be a proper family of manifolds, and let $\mathcal{E}$ be an S-family of elliptic differential moduli problems on M. Then*

$$\mathrm{Sol}(\mathcal{E}) \longrightarrow S$$

is representable by quasi-smooth derived C^∞-schemes locally of finite presentation.

This is [Ste24, Theorem 3.4.3]. Properness makes the fibre manifolds compact. The submersion hypotheses in a differential moduli problem give local coordinates on its section stacks. Ellipticity will provide the finite-dimensional analytic reduction in Chapter 13.

Relative representability means that every base change to a derived test object over S is represented by a derived scheme. When $S = *$, it says that the whole solution stack is a quasi-smooth derived scheme. A nontrivial parameter stack can retain automorphisms in the total moduli stack.

The proof has three steps, with a different output at each stage.

Step	Output
Coordinates	An exact local nonlinear equation between sections of fixed vector bundles (this chapter).
Augmentation	A criterion identifying the solution stack with $Z \times^{\mathrm{der}}_V \{0\}$, where Z is an ordinary finite-dimensional manifold (Chapter 12).
Analysis	A proof of that criterion for arbitrary smooth families of right-hand sides (Chapter 13).

Each local identification concerns the solution functor on derived parameters. In the last step, smoothness of a single ordinary zero set would prove too little: the analytic family criterion is what permits comparison with the derived pullback. The final chapter of the proof assembles these local identifications into relative representability.

11.2 Coordinates on section stacks

We first work over a point. The coordinate neighbourhoods come from local additions on the source and target submersions. Compactness of the domain then turns them into open neighbourhoods of entire sections.

Let N be a closed manifold, let $X, Y \to N$ be submersions, and let

$$p\colon J^k_N(X) \longrightarrow Y$$

define an operator $P\colon \operatorname{Sec}_N(X) \to \operatorname{Sec}_N(Y)$. Choose a section b of Y and a solution u_0 of $P(u) = b$. The source and target need not be vector bundles.

The bundles of infinitesimal variations at the chosen sections are

$$F = u_0^* T_{X/N}, \qquad E = b^* T_{Y/N}.$$

Both are finite-rank vector bundles on N. The reduction will produce an open substack $Q \subseteq \operatorname{Sec}_N(F)$ containing its zero section and an operator

$$\widetilde{P}\colon Q \longrightarrow \operatorname{Sec}_N(E), \qquad \widetilde{P}(0) = 0,$$

whose derived zero locus is an open neighbourhood of u_0 in $\operatorname{Sol}(P; b)$. It is still an infinite-dimensional section problem. The gain is that unknowns and outputs now admit vector addition.

For a submersion $q\colon X \to N$, its vertical tangent bundle is $T_{X/N} = \ker dq$. Choose a Riemannian metric along the fibres. Its fibrewise exponential map is defined near the zero section:

$$\operatorname{Exp}\colon V \subseteq T_{X/N} \longrightarrow X.$$

For a point $x \in X$, the restriction to V_x takes values in the same fibre $X_{q(x)}$ and has derivative the identity at zero.

More precisely, the map

$$(\pi, \operatorname{Exp})\colon V \longrightarrow X \times_N X$$

can be restricted to a neighbourhood on which it is a diffeomorphism onto a neighbourhood of the relative diagonal. Such a map is called a *local addition*. The existence of local additions is a relative form of the tubular neighbourhood construction.

Pulling back along u_0 gives

$$e_X\colon V_F \subseteq F \longrightarrow X$$

over N, carrying the zero section to u_0 and identifying V_F with an open neighbourhood of its image. Similarly,

$$e_Y\colon V_E \subseteq E \longrightarrow Y$$

is an open neighbourhood of b. These are the relative exponential coordinates used in the proof.

For the special case $X = N \times Z$, a section is a map $u_0 \colon N \to Z$. The first coordinate map has the concrete formula

$$v(x) \longmapsto \exp_{u_0(x)} v(x), \qquad v \in \Gamma(N; u_0^* TZ)$$

sufficiently small. It converts a nearby map into a vector field along u_0, without choosing a coordinate chart of Z containing the whole image of u_0.

An open embedding $V_F \to X$ induces a map of section stacks $\mathrm{Sec}_N(V_F) \to \mathrm{Sec}_N(X)$. For compact N this is an open inclusion [Ste24, Proposition 3.2.19]. On ordinary sections, the condition is that the image of a section lie in V_F.

Compactness explains the openness. If a family u_t lies in V_F at t_0, each point of N has a neighbourhood on which that remains true for nearby t. A finite subcover gives one parameter neighbourhood working on all of N. The source proposition establishes this assertion for section stacks, including derived parameters.

Finite-order operators require control of jets as well as values. The coordinate changes induce maps on k-jets. Choose an open neighbourhood $W \subseteq J^k_N(F)$ of the jet of the zero section on which the transformed output lies in V_E. Define

$$Q = \{v \in \mathrm{Sec}_N(F) : j^k v \text{ factors through } W\}.$$

Formally this is the pullback of the open inclusion $\mathrm{Sec}_N(W) \to \mathrm{Sec}_N(J^k_N(F))$ along jet prolongation. Thus Q is open. Its ordinary sections form an open set in the Whitney C^k topology.

11.3 The equation in local coordinates

The coordinate maps give a new finite-order operator. To use it in the representability proof, we check both that ellipticity is preserved and that its derived zero locus is the required open substack. The same construction then treats a matching problem.

On Q, set

$$\widetilde{P}(v) = e_Y^{-1}\big(P(e_X(v))\big). \tag{11.1}$$

The notation means composition on sections. It is defined because Q was chosen so that the output lies in the target chart. At $v = 0$ its value is zero.

This is again a finite-order operator. The change e_X depends only on the value of the section, so its k-jet depends on derivatives of v through order k. Postcomposing with e_Y^{-1} also uses no additional derivatives. We therefore obtain a smooth jet map $\widetilde{p} \colon W \to E$ inducing Equation (11.1).

Ellipticity is preserved. At a section, the source and target coordinate changes induce vector-bundle isomorphisms on variations. The highest-order symbol of the transformed operator is the original symbol composed with these isomorphisms. Derivatives of their coefficients contribute only lower-order terms. Hence invertibility of the principal symbol is unchanged.

The derived solution stacks also agree on this neighbourhood.

Lemma 11.2 (Open restriction of a pullback)**.** *Suppose a map between two cospans of derived stacks has open inclusions on all three objects. The induced map between their pullbacks is an open inclusion.*

This is the open-inclusion case of [Ste24, Lemma 3.4.4]. It follows by expressing the induced map as pullbacks and composites of open inclusions. The target inclusion is a monomorphism, so a matching homotopy between points in the restricted target lifts uniquely.

Apply the lemma to the source and target coordinate neighbourhoods and to the chosen right-hand side. First compute the pullback with target $\mathrm{Sec}_N(V_E)$. Its inclusion into $\mathrm{Sec}_N(E)$ is open, and both the transformed operator and zero section factor through it, so computing the pullback in the larger target gives the same result.

Proposition 11.3 (Local vector-bundle reduction)**.** *There is an open inclusion*

$$\mathrm{Sol}(\widetilde{P}; 0) \longrightarrow \mathrm{Sol}(P; b)$$

whose image contains u_0. It identifies the first stack with an open neighbourhood of that solution.

This statement concerns the full functors on derived parameters. An equality of ordinary zero sets would not be sufficient for the representability proof. The open-cospan lemma is what promotes the coordinate construction to the asserted derived neighbourhood.

Now start with $P_1(u_1) = P_2(u_2)$ and a chosen solution $(u_{1,0}, u_{2,0})$ with common output w_0. Choose relative exponential charts around all three sections. They give fixed bundles

$$F_1 = u_{1,0}^* T_{Y_1/N}, \quad F_2 = u_{2,0}^* T_{Y_2/N}, \quad E = w_0^* T_{X/N}.$$

Shrink jet neighbourhoods so both transformed outputs lie in the same target chart.

The two transformed operators can now be subtracted:

$$\widetilde{P}(v_1, v_2) = \widetilde{P}_1(v_1) - \widetilde{P}_2(v_2).$$

Its source is an open substack of $\mathrm{Sec}_N(F_1 \oplus F_2)$. Its zero locus gives an open part of the original matching problem. The reason is the identity

$$a = b \quad \Longleftrightarrow \quad a - b = 0$$

in a vector bundle, interpreted as an identity of pullback diagrams. More categorically, the diagonal of a vector bundle is the pullback of its zero section along the difference map. Pasting these pullbacks proves the derived assertion.

The principal symbol is precisely that of the linearized difference in the definition of an elliptic differential moduli problem. This is the point at which the matching equation becomes the single vector-bundle equation used in the remaining proof.

11.4 Open covers and parameter families

The local construction must cover all solutions and remain available when the equation varies. We first check the covering property over a point, then describe the additional trivializations needed over a base.

Carry out the construction at every real solution t. It gives open substacks $\mathrm{Sol}(\mathcal{E})_t$ and a map

$$\coprod_t \mathrm{Sol}(\mathcal{E})_t \longrightarrow \mathrm{Sol}(\mathcal{E}).$$

This is an effective epimorphism. Indeed, pull it back along a test family $T \to \mathrm{Sol}(\mathcal{E})$. The resulting open subsets of T contain every real point, since the chart centred at its image contains that solution. They therefore cover T.

The open local representability theorem of Chapter 9 now reduces representability to the individual charts. A local quasi-smooth representing object for each $\mathrm{Sol}(\mathcal{E})_t$ will yield a global quasi-smooth derived scheme. The original solution stack supplies the overlaps and their identifications.

For a proper submersion $M \to S$ of ordinary manifolds, Ehresmann's theorem gives local trivializations

$$M|_{S_0} \cong S_0 \times N$$

with N compact. The fibre N is fixed in this local discussion, while the coefficients of the equation vary with $s \in S_0$.

Choose a solution over s_0. The family version of the relative exponential construction gives vector-bundle neighbourhoods over $S_0 \times N$ arising from fixed bundles on N, after shrinking S_0. This is [Ste24, Lemma 3.2.20]. The bundles on N need not be trivial. It is their variation with the parameter that has been locally trivialized.

For a matching problem, these choices reduce the equation to

$$\widetilde{P}\colon S_0 \times Q \longrightarrow S_0 \times \mathrm{Sec}_N(E), \qquad (s, v) \longmapsto (s, \widetilde{P}_s(v)),$$

with $Q \subseteq \mathrm{Sec}_N(F)$ open and F, E fixed bundles on N. The chosen solution is $(s_0, 0)$ and $\widetilde{P}_{s_0}(0) = 0$. The zero section need not solve the equation for every other s: the coordinate construction extends the ambient charts, not the chosen solution as a family of solutions.

The section-stack product identifications used here are [Ste24, Lemma 3.2.21]. Compatibility of jets with base change ensures that the transformed operator remains a relative finite-order operator. The same open-cospan argument gives an open neighbourhood in the relative solution stack.

For a smooth stack base, these coordinate reductions are made after pullback to manifold families. Their use in the global proof will be explained after the analytic criterion has been established in Chapter 13.

11.5 Further reading

Maps to a circle give an explicit version of the exponential-coordinate construction. The exercises check how these coordinates affect derivatives and why the section neighbourhoods are open.

Example 11.4 (Maps to a circle)**.** Let N be compact and consider maps $u\colon N \to S^1$. Fix u_0. For a sufficiently small real function v,

$$u_v(x) = u_0(x)\exp(iv(x))$$

defines a nearby map. Choosing $|v(x)| < \pi$ gives a unique local logarithm of u/u_0. Thus a neighbourhood of u_0 in its mapping stack is an open part of the section stack of a real line bundle (in this example the bundle is trivial).

For an operator of order two on these maps, substitute u_v and differentiate. First and second derivatives of u_v involve only v, dv, d^2v, with coefficients determined by u_0. The transformed operator still has order at most two. The same reasoning works for a general target manifold using the exponential map and the bundle u_0^*TZ.

This example also shows the locality of the construction. A single logarithm does not parametrize every map $N \to S^1$. The representability argument needs one neighbourhood at a time, and the effective open cover provides the global passage.

Exercise 11.5. In the coordinate formula above, calculate du_v and show directly that its leading term in a variation of v is multiplication by a pointwise invertible map followed by dv.

Exercise 11.6. Let $p\colon W \subseteq J^k_N(F) \to E$ be a partially defined operator. Describe the open set of sections on which it is defined, and explain where compactness of N enters the openness assertion.

The reductions in this chapter follow the first part of the proof of [Ste24, Theorem 3.4.3, pp. 66–67], using Proposition 3.2.19, Lemmas 3.2.20–3.2.21, Proposition 3.3.9, and Lemma 3.4.4.

12 Elliptic representability II: the local Kuranishi criterion

After the local reduction, the equation has the form

$$\widetilde{P}\colon Q \subseteq \mathrm{Sec}_N(F) \longrightarrow \mathrm{Sec}_N(E), \qquad \widetilde{P}(u) = 0,$$

for vector bundles on a closed manifold N. We will add finitely many parameters so that the augmented equation has an ordinary manifold of solutions. Setting the added parameters back to zero will then give a finite-dimensional derived zero locus.

The pullback identity is formal, but identifying the augmented derived zero locus with an ordinary manifold requires more than a description of its real solutions. We formulate the needed condition as submersiveness in smooth stacks: every smooth family of right-hand sides must have a finite-dimensional solution manifold submersive over its parameter manifold.

We first establish the augmentation identity, then explain how smooth families of sections connect the analytic setting to section stacks. The comparison theorem for submersions gives a local Kuranishi criterion, including its relative form. The next chapter will choose the auxiliary variables and prove the family condition by elliptic analysis.

12.1 Adding auxiliary variables

Adding a finite-dimensional space of parameters turns the original problem into a fibre of an augmented problem. We first establish this identity globally and illustrate it with x^2, then restrict it to the neighbourhoods needed in the analytic argument.

Let V be a finite-dimensional vector space and choose a linear map

$$\iota\colon V \longrightarrow \Gamma(N; E).$$

It gives a map $V \to \mathrm{Sec}_N(E)$: a family of vectors is sent to the corresponding linear combination of fixed smooth sections. Define

$$\overline{P}\colon Q \times V \longrightarrow \mathrm{Sec}_N(E), \qquad \overline{P}(u, a) = \widetilde{P}(u) + \iota(a).$$

Its solution stack has a projection $\pi\colon \mathrm{Sol}(\overline{P}) \to V$.

Proposition 12.1 (The augmentation identity)**.** *The following square is a pullback in derived stacks:*

$$\begin{array}{ccc} \mathrm{Sol}(\widetilde{P}) & \longrightarrow & \mathrm{Sol}(\overline{P}) \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ \{0\} & \longrightarrow & V. \end{array}$$

Proof. The pullback of $Q \times V \to V$ along the origin is Q. On this pullback, $\overline{P}$ restricts to $\widetilde{P}$ because $\iota(0) = 0$. Pulling back the equation $\overline{P} = 0$ along the same map therefore imposes $\widetilde{P} = 0$. Associativity of pullbacks gives the displayed isomorphism. □

This identity holds for every choice of augmentation, regardless of whether $\mathrm{Sol}(\overline{P})$ is represented by a manifold.

If it is represented by an ordinary finite-dimensional manifold Z, the projection becomes a smooth map $\kappa\colon Z \to V$, and

$$\mathrm{Sol}(\widetilde{P}) \cong Z \times_V^{\mathrm{der}} \{0\}.$$

The right side is the Kuranishi presentation $(Z, Z \times V, \kappa)$. The remaining equation $\kappa = 0$ need not be transverse. It retains the singularities of the original problem.

For example, augment $f(x) = x^2$ to

$$\overline{f}(x, a) = x^2 + a\colon \mathbb{R}^2 \longrightarrow \mathbb{R}.$$

Its derivative $(v, b) \mapsto 2xv + b$ is surjective everywhere. Its derived zero locus is therefore the ordinary parabola $Z = \{a = -x^2\}$. Using x as a coordinate on Z, the projection to the auxiliary line is $\kappa(x) = -x^2$. Its derived zero fibre is the multiple point with function ring $\mathbb{R}[\delta]/(\delta^2)$ calculated earlier. The singularity is retained by the projection κ from the smooth augmented zero manifold.

In practice we work on an open neighbourhood $U \subseteq Q \times V$ of a point $(t, 0)$. Set

$$H = \{u \in Q : (u, 0) \in U\}.$$

This is an open substack of Q. Let $\mathcal{Z}_U = \mathrm{Sol}(\overline{P}|_U)$, with its projection to V. The same argument gives

$$\mathrm{Sol}(\widetilde{P}|_H) \cong \mathcal{Z}_U \times_V^{\mathrm{der}} \{0\}. \tag{12.1}$$

If $\mathcal{Z}_U$ is represented by a finite-dimensional manifold Z_U, this is a Kuranishi chart around t. Using H records exactly which part of the original equation lies inside the chosen augmented neighbourhood.

12.2 From smooth section spaces to section stacks

The local operators can be studied analytically on Fréchet spaces of sections. To use their analytic properties in derived geometry, we need to identify the associated functors of smooth families and then their extensions to derived parameters.

For a finite-rank bundle E over compact N, the space $\Gamma(N; E)$ has seminorms measuring all derivatives:

$$p_r(u) = \max_{0 \le j \le r} \sup_{x \in N} |\nabla^j u(x)|.$$

They give a complete metrizable locally convex topology, hence a Fréchet space. Choices of connection and metrics give the same topology. There is generally no single norm controlling all derivatives.

Fréchet spaces are examples of *convenient vector spaces*. One characterization of convenience is that every smooth curve admits a smooth antiderivative. Smooth maps in the convenient calculus can be tested on smooth curves. For Fréchet spaces this agrees with the usual smooth calculus based on jointly continuous iterated directional derivatives.

A convenient manifold is locally modelled on suitable open subsets of convenient vector spaces, with smooth transition maps. Only Fréchet section spaces, their open subsets, and Banach Sobolev manifolds will occur below.

A convenient manifold B determines a smooth stack

$$j_{\mathrm{Con}}(B)(T) = C^\infty(T, B), \qquad T \in \mathrm{Mfd}.$$

This is the restricted Yoneda functor. It is fully faithful and preserves all limits that exist in convenient manifolds [Ste24, Proposition 3.1.7]. These limits are preserved in *smooth stacks*. After further inclusion into derived stacks, arbitrary pullbacks need not be preserved.

The exponential law says that a smooth map $T \to \Gamma(N; E)$ is exactly a jointly smooth section on $T \times N$. Consequently,

$$j_{\mathrm{Con}}\Gamma(N; E) \cong \mathrm{Sec}_N(E)_{\mathrm{Sm}}.$$

This identifies the analytic notion of a smooth family of sections with the smooth mapping stack [Ste24, Proposition 3.1.9].

We also need the identification on derived parameters:

$$j_! j_{\mathrm{Con}}\Gamma(N; E) \cong \mathrm{Sec}_N(E). \tag{12.2}$$

It uses the additional theorem that this derived section stack is smooth [Ste24, Lemma 3.2.11]. The ordinary exponential law alone would not imply that assertion.

The same comparison applies to the open sets of sections produced in Chapter 11. Thus Q, U, and their operators can be discussed both as convenient manifolds and as smooth stacks, with the derived incarnations specified by Equation (12.2).

12.3 Submersions and the comparison of pullbacks

The inclusion of smooth stacks into derived stacks does not preserve every pullback. Submersiveness gives a sufficient condition. Its definition is phrased in terms of families so that the analytic argument can verify it using ordinary parameter manifolds.

For ordinary manifolds, a smooth map is a submersion if its differential is surjective everywhere. The implicit function theorem identifies it locally with a projection. Its pullback along any smooth map from another manifold is again a manifold, and the resulting projection is a submersion.

Definition 12.2 (Submersiveness in smooth stacks)**.** A morphism $\mathcal{A} \to \mathcal{B}$ of smooth stacks is submersive if, for every ordinary manifold T and map $T \to \mathcal{B}$, the pullback

$$T \times^{\mathrm{Sm}}_{\mathcal{B}} \mathcal{A}$$

is represented by an ordinary finite-dimensional manifold whose map to T is a submersion.

The superscript indicates that the pullback is computed in smooth stacks. It will be useful when comparing categories. A submersion need not be surjective; the definition permits an empty fibre. The finite-dimensionality requirement is substantial when $\mathcal{A}$ and $\mathcal{B}$ arise from infinite-dimensional manifolds.

There is an analogous definition for derived stacks. After base change to a derived affine test T, a submersive morphism must be represented by a derived submersion over T, locally a projection

with a finite-dimensional smooth manifold factor. The following comparison makes it possible to test this property using ordinary parameters.

Theorem 12.3 (Comparison along submersions)**.** *A morphism of smooth stacks is submersive in smooth stacks if and only if it is submersive after inclusion into derived smooth stacks. If $\mathcal{A} \to \mathcal{B}$ is such a morphism and $C \to \mathcal{B}$ is a map of smooth stacks, then*

$$j_!\big(\mathcal{A} \times^{\mathrm{Sm}}_{\mathcal{B}} C\big) \cong j_!\mathcal{A} \times^{\mathrm{der}}_{j_!\mathcal{B}} j_!C.$$

These are the relevant consequences of [Ste24, Propositions 2.2.17–2.2.18 and Corollary 2.2.19]. Locally a submersion is a projection, and inclusion of smooth stacks preserves products. Open descent then permits these local comparisons to be glued. The cited propositions carry this reasoning out for stack bases and verify the test-object criterion.

In particular, the derived fibre of a submersive map over an ordinary point is the ordinary smooth fibre. This is the mechanism that will make the augmented zero locus finite-dimensional and ordinary. The final fibre over $0 \in V$ is a different pullback, for which submersiveness of κ is not assumed.

12.4 The local Kuranishi criterion

The section-stack identification and the comparison along submersions now combine with the augmentation identity. We state the criterion over a point, spell out the analytic condition it requires, and then retain a parameter base in the same construction.

Consider the convenient smooth map

$$G = \overline{P}|_U \colon U \longrightarrow \Gamma(N; E).$$

Via the preceding identifications it induces the augmented operator on section stacks.

Theorem 12.4 (Local Kuranishi criterion)**.** *Suppose that the morphism*

$$j_{\mathrm{Con}}G \colon j_{\mathrm{Con}}U \longrightarrow j_{\mathrm{Con}}\Gamma(N; E)$$

is submersive as a map of smooth stacks. Then its derived zero fibre is represented by an ordinary finite-dimensional manifold Z_U. Projection onto V defines a Kuranishi map $\kappa \colon Z_U \to V$, and

$$\mathrm{Sol}(\widetilde{P}|_H) \cong Z_U \times^{\mathrm{der}}_V \{0\}.$$

Proof. The smooth-stack pullback along $0 \colon * \to \Gamma(N; E)$ is represented by an ordinary finite-dimensional manifold by the definition of submersiveness. The comparison theorem identifies its image in derived stacks with the derived zero fibre of G. Now apply Equation (12.1). □

This isolates the sufficient condition to be proved by analysis. Explicitly, for every ordinary manifold T and smooth family $h \colon T \to \Gamma(N; E)$, the smooth-stack pullback

$$T \times^{\mathrm{Sm}}_{\Gamma(N;E)} U$$

must be a finite-dimensional manifold submersive over T. Its equation is

$$\widetilde{P}(u) + \iota(a) = h(t).$$

Checking only $h = 0$ establishes one fibre. The universal condition on T, h is what proves submersiveness and permits the comparison with derived pullbacks.

It is also stronger than surjectivity of the derivative as a map of Fréchet spaces. General Fréchet calculus has no implicit function theorem with that hypothesis alone. The next chapter uses Sobolev completions and elliptic regularity to prove the stated family condition.

Let S now be an ordinary manifold, with local operator $\widetilde{P}_s$ as in Chapter 11. Take an open neighbourhood $U \subseteq S \times Q \times V$ and form the map over S

$$G(s, u, a) = \big(s, \widetilde{P}_s(u) + \iota(a)\big) \colon U \longrightarrow S \times \Gamma(N; E).$$

Suppose it is submersive as a map of smooth stacks. Its pullback along the zero section $S \to S \times \Gamma(N; E)$ is an ordinary manifold Z_U submersive over S. The auxiliary-variable projection gives a map over S

$$\kappa \colon Z_U \longrightarrow S \times V.$$

The local original solution stack is

$$Z_U \times^{\mathrm{der}}_{S \times V} S.$$

This is relatively quasi-smooth over S. Indeed, $Z_U \to S$ is a submersion, so its relative cotangent bundle is in degree zero. Imposing the V-valued equation gives a two-term relative cotangent complex in degrees $1, 0$. It is therefore a relative Kuranishi presentation.

The resulting zero locus identifies an open substack of the original solution stack. This is the local output needed for globalization. It remains conditional on submersiveness of the augmented operator; the next chapter proves that condition and then assembles the open atlas.

The pullback argument is the final part of the proof of [Ste24, Theorem 3.4.3, following claim $(*)$]. At this point the remaining task is to choose V, ι, U for which the submersiveness hypothesis holds.

12.5 Further reading

Two comparisons clarify the criterion: the weaker notion of stacky submersiveness explains why finite-dimensional fibres must be required, and a linear example isolates the role of the auxiliary variables.

Steffens also uses the weaker term *stacky submersive*. Its local models, after stacky étale changes, are projections away from an arbitrary smooth stack, rather than necessarily a finite-dimensional manifold [Ste24, Definition 2.2.16]. His pullback-preservation theorem holds in that greater generality. For the Kuranishi criterion, however, we need the representable submersive condition, since the augmented fibre must be an ordinary finite-dimensional manifold.

The distinction is visible in the projection $j_{\mathrm{Con}}\Gamma(N; E) \to *$. It is stacky submersive, as a projection away from a smooth stack, but an infinite-dimensional section space is not a finite-dimensional manifold. Stacky submersiveness alone cannot give the required Kuranishi chart.

Example 12.5 (The finite-dimensional linear analogue)**.** The augmentation construction has a finite-dimensional analogue. For $f \colon A \to B$ between vector spaces, choose $\iota \colon V \to B$ surjecting onto coker f. Then

$$f + \iota \colon A \oplus V \longrightarrow B$$

is surjective. Its kernel Z is a vector space, and the derived zero locus of $Z \to V$ recovers the derived zero fibre of f. The elliptic argument is a nonlinear version of this construction, with analysis needed to justify the ordinary manifold Z.

Exercise 12.6. For a linear map $f\colon A \to B$ and the surjective augmentation above, prove that the tangent complex $[Z \to V]$ is quasi-isomorphic to $[A \to B]$ in degrees $0, -1$.

Exercise 12.7. Explain why the pullback of $x^2\colon \mathbb{R} \to \mathbb{R}$ along zero in smooth stacks is different from its pullback in derived stacks. Locate exactly which hypothesis of the comparison theorem fails.

13 Elliptic representability III: the analytic input

The local Kuranishi criterion reduces representability to an analytic claim: after adding finitely many auxiliary variables, the equation must be submersive as a map of smooth stacks. We now choose those variables from the cokernel of the linearized operator and prove the required condition for every smooth family of right-hand sides.

The analytic argument uses three results. Elliptic operators on a closed manifold are Fredholm on suitable Sobolev spaces. The Banach implicit function theorem turns a surjective Fredholm derivative into a local finite-dimensional family of solutions. Elliptic regularity identifies these Sobolev solutions, and their smooth family structure, with the original smooth-section problem.

Once the analytic condition is established, the augmentation identity gives a derived chart and identifies its tangent complex. We then retain parameters and glue the charts to complete the relative representability theorem. The final calculation returns to $\Delta u + u^2 = 0$ and exhibits the multiple point that its solution stack contains near zero.

13.1 Sobolev spaces and the loss of derivatives

Fix a closed manifold N and a local equation

$$\widetilde{P}\colon Q \subseteq \Gamma(N;F) \longrightarrow \Gamma(N;E)$$

of order k, with a chosen solution $t \in Q$. The bundles F, E and the open set Q come from Chapter 11. We seek a finite-dimensional space V, a map $\iota\colon V \to \Gamma(N;E)$, and a neighbourhood U of $(t,0)$ on which $G(u,a) = \widetilde{P}(u) + \iota(a)$ is submersive. Sobolev completions provide the Banach spaces on which we will first study this augmented operator.

Choose metrics and a connection on a vector bundle E over N. For a nonnegative integer ℓ, the Sobolev norm is equivalent to

$$\|u\|_{H^\ell}^2 = \sum_{j=0}^{\ell} \int_N |\nabla^j u|^2 \, d\mathrm{vol}.$$

Completing smooth sections gives the Hilbert space $H^\ell(E;N)$. Its elements have weak derivatives through order ℓ in L^2; they need not be smooth. Different geometric choices give equivalent norms.

The Sobolev embedding theorem implies

$$H^{k+\ell}(F;N) \longrightarrow C^k(F;N) \quad \text{if } \ell > \tfrac{1}{2}\dim N.$$

Thus, at these levels, a section has a well-defined continuous k-jet. If Q was defined by requiring the jet to lie in an open $W \subseteq J^k_N(F)$, the same requirement defines an open subset $Q_{k+\ell} \subseteq H^{k+\ell}(F;N)$.

A differential operator of order k uses k derivatives. Its Sobolev extension therefore has the form

$$\widetilde{P}_\ell \colon Q_{k+\ell} \longrightarrow H^\ell(E;N). \tag{13.1}$$

For $\ell > \dim N/2$, multiplication and smooth substitution are continuous operations on H^ℓ. Applied to the finite jet formula, these results make Equation (13.1) a smooth map of Banach manifolds. We use the compatible construction in [Ste24, Lemma 3.4.5].

The extensions commute with the inclusions between Sobolev levels. Moreover,

$$\Gamma(N;E) \cong \lim_\ell H^\ell(E;N)$$

as Fréchet spaces, with the smooth-section topology. The corresponding statement holds for Q and its Sobolev domains. It is a limit statement in convenient manifolds and hence in smooth stacks after applying j_{Con}. It is not asserted to be a limit statement after passage to derived stacks.

13.2 Choosing the auxiliary variables by elliptic theory

Ellipticity makes the missing directions in the image of the derivative finite-dimensional. Linear regularity supplies smooth representatives of those directions, so a single augmentation works at every Sobolev level.

Definition 13.1. A bounded linear map between Banach spaces is Fredholm if its kernel and cokernel are finite-dimensional and its image is closed. Its index is $\dim\ker D - \dim\operatorname{coker} D$.

Theorem 13.2 (Linear elliptic theory)**.** *A smooth linear elliptic differential operator of order k on a closed manifold extends to Fredholm operators*

$$D_\ell \colon H^{k+\ell}(F;N) \longrightarrow H^\ell(E;N).$$

Its kernel consists of smooth sections and is independent of ℓ. Its cokernel is naturally identified with $\Gamma(N;E)/D\Gamma(N;F)$, also independently of ℓ. If v is a Sobolev section and Dv is smooth, then v is smooth.

We quote this standard analytic input, used in the proof of Steffens's theorem. A basic estimate underlying it is

$$\|v\|_{H^{k+\ell}} \le C_\ell\big(\|Dv\|_{H^\ell} + \|v\|_{L^2}\big).$$

Ellipticity controls the highest derivatives through Dv. The lower-order term accommodates the kernel. A parametrix and compactness of the lower Sobolev inclusions give Fredholmness; regularity gives smoothness of kernel elements. Applying the same reasoning to the formal adjoint identifies the cokernel by smooth representatives.

At the chosen solution, put $D = T_t\widetilde{P}$ and

$$V = \operatorname{coker} D = \Gamma(N;E)/D\Gamma(N;F).$$

It is finite-dimensional. Choose a linear splitting

$$\iota \colon V \longrightarrow \Gamma(N;E)$$

of the quotient map. Concretely, choose a basis of the quotient and one smooth section representing each basis vector, and extend linearly. The phrase smooth representatives refers to these sections on N. A map from finite-dimensional V obtained in this way is also smooth as a map to the Fréchet section space.

The augmented derivative is

$$D + \iota \colon \Gamma(N; F) \oplus V \longrightarrow \Gamma(N; E).$$

It is surjective: the image of ι supplies every missing cokernel class. The same is true at every Sobolev level by the linear regularity theorem. Its kernel is finite-dimensional.

For nearby sections we restrict the jet neighbourhood so the principal symbol remains elliptic. This is possible because invertibility is open and the cosphere bundle of N is compact. It also keeps the Fredholm index fixed near the chosen solution.

13.3 The augmented operator on Sobolev spaces

The augmented derivative is surjective Fredholm at the chosen solution. The Banach implicit function theorem describes its nearby fibres. We must then choose the neighbourhoods compatibly, so that they return to a single open set of smooth sections.

The version needed here follows from the Banach inverse function theorem.

Theorem 13.3 (Implicit function theorem)**.** *Let $g \colon B \supseteq O \to C$ be smooth between Banach spaces. Suppose $d_b g$ is surjective and its kernel K has a closed complement. Then near b, there are smooth coordinates in which g is the projection from an open subset of $K \times C$ to C.*

Choose $B = K \oplus B'$ so that $d_b g|_{B'}$ is an isomorphism. The bounded inverse theorem makes its inverse bounded. The map

$$x \longmapsto \big(\mathrm{pr}_K(x - b), g(x)\big)$$

has an invertible derivative at b, so the inverse function theorem gives the required coordinates. For a surjective Fredholm operator, the finite-dimensional kernel has a closed complement. In the Hilbert spaces used here one can take its orthogonal complement.

Consequently, a smooth map with everywhere surjective Fredholm derivative has local finite-dimensional fibres. More generally, pull it back along a smooth map from a finite-dimensional manifold Z. In the local coordinates above, the pullback is an open subset of $K \times Z$, and its projection to Z is a submersion. This family statement is exactly the form required in Chapter 12.

For each sufficiently large ℓ, define

$$G_\ell(u, a) = \widetilde{P}_\ell(u) + \iota(a).$$

Its derivative at $(t, 0)$ is surjective Fredholm. Surjectivity of a split surjection is open in the operator norm, and Fredholmness is also open. We can therefore restrict to an open neighbourhood $U_\ell \subseteq Q_{k+\ell} \times V$ where dG_ℓ is everywhere surjective Fredholm.

The neighbourhoods must be chosen compatibly across Sobolev levels. At a smooth section, linear elliptic regularity identifies the kernel and cokernel of the derivative at every level. Thus the condition that dG_ℓ be onto has the same restriction to smooth sections for every ℓ. Using

these surjectivity loci, and successively intersecting higher-level neighbourhoods with the inverse images of lower-level ones, gives

$$U_{\ell+1} \longrightarrow U_\ell, \qquad U = U_\ell \cap (Q \times V)$$

with the same smooth open neighbourhood U at every level. This is the neighbourhood construction in the proof of [Ste24, Theorem 3.4.3, p. 69]. The inverse limit of the resulting domains is U in convenient manifolds.

The derivative is independent of the auxiliary variable a, since ι is linear. At the chosen section it has the form $D + \iota$. The restriction is needed because the derivative in the section variable changes with u.

13.4 Testing submersiveness on smooth families

We now verify the family condition from the preceding chapter. At each Sobolev level, the implicit function theorem gives a finite-dimensional pullback. Nonlinear regularity identifies its points with smooth solutions, and linear regularity identifies its smooth structure.

Let Z be an ordinary finite-dimensional manifold and let $h\colon Z \to \Gamma(N; E)$ be smooth. At Sobolev level ℓ, consider

$$W_\ell = \{(z, u, a) \in Z \times U_\ell : \widetilde{P}_\ell(u) + \iota(a) = h(z)\}. \tag{13.2}$$

The Banach implicit function theorem makes W_ℓ an ordinary finite-dimensional manifold with a submersion $W_\ell \to Z$. The dimension of a fibre is the Fredholm index of G_ℓ, namely $\operatorname{ind} D + \dim V$.

It remains to identify these manifolds with the smooth-stack pullback.

We use the following nonlinear regularity statement from [Ste24, Lemma 3.4.6]: above a sufficiently high Sobolev threshold, a solution of the nonlinear elliptic equation with smooth right-hand side is smooth. The threshold includes $\ell > \dim N/2$ and the additional regularity needed for the nonlinear elliptic bootstrap. No optimal numerical threshold is required for the argument.

At a point of W_ℓ the equation reads

$$\widetilde{P}_\ell(u) = h(z) - \iota(a).$$

The right side is a smooth section. For ℓ above the regularity threshold, u is therefore smooth. The common smooth trace of the U_ℓ then shows that all sufficiently high W_ℓ have the same underlying set. Their transition maps are bijections.

This only compares points. A bijective smooth map between manifolds need not be a diffeomorphism, so one further step is necessary.

At a smooth solution (z, u, a), the tangent space along the fibre of $W_\ell \to Z$ is

$$\ker\bigl(T_u\widetilde{P}_\ell + \iota\bigr).$$

If (v, b) lies in this kernel, then $T_u\widetilde{P}_\ell(v) = -\iota(b)$ is smooth. Linear elliptic regularity makes v smooth. It follows that the inclusions between Sobolev levels induce isomorphisms on these vertical tangent spaces.

Each $W_\ell \to Z$ is a submersion, and the transition maps are over Z. The short exact sequences of tangent spaces therefore show that $W_{\ell+1} \to W_\ell$ induces an isomorphism on the full tangent space.

The finite-dimensional inverse function theorem makes it a local diffeomorphism. Combined with bijectivity, this makes it a diffeomorphism.

Thus the tower is eventually constant as a tower of manifolds. Write W for its limiting manifold; $W \to Z$ is a submersion.

The Sobolev tower for smooth sections is a limit in convenient manifolds, and j_{Con} preserves such limits in smooth stacks. Since limits commute with pullbacks, the preceding identification gives

$$j_{\mathrm{Con}} W \cong Z \times^{\mathrm{Sm}}_{j_{\mathrm{Con}}\Gamma(N;E)} j_{\mathrm{Con}} U.$$

This identifies the entire smooth family functor, including its smooth dependence on parameters.

The test manifold Z and map h were arbitrary. By the criterion of Chapter 12, the augmented map $j_{\mathrm{Con}} G$ is submersive. The comparison theorem then makes it submersive in derived stacks too. This proves claim $(*)$ in Steffens's argument.

13.5 The derived chart and its tangent complex

The derived zero fibre of G is now an ordinary finite-dimensional manifold Z_U. Its projection onto V gives $\kappa \colon Z_U \to V$, and the augmentation identity identifies

$$Z_U \times^{\mathrm{der}}_{V} \{0\}$$

with an open part of $\mathrm{Sol}(\widetilde{P})$.

The tangent calculation also follows directly from the augmentation. At a zero $(u, 0)$, put

$$K = \ker(T_u \widetilde{P} + \iota) = T_{(u,0)} Z_U.$$

The derivative of κ is projection $K \to V$. There is a chain map

$$[K \longrightarrow V] \longrightarrow [\Gamma(N;F) \xrightarrow{T_u \widetilde{P}} \Gamma(N;E)]$$

given by projection onto $\Gamma(N;F)$ in degree zero and $-\iota$ in degree minus one. It is a chain map because $T_u \widetilde{P}(v) = -\iota(a)$ for $(v, a) \in K$. Surjectivity of the augmented derivative makes this a quasi-isomorphism: its degree-zero homology is the original kernel, and its degree-minus-one homology is the original cokernel.

Thus the finite-dimensional derived tangent complex is represented by the linearized elliptic equation. Its virtual dimension is

$$\dim Z_U - \dim V = \mathrm{ind}(T_u \widetilde{P}).$$

At the centre t, our choice $V = \operatorname{coker} D$ implies $K \cong \ker D$ and $d\kappa = 0$.

13.6 Relative charts and completion of the proof

To finish the theorem, we carry out the same analytic construction with parameters and apply the local representability results. The relative chart is the analytic output; gluing uses the solution stack already defined before making any coordinate choices.

For a manifold base, the locally reduced operator is

$$(s, u) \longmapsto (s, \widetilde{P}_s(u)).$$

At a chosen solution $t = (s_0, u_0)$, use the cokernel of the vertical operator $T_{u_0}\widetilde{P}_{s_0}$. It is also the cokernel of the total derivative: the total derivative has the block form

$$(\dot{s}, v) \longmapsto \big(\dot{s}, \partial_s \widetilde{P}(\dot{s}) + T_{u_0}\widetilde{P}_{s_0}(v)\big),$$

and subtracting the parameter term in the target identifies its kernel and cokernel with those of the vertical operator.

Choose smooth representatives ι at this solution and repeat the Sobolev construction on $S \times Q \times V$. A test family is now a map $Z \to S \times \Gamma(N; E)$, written $(s(z), h(z))$. Its equation is

$$\widetilde{P}_{s(z),\ell}(u) + \iota(a) = h(z).$$

The Banach argument and both regularity arguments work with these additional finite-dimensional parameters. They prove that the augmented operator is submersive over S in the required sense.

Chapter 12 therefore produces relative Kuranishi charts. Chapter 11 showed that their neighbourhoods cover all solutions, and Chapter 9 glues this effective open atlas. For a smooth stack base S, choose an effective covering by ordinary manifolds and pull the differential moduli problem back to them. The argument above proves relative representability on each such family. Locality on the base, proved in Corollary 9.10, descends this conclusion to S. The globally defined solution stack supplies the overlaps and their coherence, so no independent choices of chart transitions are needed. This completes the proof of Theorem 11.1, subject to the analytic and geometric input theorems explicitly quoted above.

13.7 The running example as a multiple point

Return to $P(u) = \Delta u + u^2$ on closed connected N. At zero, the cokernel of Δ is represented by constants. Take $V = \mathbb{R}$ and $\iota(a) = a$. The augmented equation is

$$\Delta u + u^2 + a = 0.$$

Its derivative at $(0, 0)$ is $(v, b) \mapsto \Delta v + b$, which is surjective.

Write $u = c + w$, where c is constant and w has mean zero. Let $\langle g \rangle$ denote the average of a function over N. The mean-zero part of the augmented equation is

$$\Delta w + 2cw + w^2 - \langle w^2 \rangle = 0.$$

On mean-zero functions, Δ is an isomorphism between the relevant Sobolev spaces. The implicit function theorem, with parameter c, gives a unique small solution $w = w(c)$. Since $w = 0$ solves the equation for every c, uniqueness implies $w(c) = 0$.

The remaining average equation is $a = -c^2$. Hence near zero the augmented solution manifold is the curve

$$c \longmapsto (u = c,\ a = -c^2).$$

Its Kuranishi map is $\kappa(c) = -c^2$. Multiplying the target by -1 identifies its derived zero locus with that of c^2.

The local function ring is therefore

$$C^\infty(\mathbb{R})/(c^2) \cong \mathbb{R}[\delta]/(\delta^2).$$

The real solution set is one point, but it has a nontrivial infinitesimal direction. Its tangent complex is $[\mathbb{R} \xrightarrow{0} \mathbb{R}]$ in degrees $0, -1$, agreeing with the kernel and cokernel of Δ. This is the finite-dimensional singularity encoded by the original infinite-dimensional equation.

13.8 Further reading

The construction used auxiliary sections and Sobolev completions. We explain how these choices affect the presentations, and distinguish the representability theorem from the further compactness questions needed for enumerative applications.

The auxiliary space is not unique. Any finite-dimensional collection of smooth sections surjecting onto the cokernel can be used. Adding unnecessary representatives increases the dimension of the augmented zero manifold and the number of auxiliary equations together. The resulting presentations still describe open parts of the same intrinsic solution stack.

A change of Sobolev exponent is also auxiliary. The regularity argument identifies the resulting finite-dimensional manifolds at sufficiently high levels and compares their universal smooth families. This is stronger than observing that their real solution sets agree. It explains why the theorem belongs to the original smooth equation, rather than to a chosen Banach completion.

The analytic input proves representability and quasi-smoothness. Compactness of the domain N does not imply compactness of its solution moduli object. Compactifying that object and constructing enumerative invariants require further geometric arguments.

Exercise 13.4. For a Fredholm map D and a cokernel splitting ι, verify directly that $\ker(D + \iota) \cong \ker D$. Then replace V by a larger auxiliary space and calculate the change in the kernel dimension.

Exercise 13.5. In the running example, replace u^2 by u^3. Use the mean-zero equation to compute the local Kuranishi map. Compare the resulting classical function ring and tangent complex.

The detailed analytic construction is [Ste24, Lemmas 3.4.5–3.4.6 and the proof of Theorem 3.4.3, pp. 67–70]. The distinction between limits in smooth stacks and derived stacks is discussed in its Remark 3.4.7.

14 Pseudo-holomorphic curves as a derived moduli problem

Let (Σ, j) be a closed Riemann surface and (Z, J) an almost complex manifold. A map $u\colon \Sigma \to Z$ is pseudo-holomorphic when its derivative is complex linear. This first-order equation provides an application of the elliptic representability argument of the preceding three chapters.

We first write the equation as a section of its bundle of complex-antilinear forms and put it into the local coordinates required by the representability proof. The main calculation is its principal symbol. Ellipticity then gives a quasi-smooth derived scheme of maps from a fixed domain, with tangent complex given by the linearized Cauchy–Riemann equation. Constant maps provide an explicit example of its deformation and obstruction groups.

Allowing the domain to vary produces a derived moduli stack over the smooth stack of Riemann surfaces. This relative formulation keeps deformations of the map distinct from variations and automorphisms of its domain. We conclude by explaining what further input would be needed for nodal compactifications and enumerative invariants.

14.1 The Cauchy–Riemann equation

The condition that a derivative be complex linear is expressed by the vanishing of its complex-antilinear part. We recall this decomposition and apply it pointwise to a smooth map.

An almost complex structure on a smooth manifold Z is a vector-bundle endomorphism

$$J\colon TZ \longrightarrow TZ$$

satisfying $J^2 = -\operatorname{id}_{TZ}$. Each tangent space T_xZ thereby becomes a complex vector space, with multiplication by i given by J_x. The adjective *almost* records that these pointwise complex structures need not arise from complex coordinates on Z.

A Riemann surface can be described similarly as a smooth real surface Σ equipped with an endomorphism $j\colon T\Sigma \to T\Sigma$ satisfying $j^2 = -1$. In real dimension two every almost complex structure is integrable, but this fact will not be needed. The tangent-bundle formulation puts the differential equation directly in view.

We begin with the relevant linear algebra. Let (V, j) and (W, J) be complex vector spaces regarded as real vector spaces with chosen complex structures. For a real-linear map $L\colon V \to W$, define

$$L^{1,0} = \frac{1}{2}(L - JLj), \qquad L^{0,1} = \frac{1}{2}(L + JLj). \tag{14.1}$$

Lemma 14.1 (Complex-linear decomposition)**.** *The map $L^{1,0}$ is complex linear, the map $L^{0,1}$ is complex antilinear, and*

$$L = L^{1,0} + L^{0,1}.$$

Consequently, L is complex linear if and only if $L^{0,1} = 0$.

Proof. Using $j^2 = J^2 = -1$ gives

$$L^{1,0} j = \tfrac{1}{2}(Lj + JL) = JL^{1,0}, \qquad L^{0,1} j = \tfrac{1}{2}(Lj - JL) = -JL^{0,1}.$$

The sum of the two parts is L. Thus $L^{0,1} = 0$ implies that L is complex linear; conversely, $Lj = JL$ gives $JLj = -L$ and hence $L^{0,1} = 0$. □

For $p \in \Sigma$ and $x \in Z$, write

$$\operatorname{Hom}^{0,1}_{j,J}(T_p\Sigma, T_xZ)$$

for the real-linear maps $A \colon T_p\Sigma \to T_xZ$ satisfying $A \circ j = -J \circ A$. These vector spaces form a vector bundle over $\Sigma \times Z$. Restriction along the graph of a map u gives the bundle of complex-antilinear one-forms with values in u^*TZ, whose sections are denoted

$$\Omega^{0,1}(\Sigma, u^*TZ) = \Gamma\big(\operatorname{Hom}^{0,1}_{j,J}(T\Sigma, u^*TZ)\big).$$

Definition 14.2 (Nonlinear Cauchy–Riemann operator)**.** For a smooth map $u \colon \Sigma \to Z$, define

$$\bar\partial_J u = \frac{1}{2}\big(du + J \circ du \circ j\big) \in \Omega^{0,1}(\Sigma, u^*TZ). \tag{14.2}$$

The map u is *J-pseudo-holomorphic*, or *J-holomorphic*, if $\bar\partial_J u = 0$.

By Lemma 14.1, the equation is equivalent to

$$du \circ j = J \circ du.$$

When $Z = \mathbb{C}^n$ with its standard complex structure, this is the ordinary Cauchy–Riemann equation in local coordinates.

The dependence on the value $u(p)$ makes this a nonlinear equation in general: varying u changes the endomorphism $J_{u(p)}$ as well as du.

The equation is invariant under reparametrization. If $\varphi \colon (\Sigma', j') \to (\Sigma, j)$ is biholomorphic, then

$$\bar\partial_J(u \circ \varphi) = 0$$

whenever $\bar\partial_J u = 0$. Indeed, $d\varphi \circ j' = j \circ d\varphi$, and substitution in Equation (14.2) proves the claim. This elementary invariance is the first indication that a moduli object for varying domains must retain automorphisms.

Steffens formulates the application for a symplectic manifold (Z, ω) equipped with an ω-tame almost complex structure, meaning

$$\omega(v, Jv) > 0$$

for every nonzero tangent vector v. The symplectic form is fundamental for energy estimates and compactness, but the equation, the symbol calculation, and the local representability argument below use only J. We will state the representability theorem first for an almost complex target and return to the role of ω at the end.

14.2 The fixed-domain solution stack

The output of the Cauchy–Riemann operator lies in a bundle depending on the map. First jets define the global section and its derived zero locus. Near any chosen map, trivializing the equation bundle gives a single operator between sections of fixed vector bundles.

The equation depends only on the value and first derivative of u. A first jet at $p \in \Sigma$ can be represented by a pair

$$(x, L), \qquad x \in Z, \quad L\colon T_p\Sigma \longrightarrow T_xZ.$$

The complex-antilinear projection defines a smooth map on first jets,

$$(x, L) \longmapsto \left(x, \frac{1}{2}(L + J_x L j_p)\right). \tag{14.3}$$

Thus pseudo-holomorphicity is a first-order differential equation in the sense of Chapter 10.

For the moment keep the domain fixed. The smooth mapping object

$$\mathcal{B} = C^\infty(\Sigma, Z)$$

has tangent space

$$T_u\mathcal{B} = \Gamma(\Sigma, u^*TZ)$$

at u. Over $\mathcal{B}$ there is a bundle $\mathcal{F} \to \mathcal{B}$ whose fiber is

$$\mathcal{F}_u = \Omega^{0,1}(\Sigma, u^*TZ).$$

The nonlinear Cauchy–Riemann operator is a section

$$\bar\partial_J\colon \mathcal{B} \longrightarrow \mathcal{F}.$$

To take its derived zero locus, regard these as derived section stacks:

$$\mathcal{B} = \mathrm{Sec}_\Sigma(\Sigma \times Z), \qquad \mathcal{F} = \mathrm{Sec}_\Sigma(H^{0,1}),$$

where $H^{0,1} \to \Sigma \times Z$ is the bundle defined above and the second section stack uses its composite map to Σ. The comparison in Chapter 12 identifies their smooth-stack restrictions with the indicated convenient manifolds. The derived solution object is $\mathcal{B} \times^{\mathrm{der}}_{\mathcal{F}} \mathcal{B}$, using $\bar\partial_J$ and the zero section.

Fix a map u. Exponential coordinates identify nearby maps with a neighbourhood Q of zero in $\Gamma(\Sigma; F)$, where $F = u^*TZ$. Choose a connection preserving J. Parallel transport along the short exponential paths gives complex-linear isomorphisms $\tau_v\colon u^*TZ \to (\exp_u v)^*TZ$. These identify the nearby bundles of complex-antilinear forms with

$$E = \mathrm{Hom}^{0,1}_{j,J}(T\Sigma, u^*TZ).$$

The transformed equation is

$$P(v) = \tau_v^{-1}\bar\partial_J(\exp_u v)\colon Q \longrightarrow \Gamma(\Sigma; E). \tag{14.4}$$

All coordinate changes depend pointwise on v. Thus P is still a first-order differential operator between sections of fixed vector bundles. The local section-stack comparison identifies its derived zero fibre with an open part of the original solution stack.

This is the form needed for the argument of Chapters 11 to 13. The first-jet presentation in Steffens's Construction 4.0.1 requires a correction to its output bundle and a local reduction before applying the fixed-order ellipticity criterion. The correction and the local reduction are discussed in Section 14.7.

14.3 Linearization and ellipticity

We differentiate at a solution and calculate the principal symbol of the resulting operator. The symbol is invertible on every nonzero covector, so the elliptic theory of the preceding chapter applies. The index formula and the constant-map example describe the resulting finite-dimensional linear data.

Let u be pseudo-holomorphic. Because $\bar{\partial}_J u = 0$, the vertical derivative of the section at u is intrinsically defined:

$$D_u \colon \Gamma(\Sigma, u^*TZ) \longrightarrow \Omega^{0,1}(\Sigma, u^*TZ).$$

Choose a torsion-free connection ∇ on TZ in order to write it.

Proposition 14.3 (Linearized Cauchy–Riemann operator)**.** *For $\xi \in \Gamma(\Sigma, u^*TZ)$, the linearization is*

$$D_u\xi = \frac{1}{2}\big(\nabla\xi + J \circ \nabla\xi \circ j\big) + \frac{1}{2}(\nabla_\xi J) \circ du \circ j. \tag{14.5}$$

In particular, D_u is a real Cauchy–Riemann type operator: its first-order part is a Cauchy–Riemann operator, while the second summand is zeroth order.

Proof. Choose a smooth family u_t with $u_0 = u$ and $\frac{d}{dt}|_{t=0} u_t = \xi$. Covariant differentiation gives

$$\left.\frac{D}{dt}\right|_{t=0} du_t = \nabla\xi.$$

Differentiating the target almost complex structure gives

$$\left.\frac{D}{dt}\right|_{t=0} J(u_t) = \nabla_\xi J.$$

The complex structure j on the fixed domain does not vary. Applying the product rule to Equation (14.2) proves Equation (14.5).

The connection is used only to compare the bundles u_t^*TZ for varying t. The vertical derivative of a bundle section at a zero is independent of this choice, so D_u is intrinsic. □

The first summand differentiates ξ. The term involving $\nabla_\xi J$ contains no derivative of ξ, and hence contributes nothing to the principal symbol. Consequently the principal symbol is the same as that of the first summand.

For a first-order differential operator, the principal symbol is obtained by retaining only the coefficient of the first derivative. Fix $p \in \Sigma$, $0 \neq \zeta \in T_p^*\Sigma$, and $v \in T_{u(p)}Z$.

Theorem 14.4 (Ellipticity of the Cauchy–Riemann operator)**.** *The principal symbol of D_u is*

$$\sigma_\zeta(D_u)(v)(Y) = \frac{1}{2}\big(\zeta(Y)v + \zeta(jY)Jv\big). \tag{14.6}$$

For every nonzero ζ, this is an isomorphism

$$T_{u(p)}Z \xrightarrow{\sim} \operatorname{Hom}^{0,1}_{j,J}(T_p\Sigma, T_{u(p)}Z)$$

of real vector spaces. Consequently, D_u is elliptic.

Proof. The first-order part of D_u is

$$\xi \longmapsto \frac{1}{2}(\nabla\xi + J \circ \nabla\xi \circ j).$$

Replacing $\nabla_Y \xi$ by $\zeta(Y)v$ gives Equation (14.6).

First observe that the resulting map is complex antilinear in Y. Indeed,

$$\begin{aligned}\sigma_\zeta(D_u)(v)(jY) &= \frac{1}{2}\big(\zeta(jY)v - \zeta(Y)Jv\big) \\ &= -J\sigma_\zeta(D_u)(v)(Y).\end{aligned}$$

Choose a j-compatible inner product on $T_p\Sigma$ and let Y be the metric dual of ζ. Then

$$\zeta(Y) > 0, \qquad \zeta(jY) = 0.$$

If $\sigma_\zeta(D_u)(v) = 0$, evaluation at Y gives

$$0 = \frac{1}{2}\zeta(Y)v,$$

and therefore $v = 0$. The symbol is injective. Since $T_p\Sigma$ has complex dimension one, a complex-antilinear map out of $T_p\Sigma$ is determined by its value on one nonzero real direction. Hence the source and target of the symbol have the same real dimension, so the symbol is an isomorphism. □

This proof is pointwise. It uses neither a symplectic form on Z nor integrability of J. It also works uniformly when the domain and the map vary in smooth families.

Since Σ is closed, after completing in Sobolev norms, the elliptic Fredholm theorem applies.

Corollary 14.5 (Fredholm deformation operator)**.** *For sufficiently large k, the extension*

$$D_u \colon H^{k+1}(\Sigma, u^*TZ) \longrightarrow H^k\Omega^{0,1}(\Sigma, u^*TZ)$$

is Fredholm. Its kernel and cokernel are finite-dimensional and are represented by smooth sections.

Proof. Fredholmness follows from Theorem 14.4 and the elliptic Fredholm theorem used in Theorem 13.2. Elliptic regularity identifies the Sobolev kernel and cokernel with their smooth counterparts. □

If Z has complex dimension n, Riemann–Roch gives the real index

$$\operatorname{ind}_{\mathbb{R}}(D_u) = n\chi(\Sigma) + 2\,\langle c_1(u^*TZ), [\Sigma]\rangle\,. \tag{14.7}$$

This is the Riemann–Roch formula [Wen14, Theorem 3.22]. It is the virtual dimension of the moduli problem relative to the fixed domain. For an unparametrized problem with varying domains, the deformation theory of the domain must also be included.

Example 14.6 (Constant maps)**.** Fix $x \in Z$ and let $u_x \colon \Sigma \to Z$ be the constant map with value x. Since $du_x = 0$, it is pseudo-holomorphic, and the zeroth-order term in Equation (14.5) vanishes. Thus D_{u_x} is the Dolbeault operator on the trivial complex vector bundle with fiber T_xZ:

$$\bar\partial \colon \Omega^0(\Sigma) \otimes_{\mathbb{C}} T_xZ \longrightarrow \Omega^{0,1}(\Sigma) \otimes_{\mathbb{C}} T_xZ.$$

For connected Σ this gives

$$\ker D_{u_x} \cong T_x Z, \qquad \operatorname{coker} D_{u_x} \cong H^{0,1}(\Sigma) \otimes_{\mathbb{C}} T_x Z. \tag{14.8}$$

Indeed, holomorphic functions on a closed connected Riemann surface are constant, giving the kernel. Dolbeault cohomology is, by definition, the cokernel of the scalar $\bar\partial$ operator on a surface, giving the second formula. If Σ has genus g and $\dim_{\mathbb{C}} Z = n$, these groups have real dimensions $2n$ and $2ng$. The index is $2n(1-g)$.

For a sphere, the cokernel vanishes. For a positive-genus domain, constant maps have nonzero obstruction space, although the visible family of constant maps is the ordinary manifold Z. The calculation concerns the full mapping problem at a constant map; it does not identify the entire moduli object with the constant-map locus.

These kernel and cokernel calculations give the linear data. The Kuranishi construction identifies their two-term complex with the intrinsic derived tangent complex, as we now explain.

14.4 Deformation theory with the domain fixed

The local Kuranishi construction turns the elliptic deformation complex into an intrinsic tangent complex. We use this identification to interpret regularity, obstruction groups, and virtual dimension while keeping the domain fixed.

Let

$$\mathcal{M}_{(\Sigma,j)}(Z,J)$$

denote the derived solution stack for the fixed domain (Σ, j). On ordinary test manifolds, a T-point is a smooth map

$$u\colon T \times \Sigma \longrightarrow Z$$

whose restriction to every slice is pseudo-holomorphic. On a derived test object, the equation is imposed by a derived zero locus and hence retains its infinitesimal intersection data.

The analytic deformation complex is the two-term complex with differential D_u. Its identification with a derived tangent complex follows from the finite-dimensional Kuranishi reduction.

Theorem 14.7 (Fixed-domain tangent complex)**.** *At a pseudo-holomorphic map $u\colon (\Sigma, j) \to (Z, J)$, the tangent complex of the fixed-domain derived solution stack is*

$$\mathbb{T}_u \mathcal{M}_{(\Sigma,j)}(Z,J) \simeq \left[\Gamma(\Sigma, u^*TZ) \xrightarrow{D_u} \Omega^{0,1}(\Sigma, u^*TZ)\right] \tag{14.9}$$

in homological degrees 0 *and* -1.

Proof. Use the local operator of Section 14.2. Its derivative at the chosen zero is D_u, which is elliptic. The construction of Chapter 13 gives a finite-dimensional chart $\kappa\colon Z_U \to V$ and a quasi-isomorphism

$$[T_u Z_U \longrightarrow V] \longrightarrow [\Gamma(\Sigma, u^*TZ) \xrightarrow{D_u} \Omega^{0,1}(\Sigma, u^*TZ)].$$

The left side is the tangent complex of the finite-dimensional derived zero locus, by the calculation in Chapter 6. This proves the formula without applying a finite-dimensional cotangent theorem directly to the infinite-dimensional mapping object. □

Consequently,

$$H_0\mathbb{T}_u \cong \ker D_u,$$
$$H_{-1}\mathbb{T}_u \cong \operatorname{coker} D_u.$$

The first group is the space of infinitesimal deformations of the map with the domain fixed. The second is its obstruction space. Both are finite-dimensional by Corollary 14.5.

Definition 14.8 (Regular pseudo-holomorphic map)**.** A pseudo-holomorphic map u is *regular* if D_u is surjective.

At a regular map the obstruction group vanishes. The Banach implicit function theorem then identifies the nearby ordinary solution space with a smooth manifold of dimension $\operatorname{ind}_{\mathbb{R}}(D_u)$. Derivedly, the tangent complex is concentrated in degree 0.

If the cokernel is nonzero, it receives obstructions to extending an infinitesimal deformation to an actual family. A particular obstruction class can still vanish. The Kuranishi map describes these nonlinear conditions; its derivative alone need not determine them.

The virtual dimension at u is

$$\dim H_0\mathbb{T}_u - \dim H_{-1}\mathbb{T}_u = \operatorname{ind}_{\mathbb{R}}(D_u),$$

which is given by Equation (14.7).

14.5 Varying domains and relative representability

We now retain the complex structure and automorphisms of the domain as part of the moduli problem. The stack of compact Riemann surfaces supplies a universal proper family. The relative Cauchy–Riemann section defines the solution stack over this base, and the local elliptic argument proves its relative representability.

The fixed-domain problem ignores reparametrizations and variations of the complex structure. To retain both, let

$$S = \operatorname{Surf}_{\mathbb{C}}$$

denote the smooth stack of compact Riemann surfaces. For a test manifold T, an object of $S(T)$ is a proper submersion

$$\pi\colon C \longrightarrow T$$

with compact surface fibers, together with a smooth complex structure on the vertical tangent bundle $T(C/T)$. Morphisms are Cartesian diagrams whose maps on fibers are biholomorphic.

The stack retains smooth dependence on parameters and the automorphisms of each surface. For example, the Riemann sphere has automorphism group $\mathrm{PSL}_2(\mathbb{C})$, which appears as isotropy at its point of S.

The stack carries a universal family

$$M = \widetilde{\operatorname{Surf}}_{\mathbb{C}} \longrightarrow S = \operatorname{Surf}_{\mathbb{C}}. \tag{14.10}$$

Its fiber over (Σ, j) is the surface Σ itself. In particular, $M \to S$ is a proper family of smooth compact manifolds, which is exactly the domain hypothesis in the elliptic representability theorem.

Kodaira–Spencer theory implies that S is a smooth Artin C^∞-stack. We quote this result, as Steffens does in [Ste24, Remark 4.0.3]. Its role will be to turn relative representability over S into an absolute derived Artin statement.

The Artin terminology allows smooth submersive atlases, with representable overlaps, in place of the open atlases for schemes. Such a stack can have positive-dimensional groups of automorphisms. In the derived version the atlas objects are derived schemes. Pulling a smooth atlas of S back to a relatively representable moduli stack gives an atlas of this kind. Here a derived Artin stack is called quasi-smooth when it admits a smooth atlas by quasi-smooth derived schemes.

Put $Y = Z \times M$ over M. A section of Y over a family $C \to T$ is a smooth family of maps $C \to Z$. The corresponding section stack

$$\mathcal{B} = \mathrm{Sec}_{M/S}(Y)$$

parametrizes the unknown maps.

Over Y there is a finite-rank vector bundle $H^{0,1} \to Y$ whose fiber at (c, x) is

$$\mathrm{Hom}^{0,1}_{j_c, J_x}\big(T_c(M/S), T_x Z\big).$$

Pulling this bundle back along a map-section and taking relative sections produces a bundle $\mathcal{F} \to \mathcal{B}$. Its fiber at a family u is the family of spaces

$$\Omega^{0,1}(C_t, u_t^* TZ).$$

The complex-antilinear projection of the relative first jet defines a section

$$\mathrm{CR}\colon \mathcal{B} \longrightarrow \mathcal{F}, \qquad u \longmapsto \bar\partial_J u.$$

The construction uses only the vertical tangent bundle, the fiberwise complex structure, and J. It is therefore natural under Cartesian base change of families of surfaces.

Definition 14.9 (Derived moduli stack of pseudo-holomorphic maps)**.** The *derived moduli stack of pseudo-holomorphic maps*, denoted $\mathcal{M}(Z, J)$, is the derived zero locus of the Cauchy–Riemann section over $\mathcal{B}$.

It comes with a natural morphism

$$\mathcal{M}(Z, J) \longrightarrow \mathrm{Surf}_{\mathbb{C}}\,.$$

Over an ordinary test manifold $T \to S$ classifying $C \to T$, its ordinary points are precisely smooth maps $u\colon C \to Z$ whose restrictions to the fibers are pseudo-holomorphic. The derived zero locus retains the higher intersection data of this equation.

Theorem 14.10 (Pseudo-holomorphic representability)**.** *Let (Z, J) be an almost complex manifold. The morphism*

$$\mathcal{M}(Z, J) \longrightarrow \mathrm{Surf}_{\mathbb{C}} \tag{14.11}$$

is representable by quasi-smooth derived C^∞-schemes locally of finite presentation. Consequently, $\mathcal{M}(Z, J)$ is a quasi-smooth derived Artin C^∞-stack.

Proof. Representability is local on the target. Pull back along a map $T \to \mathrm{Surf}_{\mathbb{C}}$ from a manifold. It classifies a smooth proper family of Riemann surfaces $C \to T$. Locally on T, Ehresmann triviality identifies the underlying family with $T' \times \Sigma \to T'$, although the fiberwise complex structure may still vary with the parameter.

Fix a solution u in this family. The local section-stack construction from Section 14.2 identifies a neighborhood of u with $T' \times Q$, where Q is an open subset of a smooth section space. After trivializing the nearby equation bundles, the family Cauchy–Riemann section becomes a smooth family of nonlinear first-order operators

$$P\colon T' \times Q \longrightarrow T' \times \Gamma(\Sigma; E). \tag{14.12}$$

Its vertical linearization at u is D_u, which is elliptic by Theorem 14.4.

The local Kuranishi construction of Chapters 12 and 13 applies to this operator. It produces a quasi-smooth derived C^∞-scheme locally of finite presentation representing a neighborhood of u in the solution stack. These neighborhoods cover the pullback over T, and the open-atlas criterion glues them. This proves relative representability.

Finally, $\mathrm{Surf}_{\mathbb{C}}$ is a smooth Artin C^∞-stack. Relative quasi-smooth representability over this base makes the total solution stack a quasi-smooth derived Artin C^∞-stack. □

Steffens states this conclusion in the symplectic tame setting in [Ste24, Construction 4.0.1 and Remark 4.0.3]. The proof above uses the local zero-section form of the representability argument, for the reason explained in Warning 14.11. It also shows that the local representability statement depends only on J.

In particular, given a manifold T and a specified family $C \to T$ of Riemann surfaces, the pullback

$$T \times_{\mathrm{Surf}_{\mathbb{C}}} \mathcal{M}(Z, J)$$

is represented by a quasi-smooth derived C^∞-scheme locally of finite presentation over T. Its relative tangent complex at a map u is

$$\mathbb{T}_u\big(\mathcal{M}(Z,J)/\mathrm{Surf}_{\mathbb{C}}\big) \simeq \Big[\Gamma(\Sigma, u^*TZ) \xrightarrow{D_u} \Omega^{0,1}(\Sigma, u^*TZ)\Big]. \tag{14.13}$$

The absolute tangent complex also contains deformations and infinitesimal automorphisms of the domain. The corresponding fiber sequence is recorded in Equation (14.16).

14.6 Compactification and enumerative questions

Properness of the domain family ensures compact fibres on which elliptic analysis applies. Compactness of the solution object is another issue. Even with compact target, sequences of maps can develop bubbles, and smooth domains can degenerate to nodal curves. These limits require an enlarged moduli problem, typically stable nodal maps with fixed genus, marked points, and homology class.

Representing that compactification requires analysis near its nodal objects, including gluing. An enumerative application then needs a properness theorem and appropriate orientations and virtual classes. Chapter 8 supplies a bordism class for a compact quasi-smooth derived manifold. The total moduli stack here may have isotropy, so a virtual-class construction for stacks requires additional theory.

When J is tamed by a symplectic form ω, the area $\int_\Sigma u^*\omega$ is positive for a nonconstant pseudo-holomorphic map from a connected domain, and depends only on its homology class. With the associated J-invariant metric, it equals the energy of the map. Thus fixing the class gives an energy bound, an essential input to compactness. The bound allows bubbling and explains why nodal maps enter the compactification.

14.7 Further reading

The following discussion compares the local equation used in the proof with the symmetric presentation in the source, records the absolute tangent complex when the domain varies, and describes further choices of geometric data.

Warning 14.11 (The printed five-tuple)**.** In Steffens's Construction 4.0.1, the common output of the displayed Cauchy–Riemann and zero maps is the full relative first-jet bundle. Taken literally, its linearized matching complex has complementary complex-linear jet directions in degree minus one, in addition to the Cauchy–Riemann Fredholm complex.

Replacing the common output by the bundle of complex-antilinear maps removes those extra obstruction directions. The resulting symmetric matching operator still has a zeroth-order component, equating the underlying map values, and a first-order Cauchy–Riemann component. It is therefore not elliptic in the fixed-order sense of [Ste24, Definition 3.4.2]. The general theorem cannot be invoked verbatim from that five-tuple.

The derived intersection of the Cauchy–Riemann section with the zero section can be displayed as

$$\begin{array}{ccc} \mathcal{M}(Z,J) & \longrightarrow & \mathcal{B} \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \mathrm{CR}} \\ \mathcal{B} & \xrightarrow[0]{} & \mathcal{F}. \end{array} \tag{14.14}$$

At a solution u, put

$$A = \Gamma(\Sigma, u^*TZ), \qquad B = \Omega^{0,1}(\Sigma, u^*TZ).$$

Linearizing this pullback presentation gives

$$\Big[A \oplus A \xrightarrow{\delta} A \oplus B\Big], \qquad \delta(\xi_1, \xi_2) = (\xi_1 - \xi_2, D_u\xi_1). \tag{14.15}$$

The first component differentiates equality of the underlying maps, and the second differentiates the Cauchy–Riemann equation.

Lemma 14.12 (Cancellation of the repeated field)**.** *There is a chain isomorphism*

$$[A \oplus A \xrightarrow{\delta} A \oplus B] \xrightarrow{\sim} [A \xrightarrow{D_u} B] \oplus [A \xrightarrow{\mathrm{id}} A].$$

Consequently, Equation (14.15) *is quasi-isomorphic to the Cauchy–Riemann deformation complex.*

Proof. On the source use the coordinates

$$(\xi_1, \xi_2) \longmapsto (\xi_1, \xi_1 - \xi_2).$$

Reorder the target $A \oplus B$ as $B \oplus A$. In these coordinates the differential is

$$(\xi, r) \longmapsto (D_u\xi, r),$$

which is the displayed direct sum. The identity complex is contractible. □

The cancellation recovers the Cauchy–Riemann deformation complex. Its identification with the intrinsic tangent complex uses the local Kuranishi reduction proved above. The first-order symbol of the displayed matching operator is still $(\xi_1, \xi_2) \mapsto (0, \sigma(D_u)\xi_1)$, which is not invertible. Passing to the single local equation removes the repeated map variable before applying elliptic analysis.

The complex in Equation (14.13) holds the domain and its complex structure fixed. The absolute tangent complex also contains the deformation theory of (Σ, j). It fits into the relative tangent fiber sequence

$$\mathbb{T}_u\big(\mathcal{M}(Z,J)/\mathrm{Surf}_{\mathbb{C}}\big) \longrightarrow \mathbb{T}_u\mathcal{M}(Z,J) \longrightarrow \mathbb{T}_{(\Sigma,j)}\,\mathrm{Surf}_{\mathbb{C}}\,. \tag{14.16}$$

The last term records deformations and infinitesimal automorphisms of the domain. For a smooth Riemann surface, these are measured by $H^1(\Sigma, T\Sigma)$ and $H^0(\Sigma, T\Sigma)$, respectively, placed in tangent degrees 0 and 1. Thus passing from the relative complex to the absolute one includes both changes of complex structure and reparametrizations. In this example the absolute tangent complex has amplitude in $[-1, 1]$; the degree 1 term accounts for infinitesimal automorphisms. This is compatible with quasi-smoothness of the Artin stack as defined through its smooth atlases.

This distinction prevents an ambiguity in expected-dimension formulas. The index in Equation (14.7) is the relative virtual dimension for a fixed domain. An unparametrized moduli problem with varying domains must also account for $\mathbb{T}_{(\Sigma,j)}\,\mathrm{Surf}_{\mathbb{C}}$.

For connected genus-g domains and no marked points, Riemann–Roch for $T\Sigma$ gives the real virtual dimension $6g - 6$ of the domain stack. Adding this to the relative index gives

$$\mathrm{vdim}_{\mathbb{R}}\,\mathcal{M}(Z,J) = (n-3)(2-2g) + 2\langle c_1(u^*TZ), [\Sigma]\rangle.$$

In genus zero the contribution -6 accounts for the six-dimensional reparametrization group. This is an expected dimension for the stack, including its infinitesimal automorphisms.

Several variations fit the same representability argument.

(1) One may let J vary in a smooth family by enlarging the base stack.

(2) One may allow a smooth family of almost complex target manifolds.

(3) One may add marked points by adding sections of the universal curve to the objects classified by the domain stack.

(4) One may formulate equivariant problems over quotient stacks, thereby retaining isotropy rather than passing to a naive orbit set.

The first three variants are noted in [Ste24, Remark 4.0.2]. All still concern smooth compact domains.

Exercise 14.13. At a constant map from a genus-g surface to a complex n-fold, compute the dimensions of the relative deformation and obstruction spaces. Check the Riemann–Roch formula.

Exercise 14.14. Write the first-order principal symbol of the symmetric matching operator in Equation (14.15). Show why it fails to be invertible even though its complex is quasi-isomorphic to the elliptic Cauchy–Riemann complex.

The primary derived-geometric source is Steffens [Ste24]. The construction in [Ste24, Construction 4.0.1] defines the stack of smooth compact Riemann surfaces, its universal family, the jet-level Cauchy–Riemann map, and the solution stack. Varying data, marked points, and the derived Artin conclusion are discussed in [Ste24, Remarks 4.0.2–4.0.3]. The source correction needed for the printed five-tuple is recorded in Warning 14.11.

For the analytic background, Wendl's *Lectures on Holomorphic Curves in Symplectic and Contact Geometry* develops almost complex structures and pseudo-holomorphic maps in Section 2.1, real Cauchy–Riemann type operators and their linearization in Sections 2.3–2.4, regularity in Section 2.5, and Fredholm and index theory in Sections 3.3–3.4 [Wen14]. These sections support the linearization, symbol, Fredholm, and Riemann–Roch statements used here.

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